

CS201

Mid-Term

Quiz by Attiq Kundi

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CS 201

Also Added Today Quiz

Q 1; The ASCII code of null character is -----.

(000) correct answer

Q2; Which step is not performed during a proper program design?

(Write details about the performance of program) correct answer

Q3; How many elements are stored in the following ? int matrix [4][5];

(20) correct answer

Q4; Which of the following is not a valid variable name in c++?

(@studentName) correct answer

Q5; If an array has 50 elements.What is allowable range of subscripts?

(0-49) correct answer

Q6; <, <=, >, >= are called -----operators.

(Relational) correct answer

Q7; Why we use “cin “?

(To read data from keyboard) correct answer

Q8; !(x<3)means in c++ that

(x is greater than 3) correct answer

Q9; The data in the memory is -----

(volatile) correct answer

Q10; Each array declaration must provide the information about all these things except -----

The location of the first element to be stored

Q11; What will be the output of the program if x=10?

If (x!=0&& x<=10)

```
{
Cout <<"condation 1 is true";
}
If(x>=10 && x<=20)
{
```

Condation 1 is true condation 2 is true

Q12; C is widely known developpment language of_____operating system.

Windows

Q13; A repetition statement specifies that an action is to be continued while some_____remains true.

Condation

Q14;_____is a substitute of multiple if statement.

If—elseif statement

Q15; If a_____finds an error. It stops translating without generating any executable code.

Compiler

Q16; Function prototype is wrriten .

Before call of that function

Q17; The_____is an unconditional branch of execution .

Goto statement

Q18; What will be the output of the following code snippet?

```
String str1 = "Hello beautiful";
String str2 = "yes we can";
String str3 = strcpy(str1,str2);
Cout<<str3<<endl;
```

Hello beautiful Yes we can

Q20; In two dimensional array .We use_____,for oop(s)"to populate the array.

2

Q21; eof is used to check for the_____of file when a file is being read.

End

Q22; How many times the following loop will execute?

```
Int j = 3;
```

```
While(j>0)
```

```
{
```

```
Cout << "statements" << endl;
```

```
J=2;
```

```
}
```

2

Q23; Null character is used to indicate the_____of string.

End

Q24; There is a pointer named ptr of type float. Which type of variable address can be stored by ptr?

Float

Q25; How many bytes an integer type pointer intPtr will jump in memory if the statement below is executed?

```
intPtr +=2;
```

4

Q26; We must include the header file_____to convert the value of one type into another type using built-in functions.

iostream.h

Q27; Use of word processor for writing the code of a program is not good because.

It has no compiler

Q28;_____Loop executes atleast one times.

Do while

Q29; The data . contained in the structure. is defined in the_____

Curly braces

Q30; What is the use of ios::trunc mode?

To truncate an existing file to zero

CS201 Solved Quiz No-1 December 13, 2015

1. We use _____ to open a file or stream for extraction
in
2. Null character is represented by _____ in C++.
/0
3. An object of _____ class can be created for saving the current position of any opened file before reading from the file.
Streampos
pg215
4. _____ Returns true if c is a digit or a letter and false otherwise.
int isalnum(int c)
5. We use _____ to open a file or stream for insertion.
out
6. Base address is the memory address of _____ element of an array.
1st
7. _____ allow us to have a memory location and use it as int or char interchangeably
union
8. To read command-line arguments, the main() function itself must be given _____ arguments.
2
9. Individual characters in a string stored in an array can be accessed directly using array _____.
Subscript
10. To manipulate n-dimensional array _____ nested loops are required.
n-1
11. A character is stored in the memory in _____.
String
12. What will be the size of following array?

```
int arr[29];
```

29

13. An object of _____ class can be created for saving the current position of any opened file before reading from the file.

Streampos

pg215

14. We use _____ to open a file or stream for extraction.

In

15. We use _____ to open a file or stream for insertion.

Out

16. _____ are conventional names of the command line parameters of the „main()“ function.

__argc‘ and __argv‘

17. In _____, we have different data members and all of these have their own memory space.

Structures

18. How we declare an array of integer data whose size is 20 with array name “count”.

int count[20];

19. A _____ is an array of characters that can store number of character specified.

String

20. File. seekg (-10L, ios:cur)

backward direction

21. What will be the output of the following while loop? `int counter = 0 ; while(counter < 15) cout << counter << “ ”; counter++;`

1 2 3 4

1 2 3 4 5

Compile time error

Infinite loop

22. We can access a global variable _____

From the functions only

From the loops only

From the main() function only

From anywhere in the program

23. A repetition statement specifies that an action is to be continued while some _____ remains true.

Condition

Control

Assignment

Operator

24. Loops are _____ Structure.

Decision

Sequential

Repetition

None of the given options

25. _____ operator is used to pass the address of a variable in call by reference method.

%

+

@

&

26. How many times the following do-while loop will execute? `int k = 10; do { cout << "Statements" << endl; k -= 2; } while(k>0);`

4

5

6

7

27. What is the output of the following code if the 3rd case is true `switch (var) { case „a“: cout<<”apple”<<endl; case „b”: cout<<”banana”<<endl; case „m”: cout<<”mango”<<endl; default: cout<<”any fruit”<<endl; }`

mango

mango any fruit

Apple

CS201 SOLVED QUIZ NO 1 DECEMBER 09, 2019

Q1; The ASCII code of null character is_____

010

Q2; Each array declaration must provide the information about all these things except_____

The location of the first element to be stored

Q3; A record is a group of related_____

Data

Q4; Pointers store the_____

Value of the variable

Q5; We can say that a pointer is a variable that holds the____address.

Source code

Q6; `file.seekg(-10-ios::cur`

Backward direction

Q7; Dereferencing operator is represented by_____

***,+,- non of the given**

Question No: 1 (Marks: 1) - Please choose one

A precise sequence of steps to solve a problem is called

- Statement
- ☐ **Program**
- Utility
- Routine

Question No: 2 (Marks: 1) - Please choose one

The Compiler of C language is written in

- Java Language
- UNIX
- FORTRAN Language
- ☐ **C Language**

Question No: 3 (Marks: 1) - Please choose one

Initialization of variable at the time of definition is,

- Must
- Necessary
- ☐ **Good Programming**

- None of the given options

Question No: 4 (Marks: 1) - Please choose one

In if structure the block of statements is executed only,

- When the condition is false
- When it contain arithmetic operators
- When it contain logical operators

☐ **When the condition is true**

Question No: 5 (Marks: 1) - Please choose one

Which of the following function(s) is/are included in **stdlib.h** header file?

- double atof(const char *nptr)
- int atoi(const char *nptr)
- char *strcpy (char *s1, const char *s2)

☐ **1 and 2 only**

Question No: 6 (Marks: 1) - Please choose one

Dealing with structures and functions passing by reference is the most economical method

☐ **True**

- False

Question No: 7 (Marks: 1) - Please choose one

Pointer is a variable which store,

- Data

☐ **Memory Address**

- Data Type
- Values

Question No: 8 (Marks: 1) - Please choose one

Preprocessor program perform its function before _____phase takes place.

- Editing
- Linking

☐ **Compiling**

- Loading

Question No: 9 (Marks: 1) - Please choose one

Which of the following can not be a variable name?

- area
- _area

☐ **10area**

- area2

Question No: 10 (Marks: 1) - Please choose one

Which looping process is best, when the number of iterations is known?

- for

☐ **while**

- do-while
- all looping processes require that the iterations be known

Question No: 11 (Marks: 1) - Please choose one

Which character is inserted at the end of string to indicate the end of string?

- new line
- tab

☐ **null**

- carriage return

Question No: 12 (Marks: 1) - Please choose one

How many bytes are occupied by declaring following array of characters?

char str[] = —programming\;

- 10
- 11
- ☐ **12**
- 13

11 plus one for null char (11+1= 12)

Question No: 13 (Marks: 1) - Please choose one

Which of the following header file defines the **rand()** function?

- iostream.h
- conio.h
- ☐ **stdlib.h**
- stdio.h

Question No: 14 (Marks: 1) - Please choose one

Commenting the code _____

- ☐ **Makes a program easy to understand for others.**
- Make programs heavy, i.e. more space is needed for executable.
- Makes it difficult to compile
- All of the given options.

Question No: 15 (Marks: 1) - Please choose one

What's wrong with this for loop? **for (int k = 2, k <=12, k++)**

- the increment should always be ++k
- the variable must always be the letter i when using a for loop
- there should be a semicolon at the end of the statement
- **the commas should be semicolons**

Question No: 16 (Marks: 1) - Please choose one

For which array, the size of the array should be one more than the number of elements in an array?

- int
- double
- float
- ☐ **char**

Question No: 17 (Marks: 1)

To Which category of the software “Compiler and Interpreter” belongs?

- ☐ **Solution:**
They belong to system software.

Question No: 18 (Marks: 1)

What is the result of the expression $x = 2 + 3 * 4 - 4 / 2$

- ☐ **Solution:**
12

Question No: 19 (Marks: 2)

Write a declaration statement for an array of 10 elements of type float. Include an initialization statement of the first four elements to 1.0, 2.0, 3.0 and 4.0.

Solution:

float tmp [10] = {1.0,2.0,3.0,4.0};

Question No: 20 (Marks: 3)

Write down the output of the following code? int array[7], sum = 0;
for(int i=0;i<7;i++)
{
array[i] = i; sum+= array[i];
}
cout<< " Sum = " <<sum;

Solution:

21

Question No: 21 what will be the output of the following segment of C++ code?

```
int A[5] = {1 , 2, 3, 4};  
int i;  
for (i=0; i<5; i++)  
{  
A[i] = 2*A[i];  
cout << A[i] << " ";  
}
```

Solution:

2 4 6 8 0

Question No: 22 (Marks: 10)

Write a C++ program that will determine if a departmental store customer has exceeded the credit limit on a charge account.

Program should input the following facts in five variables

1. Account number
2. Balance at the beginning of month (Beginning balance)
3. total of all items charged by customer this month (charges)
4. total of all credits (credits)
5. allowed credit limit Calculate the new balance

New balance = Beginning balance + charges – credits

Determine if **new balance** exceeds the **allowed credit limit**. For those customers whose credit limit is exceeded. The program should display the message "Credit Limit exceeded."

Question No: 1 (Marks: 1) - Please choose one

What is the output of the following statement?

```
int i = 2.5; do { cout << i * 2; } while (i > 3 && i < 10);
```

- (a) 510
- (b) 5
- (c) 48
- (d) error

Question No: 2 (Marks: 1) - Please choose one

What does !(7) evaluate to in decimal where ! is a NOT operator?

- (a) 7
- (b) 8
- (c) 9
- (d) 10

Question No: 3 (Marks: 1) - Please choose one

The condition in while loop may contain logical expression but not relational expression.

- (a) True
- (b) False

Question No: 4 (Marks: 1) - Please choose one

Searching is easier when an array is already sorted

- (a) True
- (b) False

Question No: 5 (Marks: 1) - Please choose one

If an array has 100 elements, what is allowable range of subscripts?

- (a) - 99
- (b) 1 - 99
- (c) 0 - 100
- (d) 1 - 100

Question No: 6 (Marks: 1) - Please choose one

What will be the value of „a“ and „b“ after executing the following statements? a = 3;

b = a++;

- (a) 3, 4
- (b) 4, 4
- (c) 3, 3
- (d) 4, 3**

Question No: 7 (Marks: 1) - Please choose one

What will be the correct syntax to initialize all elements of two-dimensional array to value 0?

- (a) `int arr[2][3] = {0,0} ;`
- (b) `int arr[2][3] = {{0},{0}} ;`**
- (c) `int arr[2][3] = {0},{0} ;`
- (d) `int arr[2][3] = {0} ;`

Question No: 8 (Marks: 1) - Please choose one

Which of the following function returns the size of a string variable?

- (a) `strlen()`
- (b) `stringlen()`
- (c) `strlen()`**
- (d) `strLength()`

Question No: 9 (Marks: 1) - Please choose one

What will be the range of numbers generated by function `rand () % 9`?

- (a) 0 to 9
- (b) 1 to 9
- (c) 0 to 8
- (d) 1 to 8

Question No: 11 (Marks: 1) - Please choose one

Computer can understand only machine language code.

- (e) True**
- (c) False**

QuestionNo:13 (Marks: 1) - Please choose one

What does $5 \wedge 6$, evaluate to in decimal where „ $\wedge$ “ is Exclusive OR operator?

- (a) True
- (b) Fals**

Detail:-

It mean

5 = 0101

6 = 0110

$5 \wedge 6 = 0011$

If both input is same then the output is 0 and if different then output is 1

Question No: 14 (Marks: 1) - Please choose one

If the file is not properly closed in the program, the program_____.

- (a) **Terminate normally**
- (b) Indicate Runtime error
- (c) Indicate Compile time error
- (d) Crashes

Question No: 15 (Marks: 1) - Please choose one

Which of the following header file include string conversion functions?

- (a) **string.h**
- (b) stdlib.h
- (c) ctype.h
- (d) sconvert.h

Question No: 16 (Marks: 1) - Please choose one

In Program commenting the code liberally is

Solution:-

It need to be self-contained and understandable. Comments should be placed liberally. The comments should explain the logic, not the mechanics. Try to avoid fancy programming.

Question No: 17 (Marks: 1)

Which header file must be included while handling files?

Solution:-

<fstream.h>

Question No: 18 (Marks: 1)

What is meant by C++ statement: `const int *ptr = &x;`

Solution:-

ptr is a pointer to data of type `const int` type. And to assign the address of x to pointer ptr

Question No: 19 (Marks: 2)

What is a truth Table?

Solution:-

We know the concept of truth table. The truth tables are very important. These are still a tool available for analyzing logical expressions. We will read logic design in future, which is actually to do with chips and gate. We find it difficult to evaluate a complicated logical expression. Sometimes the logic becomes extremely complicated so that even writing it as a simple syntax statement in any language.

QuestionNo:20 (Marks: 3)

(1) An array day is declared as: `int day[] = {1, 2, 3, 4, 5, 6, 7};`

How many elements does array 'day' has?

Solution:-

7 elements

(2) If the declaration is changed as: `int day[7] = {1, 2, 3, 4, 5, 6, 7};`

How many elements does array 'day' has?

Solution:-

7 elements

QuestionNo:21 (Marks: 5)

What are similarities and differences between Structures and Unions?

In structures, we have different data members and all of these have their own memory space. In union, the memory location is same while the first data member is one name for that memory location. However, the 2nd data member is another name for the same location and so on. Consider the above union (i.e. `intOrChar`) that contains an integer and a character as data members. What will be the size of this union? The answer is the very simple. The union will be allocated the memory equal to that of the largest size data member. If the `int` occupies four bytes on our system and `char` occupies one byte, the union *intOrChar* will occupy four bytes

Question No: 22 (Marks: 10)

Write a **void function()**; that takes integer numbers from the user and then displays the sum of odd and even numbers entered by the user. Your program should terminate if user enters a negative number

Solution:-

```
#include<iostrea
m.h>
#include<conio.h
> void
function(void);
main()
{
function();

getche();
}
void function(void)
{
int n[5];
```

```

for(int i=0; i<=5; i++)
{
    cout <<"Enter Element = ";
    cin >>n[i];
}
cout <<"\nODD Inputs are = \n";
for(int i=0; i<=5; i++)
{

    if(n[i]%2==0)
    {
        cout <<n[i]<<endl;
    }

}
cout <<"\nEVEN Inputs are = \n";
for(int i=0; i<=5; i++)
{
    if(n[i]%2==1)
    {
        cout <<n[i]<<endl;
    }

}
}
}

```

MIDTERM EXAMINATION
Spring 2009
CS201- Introduction to Programming

Question No: 1 (Marks: 1)
- Please choose one

The function of cin is

- (a) To display message
To read data from keyboard
- (b) To display output on the screen
- (c) To send data to printer.

Question No: 2
(Marks: 1) - Please choose one

In C/C++ language the header file which is used to perform useful task and manipulation of character data is

- (a) cplex.h
- (b) ctype.h**
- (c) stdio.h

(d) delay.h

Question No: 3 (Marks: 1) - Please choose one

How many parameter(s) function getline() takes?

(a) 0

(b) 1

(c) 2

(d) **3**

(e) Question No: 4 (Marks: 1) - Please choose one

Word processor is

(a) Operating system

(e) **Application software**

(b) Device driver

(c) Utility software

Question No: 5 (Marks: 1) - Please choose one

For which values of the integer _value will the following code becomes an infinite loop?

```
int
number
=1;
while
(true) {
    cout << number;
    if (number == 3) break;
    number += integer_value; }
```

(a) any number other than 1 or 2

(b) **only 0**

(c) only 1

(d) only 2

Question No: 6 (Marks: 1) - Please choose one

Each pass through a loop is called a/an

(a) enumeration

(f) **Iteration**

(b) culmination

(c) pass through

Question No: 7 (Marks: 1) - Please choose one

- A continue statement causes execution to skip to
- (a) the return 0; statement
 - (b) the first statement after the loop
 - (c) the statements following the continue statement**
 - (d) the next iteration of the loop

Question No: 8 (Marks: 1) - Please choose one

- What is the correct syntax to declare an array of size 10 of int data type?
- (a) int [10] name ;
 - (b) name[10] int ;
 - (g) int name[10] ;**
 - (c) int name[] ;

Question No: 9 (Marks: 1) - Please choose one

Consider the following code segment. What will the following code segment display?

```
int main(){
int age[10]
= {0}; cout
<< age ;
}
```

- (a) Values of all elements of array
- (b) Value of first element of array**
- (c) Starting address of array
- (d) Address of last array element

Question No: 10 (Marks: 1) - Please choose one

- What will be the correct syntax to initialize all elements of two-dimensional array to value 0?
- (a) int arr[2][3] = {0,0} ;**
 - (b) int arr[2][3] = {{0},{0}} ;
 - (c) int arr[2][3] = {0},{0} ;
 - (d) int arr[2][3] = {0} ;

Question No: 11 (Marks: 1)- Please choose one

- How many bytes will the pointer *intPtr* of type int move in the following statement?
- ```
intPtr += 3 ;
```
- (a) 3 bytes
  - (b) 6 bytes
  - (c) 12 bytes**

(d) 24 bytes

Question No: 12 ( Marks: 1 ) - Please choose one

---

If there are  $2^{(n+1)}$  elements in an array then what would be the number of iterations required to search a number using binary search algorithm?

- (a) n elements
- (e) (n+1) elements**
- (b)  $2(n+1)$  elements
- (c)  $2^{(n+1)}$  elements

Question No: 13 ( Marks: 1 ) - Please choose one

---

Which of the following operator is used to access the value of variable pointed to by a pointer?

- (a) \* operator**
- (b) -> operator
- (c) && operator
- (d) & operator

Question No: 14 ( Marks: 1 ) - Please choose one

---

The\_\_\_\_statement interrupts the flow of control.

- (a) switch
- (b) continue
- (c) goto
- (e) break**

Question No: 15 ( Marks: 1 ) - Please choose one

---

Analysis is the -----step in designing a program

- (a) Last
- (b) Middle
- (c) Post Design
- (d) First**

Question No: 16 ( Marks: 1 ) - Please choose one

---

Paying attention to detail in designing a program is \_\_\_\_\_

- (a) Time consuming
- (b) Redundant

**(e) Necessary**

(a) Somewhat Good

Question No: 17 ( Marks: 1 )

---

Which programming tool is helpful in tracing the logical errors?

**Debugger tool is helpful in tracing the logical errors.**

Question No: 18 ( Marks: 1 )

---

Give the syntax of opening file „myFile.txt“ with „app“ mode using ofstream variable „out“.

ofstream outfile;

outfile.open ("myFile.txt "); // Open the file

Question No: 19 ( Marks: 2 )

---

What is the difference between switch statement and if statement.

In switch statement only one variable can be tested on various condition but using if we can tested multi variables in single statement.

Question No: 20 ( Marks: 3 )

---

Identify the errors in the following code segment and give the reason of errors.

```
ma
in(
){
int
x =
10
const int *ptr = &x ;
*ptr = 5 ;
}
```

```
main()
{
int x = 10;
const int *ptr

= &x ; cout

<<ptr;

}
```

Question No: 21

---

( Marks: 5 )

If int array[10]; is an integer array then write the statements which will store

values at Fifth and Ninth location of this array,

```
cout <<"Enter fifth
postion "; cin >>
array[4];
cout <<"Enter Ninth
postion "; cin >>
array[8];
```

Question No: 22

( Marks: 10

) \_\_\_\_\_

Write a function BatsmanAvg which calculate the average of a player (Batsman), Call this function in main program (Function). Take the input of Total Runs made and Total number of matches played from the user in main function

Q1

When the if statement consists more than one statement then enclosing these statement in curly braces is,

- (a) Not required
- (b) Good programming
- (c) Relevant
- (d) Must**

Q2

The while loop becomes infinite,

- (a) When the condition is always false
- (b) . When the condition is less than zero
- (e) When the condition is always true**
- (c) When the condition contains a logical operator

Q3

Which of the following function(s) is/are included in **stdlib.h** header file?

- (a) double atof(const char \*nptr)
- (b) int atoi(const char \*nptr)
- (c) char \*strcpy ( char \*s1, const char \*s2)

**1 and 2 only**

Q4

If we want to store a string "abc" in an array str then the size of this array must be at least,

**2**

- (a) 3
- (b) 4
- (c) 5

Q5

No executable code will be generated if error is found during translation of the program using interpreter.

- (a) **True**
- (b) False

Q6

Word processor is

- (a) Operating system
- (b) **Application software**
- (c) Device driver
- (d) Utility software

Q7

Which of the following is the correct syntax to print multiple values or variables in a single command using cout?

- (a) `cout << "Hello" + x + "\n";`
- (b) **`cout << "H" << x << "\n";`**
- (c) `cout << "H", x, "\n";`
- (d) `cout << ("H" & x & "\n");`

Q8

Which of the following is correct way to initialize a variable x of int type with value 10?

- (a) `int x ; x = 10 ;`
- (b) **`int x = 10 ;`**
- (c) `int x, x = 10;`
- (d) `x = 10 ;`

Q9

If there is more than one statement in the block of a for loop, which of the following must be placed at the beginning and the ending of the loop block?

- (a) parentheses ( )
- (b) **braces { }**
- (c) brackets [ ]
- (d) arrows < >

Q10

Name of an array is a constant pointer.

**True**

(a) False

Q11

How many bytes will the pointer *intPtr* of type **int** move in the following statement?

*intPtr* += 3 ;

(a) 3 bytes

(b) 6 bytes

**(c) 12 bytes**

(d) 24 bytes

Q12

What will be the value of 'a' and 'b' after executing the following statements?

a = 3;

b = a++;

**(a) 3, 4**

(b) 4, 4

(c) 3, 3

(d) 4, 3

Q13

Loader loads the executable code from hard disk to main memory.

**(a) True**

(b) False

Q14

Which of the following is used with bit manipulation?

(a) Signed integer

(b) Un-signed integer

**(c) Signed double**

(c) Un-signed double

Q15

Which of the following values C++ use to represent true and false?

**(d) 1 and 0**

(a) 1 and -1

(b) 11 and 00

(c) Any numerical value

Q16

The argument of the isdigit() function is \_\_\_\_\_

- (a) **a character**,
- (b) a C-string,
- (c) a C++ string class variable
- (d) None of the given options.

Q17

Which data type should be used to store the value 50.7 ?

**Float**

Q18

Why should goto statement be avoided in C/C++?

When structured programming was started, it was urged not to use the *goto* statement. Though *goto* is there in C language but we will not use it in our programs. It will adopt the structured approach. All of our programs will consist of sequences, decisions and loop. Because loop provide best platform to manipulate the data.

Q19

What operator do you use to assign a pointer the address of another variable or constant? Marks: 2

&sig

i.e.

**int i;**

**int \***

**ptri; ptri**

**= &i;**

Q20

If there are  $2^n$  elements in an array then what would be the number of iterations required to search a number using binary search and linear search? Marks: 3

Q21

Convert the following switch statement into if statements. Marks: 5

```
switch
(operator) {
case '+':
 result = op1 + op2;
 break;
case '-':
 result = op1 - op2;
```

```

 break;
 case 'x':
 case '*':
 result = op1 * op2;
 break;
 case '/':
 result = operand1 / operand2;
 break;
 default:
 cout << "Unknown operator" ;
 }

```

```

if(operator==' +')
{
 result = op1 + op2;
}

```

```

else if(operator=='
-')
{
 result = op1 - op2;
}

```

els  
e

```

if (operator==' *')
{
 result = op1 * op2;
}

```

else

```

if (operator==' /')
{
 result = op1 / op2;
}

```

else

```

{

```

```
cout << "Unknown operator" ;
```

```
}
```

Q22

Write a recursive function that takes character array and starting subscript as arguments. In each recursive call, the function should display the string from subscript to the end of string. The starting subscript in first call should be 0. In each successive call, the subscript should increase by one and function should print the array from subscript to the end of string. The function should stop processing and return when null character encounters.

Suppose the char string passed to the function is, "SampleString", then the function will print output as follows,

Sample  
String  
ampleSt  
ring  
mpleStri  
ng  
pleStrin  
g  
so on....

Marks: 10

## MIDTERM EXAMINATION

Spring 2009

CS201- Introduction to Programming

Question No: 1

( Marks: 1 )

- Please choose one

A precise sequence of steps to solve a problem is called

€ Statement

€ **Program**

€ Utility

€ Routine

Question No: 2

( Marks: 1 )

- Please choose one

The Compiler of C language is written in

€ JavaLanguage

€ UNIX

€ FORTRONLanguage

## € C Language

Question No: 3 ( Marks: 1 )

- Please choose one

---

Initialization of variable at the time of definition is,

€ Must

€ Necessary

## € Good Programming

€ None of the given options

Question No: 4 ( Marks: 1 )

- Please choose one

---

In if structure the block of statements is executed only,

€ When the condition is false

€ When it contain arithmetic operators

€ When it contain logical operators

## € When the condition is true

Question No: 5 ( Marks: 1 )

- Please choose one

---

Which of the following function(s) is/are included in stdlib.h header file?

€ double atof(const char \*nptr)

€ int atoi(const char \*nptr)

€ char \*strcpy ( char \*s1, const char \*s2)

€ **1 and 2 only**

Question No: 6 ( Marks: 1 )

- Please choose one

---

Dealing with structures and functions passing by reference is the most economical method

€ **True**

€ False

Question No: 7 ( Marks: 1 ) - Please choose one

---

Pointer is a variable which store,

€ Data

## € Memory Address

€ Data Type

€ Values

Question No: 8 ( Marks: 1 )

---

- Please choose one

---

Preprocessor program perform its function before phase takes place.

€ Editing

€ Linking

€ **Compiling**

€ Loading

Question No: 9

( Marks: 1 )

- Please choose one

---

Which of the following can not be a variable name?

€ area

€ \_area

€ **10area**

€ area2

Question No: 10

( Marks: 1 )

- Please choose one

---

Which looping process is best, when the number of iterations is known?

€ for

€ while

€ do-while

€ **all looping processes require that the iterations be known**

Question No: 11 ( Marks: 1 ) - Please choose one

---

Which character is inserted at the end of string to indicate the end of string?

€ new line

€ tab

€ **null**

€ carriage return

Question No: 12

( Marks: 1 )

- Please choose one

---

How many bytes are occupied by declaring following array of characters?

char str[] = "programming";

€ 10

€ 11

€ **12**

€ 13

Question No: 13 ( Marks: 1 ) - Please choose one

---

Which of the following header file defines the rand() function?

€ iostream.h

€ conio.h

€ **stdlib.h**

€ stdio.h

Question No: 14

( Marks: 1 )

- Please choose one

---

Commenting the code

€ **Makes a program easy to understand for others.**

€ Make programs heavy, i.e. more space is needed for executable.

€ Makes it difficult to compile

€ All of the given options.

Question No: 15

( Marks: 1 )

- Please choose one

---

What's wrong with this for loop?

for (int k = 2, k <=12, k++)

€ the increment should always be ++k

€ the variable must always be the letter i when using a for loop

€ there should be a semicolon at the end of the statement

€ **the commas should be semicolons**

Question No: 16

( Marks: 1 ) - Please choose one

---

For which array, the size of the array should be one more than the number of elements in an array?

€ int

€ double

€ float

€ **char**

Question No: 17

( Marks: 1 )

To Which category of the software “Compiler and Interpreter” belongs?

Question No: 18

( Marks: 1 )

What is the result of the expression  $x = 2 + 3 * 4 - 4 / 2$

**12**

first multiplies  $3*4 = 12$  then Division  $4/2 = 2$

$2+12-2 = 12$

Question No: 19

( Marks: 2 )

---

Write a declaration statement for an array of 10 elements of type float. Include an initialization statement of the first four elements to 1.0, 2.0, 3.0 and 4.0.

Question No: 20

( Marks: 3 )

---

Write down the output of the following code?

```
int array[7], sum = 0;
for(int i=0;i<7;i++)
{
 array[i] = i;
 sum+= array[i];
}

cout<< " Sum = " <<sum;
```

ANS

**21**

Loop will run times starts from zero and add values from 1 to 6 which is equal to 21

What will be the output of the following segment of C++ code?

```
int A[5] = {1 , 2, 3, 4};
int i;
for (i=0; i<5; i++)
{
 A[i] = 2*A[i];
 cout << A[i] << " ";
}
}
```

ANS

**2 4 6 8 0**

Loops will run 5 times as its starting from zero. It will multiply the value of each item in array as last time is not initialized so it will multiply it with zero to give zero as

output

Question No: 22 ( Marks: 10 )

Write a C++ program that will determine if a departmental store customer has exceeded the credit limit on a charge account.

Program should input the following facts in five variables

1. Account number
2. Balance at the beginning of month (Beginning balance)
3. total of all items charged by customer this month (charges)
4. total of all credits (credits)
5. allowed credit limit

Calculate the new balance

New balance = Beginning balance + charges – credits

Determine if new balance exceeds the allowed credit limit. For those customers whose credit limit is exceeded. The program should display the message "Credit

Limit exceeded.”

Q1

In C/C++ the string constant is enclosed

- (a) In curly braces
- (b) In small braces
- (c) In single quotes**
- (d) In double quotes

Q2

In flow chart, the symbol used for decision making is,

- a) Rectangle
- b) Circle
- c) Arrow
- (e) Diamond**

Q3

The data type before a function name represents its,

- (a) Return Type**
- (b) Function data
- (c) Function arguments
- (d) Function name

Q4 The operator used to take the address of a variable is,

- (a) &&
- (b) ++
- (c) –
- (e) &**

Q5

The header file which is used for input and output is

- (a) maths.h
- (b) string.h
- (f) iostream.h**
- (c) ctype.h

Q6

When we are using command line argument(s), the default argument in C/C++ is/are

\_\_\_\_\_.

- a) argc
- b) argd
- c) argv
- (g) argc and argv**

Q7

In C++, Integer calculation occurs in \_\_\_\_\_ bytes.

- a) 1 byte
- (h) 2 bytes**
- b) 4 bytes
- c) 8 bytes

Q8

In the declaration of two dimensional array,

(a) **First index represents row and second represents column**

(b) First index represents column and second represents row

(c) Both indexes represent rows

(d) Both indexes represent column

Q9

The address operator (&) can be used with,

a) Statement

b) Expression

(e) **Variable**

c) Constan

t Q10

\_\_ translates high level language program into machine language code

(a) Debugger

(b) Editor

(c) **Compiler**

(d) Linke

r Q11

Which of the following data type(s) can operate on modulus operator „%“?

(e) **float, int**

a) float, double

b) int

c) cha

r Q12

What will be the result of the expression  $z = x \% y$ , if  $x = 19$  and  $y = 4$ ?

(f) **3**

a) 4

b) 15

c) 1

9 Q13

Which character is inserted at the end of string to indicate the end of string?

a) new line

b) tab

(c) **null**

d) carriage

return Q14

What will be the value of i and j in the following code segment?

```
int i, j ;
int x[5] = {2, 3, 4, 8, 9} ;
int *ptr
=&x[2];
i =
(*ptr)++;
;
j = *ptr++ ;
```

**a) i = 5, j = 5**

b) i = 5, j = 8

c) i = 4, j = 8

d) i = 5, j = 9

Q15

Syntax of union is identical to \_\_\_\_\_

**Structure**

a) Class

b) Function

c) None of the given options

Q16

Let suppose

```
Union intorDouble{
 Int ival;
 Double charvar;
};
```

```
main(){
 intorDouble VAZ;
 int size ;

 size = sizeof(VAZ);
}
```

What will be the value of variable "size", if int occupies 4 bytes and double occupies 8 bytes?

a) 2

b) 4

c) 8

d) 1

2 Q17

To Which category of the software "Compiler and Interpreter" belongs? (1)

**System  
software**

Q1

8

Give the general syntax of definition of structure. (1)

ANS

**struct Name**

```
{
// body of the structure
};
```

**Q1**

**9**

Write the General syntax for the definition of a user defined function. (2)

**ANS**

**return data type Function Name (parameter list with data type)**

```
{
// definition body
}
```

**Q20**

What will be the output of following code segment? (3)

```
int x[5] = {2, 4, 5, 7, 1} ;
```

```
int *ptr = &x[2];
```

```
cout << (*ptr)++
```

```
<<" "; cout <<
```

```
*ptr++ ;
```

**AN**

**S 5**

**6**

**Q2**

**1**

Detect and correct compile time error(s) in the following code.

**Hints: The following code pass a value to a function by Reference and print the value before and after pass. (5)**

**AN**

**S**

```
int test (int &x)
{
*x = *x + 200;
return x;
}
```

```
main()
```

```
{
```

```
int x = 100;
```

```
cout <<" x = " <<x;
```

```

 test (*x); // //invalid type of argument uniary

 cout<<endl<< " x = " <<x;

 getch();
 return 0;
}

```

ANS

The following function pass a reference to the function but in the main body the call mechanism is wrong. The \* sign is not used with it .

**X = 100 after the function completion x = 300;**

Q22

Write a C/C++ program which calculates the aggregate of a student.

The aggregate can be calculated by the formula: (10)

$$\text{Aggregate} = (\text{Matrix\_marks} \times 2 + \text{Fsc\_marks} \times 4) / 24$$

If the aggregate of a student is less than 150 then the program should display message

**“You can not be admitted to VU”** otherwise display the message

**“Congratulation! You admitted in VU “**

ANS

```
#include<iostream.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
 int m,f;
```

```
 float a;
```

```
 m=254;
```

```
 f=340;
```

```
 a = ((m*2) +
```

```
 (f*4))/24; if(a < 150)
```

```
 {
```

```
 cout <<"You can not be admitted to VU";
```

```
 }
```

```
 else
```

```

{
 cout <<"Congratulation! You admitted in VU ";
}

//Aggregate = (Matrix_marks*2 + Fsc_marks*4) / 24

getche();
}

```

Q1

What is function of **cout** ?

- (a) To send data to printer
- (b) To read data from keyboard
- (c) To display message
- (d) To display output on the screen**

Q2

In Flow chart process is represented by

**Rectangle**

- (a)** Arrow symbol
- (b)** Oval
- (c)** Circle

Q3

&& is----- operator.

- (a) An arithmetic
- (b) Logical**
- (c) Relational
- (d) Unary

Q4

An over Flow condition occurs when we try to assign a value to a variable which is.

- (a) Less than its maximum size**
- (b) Greater than its maximum size
- (c) With in its range
- (d) Equal to its size

Q5

For dereferencing an array element using pointer, we use the operator

- (a) &**
- (b) \***
- (c) /**
- (d) +**

Q6

In the declaration of two dimensional array,

- (d) First index represents row and second represents column**
- (a) First index represents column and second represents row

- (b) Both indexes represent rows
- (c) Both indexes represent column

Q7

Which of the following data type(s) can operate on modulus operator „%“?

**float, int**

- (a) float, double
- (b) int
- (c) char

Q8

Which of the following is the correct way to declare a variable x of integer type?

- (a) x int ;
- (b) integer x ;
- (c) **int x;**
- (d) x integer

Q9

Which of the following can not be a variable name?

- (a) area
- (b) \_area
- (e) **10area**
- (c) area2

Q10

Which of the function call is **call by value** for the following function prototype? float area (int);

- (a) area(&num) ;
- (f) **area(num) ;**
- (b) area(int num) ;
- (c) area(\*num)

; Q11

Recursive functions are used when there is a repetitive pattern.

(g) **True**

(a) False

Q12

What will be the range of numbers generated by function rand () % 9?

- (a) 0 to 9
- (b) 1 to 9
- (c) **0 to 8**
- (d) 1 to

8 Q13

What will be the correct syntax to declare two-dimensional array of float data type?

- (a) float arr{2}{2} ;
- (e) **float arr[2][2] ;**
- (b) float arr[2,2] ;
- (c) float[2][2] arr

; Q14

When a function finishes its execution then,

- (a) The control return to its Prototype

- (b) The control returns to its definition  
**(f) Control returns to statement following function call**  
(c) The compiler stop execution of whole program Q15

Consider the following statements to initialize a two-dimensional array.

- i. `int arr[2][3] = {4, 8, 9, 2, 1, 6} ;`  
ii. **`int arr[3][2] = {4, 8, 9, 2, 1, 6} ;`**  
**iii. `int arr[][2] = {{4,8},{9, 2},{1, 6}} ;`**

Which of the following option(s) are correct to initialize a two-dimensional array with 3 rows and 2 columns?

- (a) (ii) only  
(b) (iii) only  
**(c) (ii) and (iii)**  
(d) (i)

and Q16

Editors are used to compile the code.

- (a) True  
**(b) False**

Q17

What are global variables? (1)

ANS

**Global variables are those that are defined outside of main. It can be accessed in entire program.**

Q18

Is it possible to evaluate the size of structure, if yes then how? (1)

ANS

YES

```
#include <iostream.h>
```

```
#include <stdlib.h>
```

```
struct VehicleParts
```

```
{
```

```
int
```

```
wheels;
```

```
int seats;
```

```
VehicleParts()
```

```
{
```

```
cout << "\n VehicleParts - default constructor";
```

```
}
```

```

VehicleParts(int wheels, int seats)
{
 this->wheels = wheels; this->seats = seats; cout << "\n
VehicleParts - parameterized constructor";
}

```

Q19

Write down the general syntax of switch statement. (22)

ANS

**switch ( *variable/expression* )**

```

{
 case constant1 : statementLlist1 ; case constant2: statementLlist1 ; default: statementLlist1 ;
}

```

Q20

What will be the output of following code segment? (3)

```

int x[5] = {2, 4, 5, 7, 1} ;
int *ptr = &x[2];
cout << (*ptr)++
<<" "; cout <<
*ptr++ ;

```

AN

S 5

6

Q2

2 What is the difference between compiler and interpreter? (5)

ANS

compiler and interpreter are system software but the difference is that the compiler compile the whole program but the interpreter is compiled line by line. Compiler are efficient in performance.

Q23 (10)

Write a recursive function that takes three arguments (an integer array, starting subscript 's' and ending subscript 'e' ).

In first recursive call, the function should display the array from subscript 's' (s = 0) to 'e' (e = size of array). In each successive call, the function should print the array from index s+1 to e. The function should stop processing and return when starting subscript becomes equal to ending subscript.

For example, if user enters values for array 2, 3, 4, 5, 6 then the recursive function must display the following output.

2 3 4 5 6

3 4 5 6

4 5 6

5 6

6

### ANS

```
#include<iostream.
h>
#include<conio.h>

void recursive(int
[],int,int); void main()
{
 int array[5];
 for(int
 i=0;i<5;i++)
 {
 cout<<"\nEnter the "<<i<<" Index number
 :"; cin>>array[i];
 }
 recursive(array,0,4); //0 is the starting index and 4 is the ending subscript getch();
}

void recursive(int arr[],int s,int e)
{
 if(s!=e+1)
 {
 for(int i=s;i<=e;i++)
 cout<<arr[i]<<"\t";

 cout<<endl;
 s++;

 recursive(arr,s,e); //Recursive call
```

```
}

}
```

Q1

What is function of **cout** ?

- (a) To send data to printer
- (b) To read data from keyboard
- (c) To display message
- (d) To display output on the screen**

Q2

<, <=, >, >= are called ----- operators.

- (a) Logical
- (b) Arithmetic
- (c) Relational**
- (d) Conational

Q3

In while loop the loop counter must be initialized,

- (a) With in the loop
- (e) Before entering the loop**
- (b) At the end of the loop
- (c) None of the given options

Q4

Data Size of the file is always \_\_\_\_\_ the actual size of the file.

- (f) Greater than**
- (a) Equal to
- (b) Less than or equal to
- (c) None of the above

Q5

The precedence of \* is higher than dot operator (.)operator

- (g) True**
- (a) False

Q6

Let ptr1 and ptr2 are pointer variables that points to integer data type then which one of the following arithmetic is allowed,

- (a) ptr1 + ptr2
- (b) ptr1 - ptr2
- (h) ptr1 \* ptr2**
- (c) ptr1 / ptr2

Q7

Word processor is

- (a) Operating system
- (i) Application software**
- (b) Device driver
- (c) Utility software

Q8

What will be the range of numbers generated by function `rand () % 9`?

- (a) 0 to 9
- (b) 1 to 9
- (j) 0 to 8
- (c) 1 to 8

Q9

How many bytes will the pointer *intPtr* of type **int** move in the following statement?

`intPtr += 3;`

- (a) 3 bytes
- (b) 6 bytes
- 12 bytes
- (c) 24

bytes Q10

What will be the correct syntax to assign an array named *arr* of 5 elements to a pointer *ptr*?

- (a) `*ptr = arr ;`
- (b) `ptr = arr ;`
- (c) `*ptr = arr[5] ;`
- (d) `ptr = arr[5]`

; Q11

What will be the output of the following code segment?

`char *x = "programming" ;`

`cout << *(x+2) << *(x+3) << *(x+5) << *(x+8) ;`

- (a) prgm
- (b) rorm
- (e) ogai
- (c) ramg

Q12

There is a pointer variable named *ptr* of type **int** then address of which type of variable the *ptr* will store in it?

- (a) variable of type **char**
- (b) variable of type **short**
- (f) variable of type **int**
- (c) variable of type **double**

Q13

Flow charts explain the working of a program in pictorial format.

- (g) True
- (a) False

Q14

The object \_\_\_\_\_ may be used both for file input and file output

- (a) `fstream`,
- (h) `ifstream`,

- (b) ofstream,
  - (c) none of the given
- options. Q15

Most efficient method of dealing with structure variables is to define the structure globally

(i) **True**

(a) False

Q16

If a variable is passed by value to a function and the function makes some changes to that variable then it

(j) **does not affect the original variable**

(a) affects the original variable

(b) causes syntax error

(c) None of the given options

Q17

How does elements of two-dimensional array store into memory? (1)

ANS

Two dimensional arrays are also stored in liner order. As like one dimensional array.

Q18

What is a structure? (1)

ANS

**In structure, we introduce a new data type. "A structure is a collection of variables under a single name. These variables can different types, and each has a name that is used to select it from the structure"**

Q19 (2)

When a pointer is incremented then how many bytes will it move to change its address?

ANS

If an integer occupies four bytes in the memory, then the *yptr++*; will increment its value by four.

Q20

What happens when we increment a pointer? (3)

ANS

become 11. The increment of a pointer depends on its data type. The data type, the pointer points to, determines the amount of increment. In this case, *yptr* is an integer pointer. Therefore, when we increment the *yptr*, it points to the next integer in the memory. If an integer occupies four bytes in the memory, then the *yptr++*; will increment its value by four. This can be understood from the following example.

Q  
2  
1

What are the advantages of random access file over sequential access file? (5)

Q22

Write a C/C++ program which calculates the aggregate of a student.

The aggregate can be calculated by the formula: (10)

$$\text{Aggregate} = (\text{Matrix\_marks} \times 2 + \text{Fsc\_marks} \times 4) / 24$$

If the aggregate of a student is less than 150 then the program should display message

**“You can not be admitted to VU”** otherwise display the message

**“Congratulation! You admitted in VU “**

ANS

```
#include<iostream.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
 int m,f;
```

```
 float a;
```

```
 m=254;
```

```
 f=340;
```

```
 a = ((m*2) +
```

```
 (f*4))/24; if(a < 150)
```

```
 {
```

```
 cout <<"You can not be admitted to VU";
```

```
 }
```

```
else
{
 cout <<"Congratulation! You admitted in VU ";
}
//Aggregate = (Matrix_marks*2 + Fsc_marks*4) / 24

getche();
}
```

Q1

In C/C++ language the header file which is used to perform useful task and manipulation of character data is

- (a) cplex.h
- (b) ctype.h
- (c) stdio.h
- (d) delay.h

Q2

The header file which is used for input and output is

- (a) maths.h
- (b) string.h
- (c) iostream.h

(d) ctype.h

Q3

Suppose `int multi[5][10]`; when we are using `**multi`, it means,

(a) Single dereferencing

(b) Single referencing

**(c) Double referencing**

(d) Double dereferencing

Q4

To access the data members of structure \_\_\_\_ is used.

**(e) dot operator (.)**

(a) \* operator

(b) € operator

(c) None of given.

Q5

The precedence of \* is higher than dot operator (.) operator

**(f) True**

(a) False

Q6

Which of the following is the starting index of an array in C++?

**(g) 0**

(a) 1

(b) -1

(c) any number

Q7

When a call to function statement is encountered,

**(h) The control transfers to its Prototype**

(a) The control transfers to its definition

(b) Control returns to statement following function call

(c) The compiler stops execution of whole program

Q8

A function must always return value.

(a) True

**(i) False**

Q9

If an array has 100 elements, what is allowable range of subscripts? (a) 0 –

99

(b) 1 – 99

**(c) 0 – 100**

(d) 1

– 100

Q10

If a pointer appears on left hand side of an assignment operator then right side of that assignment operator must be,

(a) Variable name

**(b) Address of variable**

- (c) Variable value
- (d) Constant

Q11

Editors are used to compile the code.

- (a) True
- (b) False**

Q12

Which bitwise operator returns false if both bits are 1?

- (a) AND
- (b) XOR
- (c) NOT
- (c) OR**

Q13

What does !(7) evaluate to in decimal where ! is a NOT operator?

- (d) 7**
- (a) 8
- (b) 9
- (c) 1

Q14

Structures cannot be passed as Function Parameters

- (a) True
- (b) False**

Q15

When break statement is encountered in a loop body it,

- (a) Transfers the control outside from current loop**
- (b) Transfers the control outside from current program
- (c) Enforces the next iteration of loop
- (d) Generates compile time

error. Q16

A union is a user-defined data type that contains only\_\_\_from its list of members at a time.

- (a) One object
- (b) Two objects**
- (c) Three objects
- (d) None of the given options

Q17

Why programming is important? Describe in ONE line. (1)

“A program is a precise sequence of steps to solve a particular problem.”

Q18

Which bitwise operator returns true if both bits are different and returns false if both bits are same? (1)

ANS

1. !=
2. ==

Q19

Why we close a file after use? (2)

ANS

**To save our data stored on file. Also this process makes our program fast and reliable.**

Q20

Can you use an assignment operator to assign the value of one C-string to another? (3)

ANS

Yes we can assign value c-string to another

ie char a={"I Love Pakistan"};

char b={"I Love Pakistan"};

Q21

The statement int Name [2][2]; define a 2x2 array, Write the code which read data from keyboard for this array. (5)

ANS

```
for(int i=0; i<2; i++)
{
 for (w=0; w<2; w++)
 {
 cin >>Name[i][w];
 }
 cout <<endl;
}
for(int i=0; i<2; i++)
{
 for (w=0; w<2; w++)
 {
 cout<<Name[i][w]<<" ";
 }
 cout <<endl;
}
```

Q22

Write a program which reads a text file "**PlayersInfo.txt**" residing in the current directory. Open the file **PlayersInfo.txt** in read mode and assign these values to the struct **Player**; assume order of the data in the file to be exactly the same as the

order of struct attributes. The struct **Player** has following attributes

- i) Name
- ii) Height
- iii) Age
- iv) Score
- v) Game

After reading the file and assigning values to the struct, in the end close the file

PlayersInfo.txt.

(10)

Q1

The size of int data type is

- (a) 1 bytes
- (e) 2 bytes**
- (b) 3 bytes
- (c) 4 bytes

Q2

When the logical operator AND (&&) combine two expressions exp1 and exp2 then the result will be true only,

- (a) When both exp1 and exp2 are true**
- (b) When both exp1 and exp2 are false
- (c) When exp1 is true and exp2 is false
- (d) When exp1 is false and exp2 is true

Q3

The correct syntax of do-while loop is,

- (a) (condition ) while; do { statements; };
- (b) { statements; } do-while ();
- (c) while(condition); do { statements; };
- (e) do { statements; } while (condition);**

Q4

\_\_\_\_\_provide communication channels between files and program.

- (a) Streams
- (b) Language like C++
- (f) Function seekg()**
- (c) None of the above

Q5

All elements of an array must be of,

- (a) different data type
- (b) float data only
- (c) character data only
- (g) same data type**

Q6

Function seekg() takes\_\_\_\_\_parameter(s).

- (a) 0

1

(b) 2

(c) 3

Q7

Structures help to define program-specific\_\_\_\_\_.

**functions**

(a) datatypes

(b) Arithmetic operations

(c) None of the given options.

Q8

In the declaration of two dimensional array,

**(a) First index represents row and second represents column**

(b) First index represents column and second represents row

(c) Both indexes represent rows

(d) Both indexes represent column

Q9

What will be the result of arithmetic expression  $6 + 27/3 * 3$ ?

**(e) 33**

(a) 45

(b) 9

(c) 3

Q10

Which of the function call is **call by value** for the following function prototype?

`float area (int);`

(a) `area(&num) ;`

**(f) `area(num) ;`**

(b) `area(int num) ;`

(c) `area(*num)`

; Q11

How many bytes are occupied by declaring following array of characters?

`char str*+ = "programming";`

(a) 10

(b) 11

**(c) 12**

(d) 1

Q12

What will be the correct syntax to assign an array named **arr** of 5 elements to a pointer **ptr**?

(a) `*ptr = arr ;`

(b) `ptr = arr ;`

**(e) `*ptr = arr[5] ;`**

(c) `ptr = arr[5] ;`

Q13

Let ptr1 and ptr2 are pointer variables then which of the following

arithmetic operation is allowed on pointers?

- (a) Addition
- (b) Subtraction
- (c) Multiplication
- (f) All of the above**

Q14

The variables having a name, type and size are just like empty boxes.

**(g) True**

(a) False

Q15

When break statement is encountered in a loop body it,

- (a) Transfers the control outside from current loop**
- (b) Transfers the control outside from current program
- (c) Enforces the next iteration of loop
- (d) Generates compile time

error. Q16

If two programs **a** and **b** are trying to open a file **xyz.txt** at approximately same time then

**(e) Both programs will generate error**

- (a) One of them will succeed in opening that file and other will fail
- (b) Both programs will open the file
- (c) One of the program will re-

start Q17

What is meant by array manipulation? (1)

Q18

What will be the value of x after the execution of the following code segment? (1)

```
int x=10;
```

```
int y=30;
```

```
int *xptr = &x; x
```

```
= *xptr + 10;
```

Q19

(2)

What is the output of the code given below? void main()

```
{
 int a=10,b=20;
 char x=1,y=0;
 if(a,b,x,y)
 {
 cout << "EXAM";
 }
```

```
}
}
```

Q  
2  
0

What is Overflow condition? (3)

ANS

When we try to store larger information in a variable, than a data type can store, overflow condition occurs.

Q21

Write code which read a string not greater than 20 characters from keyboard stored it in an array Name and display it on the screen. (5)

ANS

```
char string[19];
cout<<"Enter a
string"; cin >>
string;
```

Q22 (10)

Write a C++ program which contains a user-define function named **convertHeight** which takes height of person in centimeter as an argument. This function converts the height in centimeter into feet and inches and displays them on the screen.

Program should prompt the user to enter height in centimeter and pass it to function **convertHeight** as an argument which displays height in feet and inches.

Hint:

1 foot = 12 inches  
1 inch = 2.5 cm

ANS

```
#include<iostrea
m.h>
#include<conio.h
>

void convertHeight
(int); main()
{
int hight;
cout <<"Enter the High of the Person = "; cin
>> hight;
```

```
convertHeight(hight);
```

```
getche();
}
```

```
void convertHeight (int f)
{
 //1 foot = 12 inches
 //1 inch = 2.5 cm

 int i,c;
 i = f*12;
 c = i*2.5;
 cout <<"high in Feet = "<<f<<endl; cout
 <<"high in inches = "<<i<<endl; cout
 <<"high in CM = "<<c<<endl;

}
```

Q1

The data type of size one byte is

- (a) char
- (b) int
- (c) long
- (d) double

Q2

If Num is an integer variable then Num\*= 4; means,

- (a) Multiply Num 4 times
- (b) Multiply 4 with Num and display
- (f) Multiply 4 with Num and assign the result to Num**
- (c) Add 4 with Num

Q3

Member function tellg() returns the current location of the \_\_\_\_\_  
pointer.

- (a) tellptr()**
- (b) write()
- (c) seekg()
- (d) get()

Q4

If we want to store a string "abc" in an array str then the size of this array must be at least,

- (e) 2**
- (a) 3
- (b) 4
- (c) 5

Q5

Pointer is a variable which store,

- (a) Values
- (b) Data
- (f) Memory Address**
- (c) Data Type

Q6

C is widely known as development language of \_\_\_\_\_ operating system.

- (a) Linux
- (g) Windows**
- (b) Unix
- (c) Mac

OS Q7

C++ is a case-sensitive language

- (h) True**
- (a) False

Q8 What is the output of the following code?

```
for (int a = 1; a <= 1; a++) cout << a++; cout << a;
```

- (a) 22
- (b) 12**
- (c) 23
- (d) 13

Q9

A continue statement causes execution to skip  
to (a) the return 0; statement

- (b) the first statement after the loop
- (c) the statements following the continue**
- statement (d) **the next iteration of the loop**

Q10

If there is more than one statement in the block of a for loop, which of the  
following must be placed at the beginning and the ending of the loop block?

- (a) parentheses ( )
- (b) braces { }**
- (c) brackets [ ]
- (d) arrows <

> Q11

What will be the correct syntax for the following function call?

float square (int&);

- (a) square (int num);**
- (e) square (&num);**
- (b) square (num);**
- (c) square**

(\*num);

Q12

Which of the following is the correct way to assign an integer value 5 to element of a matrix say „m“ at second row and third column?

**(f) (a) m[2][3] = 5 ;**

(b) m[3][2] = 5 ;

(c) m[1][2] = 5 ;

(d) m[2][3] = „5“; Q13

How many dimensions does n-dimensional array has?

**(g) n dimensions**

(a) 2n dimensions

(b) (n+1) dimensions

(c) (n-1)

dimensions Q14

Consider the following code segment. What will be the output of following code?

```
int addValue
(int *a){ int b =
(*a) + 2; return
b ;
ma
in
()
{
int
x
=6
;
cout << x << “, ” ;
cout << addValue(&x)
<< “, ” ; cout << x ;
}
```

**(a) 6,8,6**

(b) 6,6,8

(c) 6,8,8

(d)

6,6,6 Q15

If most significant bit of un-signed number is 1 then it represents a positive number.

**(a) True**

(b) Fals

e Q16

When we declare a multidimensional array the compiler store the elements of multidimensional array in the form of,

(a) Contiguous memory location

(c) **Matrix**

(b) Columns

(c) Row

s Q17

What is the functionality of the function: **char \*strncat (char \*s1, const char \*s2, size\_t n) (1)**

ANS

| of **s1**. The value of **s1** is returned. |

Q1

8

Write a piece of code that outputs three values of type int, long and float to a stream. (1)

ANS

```
main()
{
 int a;
 long b;
 float c;
 a = 10;
 b = 124568979;
 c = 6.57;
 cout <<a<<"\t"<<b<<"\t"<<c;

 getch();
}
```

Q19

Which bit of the number is used as a sign bit? (2)

ANS

**The most significant bit is used as a sign bit.**

Q20

What is difference between single-dimensional and multi-dimensional array?

ANS

**Single dimensional array used to stored lists and multi dimentional arrays used to stored value as tabler formate i.e matrix.**

Q2

1

Write down the C++ program that calculates the Zakat on the amount entered by the user

Note: Zakat is 2.5% of the given amount (5)

ANS

```
main()
{
 int amount;
 float zakat;
 cout <<"Enter the Amount = ";
 cin >>amount;
 zakat = (amount*2.5)/100;
 cout <<"Amount = "<<amount<<endl; cout
 <<"Amount = "<<zakat<<endl; getch();
}
```

Q22

What is meant by scope of identifiers? Differentiate between different scope of identifiers and explain them with examples? (10)

### *Scope of Identifiers*

An 'Identifier' means any name that the user creates in his/her program. These names can be of variables, functions and labels. Here the scope of an identifier means its visibility. We will focus Scope of Variables in our discussion.

Now this variable „i“ can be used in any statement inside the function func1(). But consider this variable being used in a different function like:

void func2()

```
{

int k = i + 4; //Compilation error
}
```

The variable „i“ belongs to func1() and is not visible outside that. In other words, „i“ is local to func1().

Q1

What is function of **cout** ?

- (a) To send data to printer
- (b) To read data from keyboard
- (c) To display message
- (d) To display output on the screen**

Q2

For one byte there are \_\_\_\_\_ combinations of values that can be stored in computer.

- (A)  $2^6$
- (B)  $2^7$
- (C)  $2^8$
- (D)  $2^4$

Q3

\_\_\_\_\_ provide communication channels between files and program.

- (a) Streams
- (b) Language like C++
- (c) Function seekg()**
- (d) None of the above

Q4

The data type before a function name represents its,

- (e) Return Type**
- (a) Function data
- (b) Function arguments
- (c) Function name

Q5

In C/C++ language when an array is passed to a function then by default its passing mechanism is,

- (a) Call by value
- (f) Call by Reference**
- (b) It depends on type of array
- (c) It depends on the return type of function.

Q6

Array is a data structure which store

- (a) Memory addresses
- (g) Variables**
- (b) Data Type
- (c) Data

Q7

If there is more than one statement in the block of a for loop, which of the

following must be placed at the beginning and the ending of the loop block?

- (a) parentheses ( )
- (b) braces { }**
- (c) brackets [ ]
- (d) arrows < >

Q8

Array is passed by value to a function by default.

- (a) True
- (e) False**

Q9

Which of the following is the correct function call having array named *student* of 10 elements as a parameter.

- (a) addRecord(student[]) ;
- (b) addRecord(student) ;
- (f) addRecord(student[10]) ;**
- (c) addRecord(\*student) ;

Q10

What will be the correct syntax for initialization of pointer *ptr* of type int with variable *x*?

- (a) int ptr = &x ;
- (b) int ptr = x ;
- (g) int \*ptr = &x ;**
- (c) int ptr\* = &x ;

Q1

1 What will be the correct syntax for initialization of pointer *ptr* with string "programming"?

- (a) char ptr = 'programming' ;
- (b) char \*ptr = "programming" ;**
- (c) char \*ptr = 'programming' ;
- (d) \*ptr = "programming"

; Q12

The condition in while loop may contain logical expression but not relational expression.

- (e) True**
- (a) False

Q13

We want to access array in random order which approach is better?

- (f) Pointers**
- (a) Array index
- (b) Both pointers and array index are better
- (c) None of the given

options. Q14

Single line comments explaining code would be preceded like in the following example.

- (a) /\*
- (g) //**
- (b) /

Q15

Function write() takes \_\_\_\_\_ as parameter(s).

- (a) String of pointer type
- (b) String and no. of bytes to be written
- (c) Pointer array of characters and delimiter

**String of variable lengths, no. of bytes to be read and flags**

Q16

Structure is a collection of \_\_\_\_\_ under a single name.

- (a) Only Functions**
- (b) Only Variables
- (c) Both Functions and Variables
- (d) None of the given options

Q17

What will be the correct syntax to initialize a pointer „ptr“ with two-dimensional array „m“? (1)

ANS

```
int m[2][2];
int * ptr;
ptr = *m;
```

Q18

Which one of the loop (while or do-while) must be used if it is necessary to execute a loop at least once? (1)

ANS

do-while loop

Q19

Identify each of the following function as string conversion function or string manipulation function. (2)

```
double atof(const char *nptr)
char *strcpy (char *s1, const char
*s2) int atoi(const char *nptr)
```

ANS

1. **double atof(const char \*nptr)** ---€Converts the string nPtr to double.
2. **char \*strcpy ( char \*s1, const char \*s2)** -€ Copies string s2 into character array s1. The value of is returned.
3. **int atoi(const char \*nptr)**--€ Converts the string nPtr to int.

Q20

What is difference between single-dimensional and multi-dimensional array? (3)

ANS

Single dimensional array used to stored lists and multi dimentional arrays used to stored value as tabler formate i.e matrix.

Q21

What will be the output of following code segment?(5)

```
int num[10] = {2, 3, 5, 8, 9, 10, 12, 15, 19, 20} ;
int *ptr = num ;
for (int i=0; i<10; i+=2){
cout << *(ptr+i) << “, ”;
}
```

ANS

2, 5, 9, 12,

19, Q22

Write a C++ program which contains a user-define function named **convertHeight** which takes height of person in centimeter as an argument. This function converts the height in centimeter into feet and inches and displays them on the screen.(10)

Program should prompt the user to enter height in centimeter and pass it to function **convertHeight** as an argument which displays height in feet and inches.

Hint:

1 foot = 12 inches  
1 inch = 2.5 cm

ANS

```
#include<iostrea
m.h>
#include<conio.h>

void convertHeight
(int); main()
{
int hight;
cout <<"Enter the High of the Person = "; cin
>> hight;
convertHeight(hight);

getche();
}

void convertHeight (int f)
{
//1 foot = 12 inches
```

```
//1 inch = 2.5 cm
```

```
int i,c;
i = f*12;
c = i*2.5;
cout <<"high in Feet = "<<f<<endl; cout
<<"high in inches = "<<i<<endl; cout
<<"high in CM = "<<c<<endl;

}
```

Q1

Compiler is a

- (a) **System software**
- (b) Application Software
- (c) Driver
- (d) Editor

Q2

In while loop the loop counter must be initialized,

- (a) With in the loop
- (e) **Before entering the loop**
- (b) At the end of the loop
- (c) None of the given options

Q3

If Num is an integer variable then Num\*= 4; means,

- (a) Multiply Num 4 times
- (b) Multiply 4 with Num and display
- (f) **Multiply 4 with Num and assign the result to Num**
- (c) Add 4 with Num

Q4

In C/C++ ,the arguments are passed by\_\_\_\_\_to a function by default .

- (a) reference
- (g) **value**
- (b) data
- (c) type

Q5

Disks is divided into\_\_\_\_\_with power of\_\_\_\_\_.

- (a) **Chunks,  $2n$**
- (b) Blocks,  $n^2$
- (c) Blocks,  $2^n$
- (d) Chunks,  $n^2$

Q6

C is widely known as development language of\_\_\_\_\_operating system.

- (a) Linux
- (h) **Windows**
- (b) Unix

(c) Mac OS

Q7

Assignment operator „=“ is a

(a) Unary operator

**(b) Binary operator**

(c) Ternary operator

(d) None of the given options

Q8

Consider the following code segment. What will be the output of the following program?

```
int func(int) ;
int num = 10 ;
```

```
int
main(){
int num ;
num = 5 ;

cout << num ;

cout << func(num) ;

}

int func(int
x){ return
num ;

}
```

**(a) 5, 5**

(b) 10, 5

(c) 5, 10

(d) 10, 10

Q9

Name of an array is a constant pointer.

**(a) True**

(b) False

Q10

What will be the correct syntax to assign an array named **arr** of 5 elements to a pointer **ptr**?

(a) \*ptr = arr ;

(b) ptr = arr ;

**(c) \*ptr = arr[5] ;**

(d) ptr = arr[5]

Q11

If there are  $2^{(n+1)}$  elements in an array then what would be the number of iterations required to search a number using binary search algorithm?

- (a) n elements
- (b) (n+1) elements
- (c) 2(n+1) elements
- (d)  $2^{(n+1)}$

elements Q12

In C/C++, null character is represented as

- (e) **\n**
- (a) \0
- (b) \t
- (c) \

r Q13

How many nested loop would be required to manipulate n-dimensional array?

- (a) 2n
- (b) **n**
- (c) n +1
- (d) n -

1 Q14

What will be the correct syntax to access the value of fourth element of an array using pointer ptr?

- ptr[3]**
- (a) (ptr+3)
- (b) \*(ptr+3)
- (c) Both 1 and

3 Q15

Single line comments explaining code would be preceded like in the following example.

- (a) /\*  
**//**
- (b) /
- (c) //

Q16

If a variable is passed by value to a function and the function makes some changes to that variable then it

- does not affect the original variable**
- (a) affects the original variable
- (b) causes syntax error
- (c) None of the given options

Q1

7 What is meant by \*num and &num? [1]

**ANS** \* num is a pointer and &num is a reference to that pointer. Q18

Suppose there is a pointer to structure **\*sPtr**. How can we access the data member „name“ with **sPtr**? [1]

ANS

Structure data members using pointers Using the \* operator;  
(\*sptr).name

Q19

Why we close a file after use? [2]

ANS

**To save our data stored on file. Also this process makes our program fast and reliable**

Q20

Define Flow  
chart. [3] ANS

**Flow Chart**

A flow chart is a pictorial representation of a program. There are labeled geometrical symbols, together with the arrows connecting one symbol with other. A flow chart helps in correctly designing the program by visually showing the sequence of instructions to be executed.

Q21

Write down the function definition if we want to pass the arguments to a function by reference without changing the values stored at that addresses. [5]

Q22

From writing to execution of the program following software are used explain for what purpose each is used. [10]

Editor  
Compiler/Interpreter  
Linker  
Loader

**ANS**

**Editors** is a tool for writing the code of a program. For this purpose we used Editors in which we write our code. We can use word processor too for this, but word processors have many other features like bold the text, italic, coloring the text etc, so when we save a file written in a word processor, lot of other information including the text is saved on the disk. For programming purposes we don't need these things we only need simple text. Text editors are such editors which save only the text which we type. So for programming we will be using a text editor

**Compiler and Interpreter**

Compilers translate the English like language (Code written in C) into a language (Machine language) which computers can understand. The Compiler read the whole program and translates it into machine language completely. The difference between interpreter and compiler is that compiler will stop translating if it finds an error and there will be no executable code generated whereas Interpreter will execute all the lines before

error and will stop at the line which contains the error. So Compiler needs syntactically correct program to produce an executable code. We will be using compiler in our course As we write the code in English and we know that computers can understand only 0s and 1s. So we need a translator which translates the code of our program

into machine language. There are two kinds of translators which are known as Interpreter and Compilers. These translators translate our program which is written in C-Language into Machine language. Interpreters translates the program line by line meaning it reads one line of program and translates it, then it reads second line, translate it and so on. The benefit of it is that we get the errors as we go along and it is very easy to correct the errors. The drawback of the interpreter is that the program executes slowly as the interpreter translates the program line by line. Another drawback is that as interpreters are reading the program line by line so they cannot get the overall picture of the program hence cannot optimize the program making it efficient.

**Linker** Most of the time our program is using different routines and functions that are located in different files, hence it needs the executable code of those routines/functions. Linker is a tool which performs this job, it checks our program and includes all those routines or functions which we are using in our program to make a standalone executable code and this process is called Linking

**Loader** after a executable program is linked and saved on the disk and it is ready for execution. We need another process which loads the program into memory and then instruct the processor to start the execution of the program from the first instruction (the starting point of every C program is from the main function). This processor is known as loader. Linker and loaders are the part of development environment. These are part of system software.

Q1

There are mainly -----types of software

- (a) **Two**
- (b) Three
- (c) Four
- (d) Five

Q2

<, <=, >, >= are called ----- operators.

- (a) Logical
- (b) Arithmetic
- (e) **Relational**
- (c) Conational

Q3

In order to get 256 from the number 2568 we divide this number by 10 and take,

- (a) Its remainder
- (b) The number

**Its quotient**

(c) Its divisor

Q4

If `int x = 10;` then the value of `x/= 3;` will be,

(a) 10

**3**

(b) 13

(c) 1

Q5

How many parameter(s) function `getline()` takes?

(a) 0

(b) 1

(c) 2

**3**

Q6

Suppose `int multi[5][10];` when we are using `**multi`, it means,

(a) Single dereferencing

(b) Single referencing

**(c) Double referencing**

(d) Double dereferencing

Q7

To access the data members of structure\_\_\_\_is used.

**dot operator (.)**

(a) \* operator

(b) € operator

(c) None of given.

Q8

There is NO difference between **bitwise AND operator (&)** and **Logical AND (&&) operator**.

**True**

(a) False

Q9

Which of the following data type(s) can operate on modulus operator „%“?

**float, int**

(a) float, double

(b) int

(c) cha

r Q10

What's wrong with this while loop?

`while( (i < 10) && (i > 24))`

(a) the logical operator && cannot be used in a test condition

(b) the while loop is an exit-condition loop

**the while loop is an exit-condition loop**

(c) the test condition is always true

Q1

1 The switch structure is a \_\_\_\_\_ construct

- (a) single-selection
- (b) bi-selection

**multiple-selection**

- (c) unconditional

Q12

Keyword „array” must be used to declare an array.

**(a) True**

(b) Fals

e Q13

What will be the correct syntax to declare two-dimensional array of float data type?

- (a) float arr{2}{2} ;

**float arr[2][2] ;**

- (b) float arr[2,2] ;

- (c) float[2][2] arr

; Q14

When an array element is passed to a function then this array element is passed to the function.

**By reference**

- (a) By data type

- (b) By value

- (c) By

data Q15

Which of the following operator is used to access the value of variable pointed to by a pointer?

- (a) \* operator

- (b) -> operator

- (c) && operator

**& operator**

Q16

Paying attention to detail in designing a program is \_\_\_\_\_

- (a) Time consuming

- (b) Redundant

**Necessary**

- (c) Somewhat Good

Q1

7

How does elements of two-dimensional array store into memory? (1)

Two dimensional arrays are also stored in liner order. As like one dimensional array.

Q18

Which strategy is used by binary search algorithm to search a number? (1)

ANS

„divide and conquer” strategy is applied.

Q19

Write down the general syntax of switch statement. (2)

ANS

**switch ( *variable/expression* )**

{

*case constant1 : statementLlist1 ; case constant2: statementLlist1 ; default: statementLlist1 ;*  
}

Q20

What is a Linker? (3)

**Linker** Most of the time our program is using different routines and functions that are located in different files, hence it needs the executable code of those routines/functions. Linker is a tool which performs this job, it checks our program and includes all those routines or functions which we are using in our program to make a standalone executable code and this process is called Linking

Q21

What are similarities and differences between Structures and Unions? (5)

Structure

In structures, the data members are public by default. It means that these are visible to all and anyone can change them. Is there any disadvantage of this? Think about the date.

syntax

struct student

{

char

name[60];

char

address[100];

float GPA;

};

**Unions** We have another construct named union. The concept of union in C/C++ is: if we have something in the memory, is there only one way to access that memory location or there are other ways to access it. We have been using int and char interchangeably in our programs. We have already developed a program that prints the ACSII codes. In this program, we have stored a char inside an integer. Is it possible to have a memory location and use it as int or char interchangeably?

For such purposes, the construct union is used. The syntax of union is:

union intOrChar

{

i

n

t

```

i
;
c
h
a
r
c
;
};

```

Q22

Differentiate between C and c++. (10)

1. C was the C++ predecessor. As its name implies, a lot of C remains in C++. Although not actually being more powerful than C.
2. C++ allows the programmer to more easily manage and operate with Objects, using an OOP (Object Oriented Programming) concept
3. C++ allows the programmer to create classes, which are somewhat similar to C structures. However, to a class can be assigned methods, functions associated to it, of various prototypes, which can access and operate within the class, somewhat like C functions often operate on a supplied handler pointer.
4. Although it is possible to implement anything which C++ could implement in C, C++ aids to standardize a way in which objects are created and managed, whereas the C programmer who implements the same system has a lot of liberty on how to actually implement the internals, and style among programmers will vary a lot on the design choices made
5. In C, some will prefer the handler-type, where a main function initializes a handler, and that handler can be supplied to other functions of the library as an object to operate on/through. Others will even want to have that handler link all the related function pointers within it which then must be called using a convention closer to C++.
6. C++ applications are generally slower at runtime, and are much slower to compile than C programs. The low-level infrastructure for C++ binary execution is also larger. For these reasons C is always commonly used even if C++ has a lot of popularity, and will probably continue to be used in projects where size and speed are primary concerns, and portable code still required (assembly would be unsuitable then).

Q1

The remainder (%) operator is

- (a) A logical operator
- An arithmetic operator**
- (b) A relational operator
- (c) A division operator

Q2

If `int sum = 10;` then the value of the statement `sum = sum + 3 ;` is ,

- (a) 7
- (b) Illegal statement
- (c) Garbage value

**13**

Q3

Which of the following function(s) is/are included in **ctype.h** header file?

- (a) `isdigit(int c)`
- (b) `isxdigit(int c)`
- (c) `tolower(int c)`

**All of the above**

Q4

In C/C++ which of the following header file is used for string manipulation?

**stdlib.h**

- (a) `string.h`
- (b) `strings.h`
- (c) `stype.h`

Q5

\_\_\_\_\_ provide communication channels between files and program.

- (a) Streams
- (b) Language like C++

**Function seekg()**

- (c) None of the above

Q6

\_\_\_ translates high level language program into machine language code

- (a) Debugger
- (b) Editor

**Compiler**

- (c) Linker

Q7

Which of the following data type(s) can operate on modulus operator „%“?

**float, int**

- (a) float, double
- (b) int
- (c) char

Q8

C++ is a case-sensitive language

**True**

- (a) False

Q9

To include code from the library in the program, such as `iostream`, a directive would be called up using this command.

- (a) `#include "iostream.h"`
- (b) `include <iostream.h>`
- (c) `include <iostream.h>`

**#include <iostream.h>**

Q10

What will be the range of numbers generated by function `rand () % 9`?

(a) 0 to 9

(b) 1 to 9

**0 to 8**

(c) 1 to 8

Q1

1 An array stores the numbers into consecutive memory locations.

**True**

(a) Fals

e Q12

Which of the following is the correct statement for the following declaration? `const int *ptr`.

(a) `ptr` is a constant

pointer **`ptr` is constant**

**integer pointer**

(b) `ptr` is a constant pointer to `int`

(c) `ptr` is a constant pointer to

int Q13

Which of the following header file defines the **`rand()`** function?

(a) `iostream.h`

(b) `conio.h`

**`stdlib.h`**

(c) `stdio`.

h Q14

Consider the following code segment. What will be the output of following code?

```
int addValue
(int *a){ int b =
(*a) + 2; return
b ;
}
main
in
()
{
int
x
=6
;
cout << x << " , " ;
cout << addValue(&x)
<< " , " ; cout << x ;
}
```

**(a) 6,8,6**

- (b) 6,6,8
- (c) 6,8,8
- (d) 6,6,

Q15

Identifier is a name that can be given to variables, labels and functions.

**True**

(a) Fal

s Q16

For which array, the size of the array should be one more than the number of elements in an array?

- (a) Int
- (b) Char
- (c) Double

**float**

Q17

Give a precise definition of **function** . (1)

**AN**

**S**

**Functions** In C/C++, functions are a way of modularizing the code. A bigger problem is broken down into smaller and more manageable parts. There is no rule of thumb for the length of each part but normally one function's length is not more than one screen.

Q18

What will be the size of array if we initialize an array with declaration: **int arr[] = {0, 0, 0, 0};**? (1)

**ANS**

```
int
arr
[3]
;
Q1
9
```

What is the difference between **switch** statement and **if** statement. (2)

**ANS**

In switch statement only one variable can be tested on various condition but using if we can tested multi variables in single statement

Q20

Evaluate the following arithmetic expressions. (3)

a)  $X = 2 + 6 * 4 - 4 * 20 / 5 + 3 * 2$

b)  $Y = (6 * 7) - (2 + 3) * (3 - 1) + 5 * (3 + 1)$

ANS

- a)  $X = 2 + (6 * 4) - ((4 * 20) / 5) + (3 * 2)$   
b)  $Y = ((6 * 7) - (2 + 3) * (3 - 1) + (5 * (3 + 1)))$   
a) 16  
b) 52

Q21

What is the difference between = in C as compared to = used in algebra. (5)

ANS

In C = sign is used to assigned the value in algebra = sign is show that the both side are equal.

In C there must be a variable on the = sign and the right side of the = sign must be a arithmetic expression, variable or a value.

In algebra both side of the equation may or may not be expression. Q22

**Write a program which consists of three variables Area, Per, Base, this program should find the area of triangle using the formula,**  
(10  
)

$$\text{Area} = (\text{Base} * \text{Per})/2$$

```
#include<iostream.h>
#include<conio.h>
main()
{
 float Area, Per, Base;
 // Area = (Base * Per)/2
 Per = 5.0;
 Base = 2.5;
 Area = (Base * Per)/2; cout
 <<"Area = "<<Area;
 getch();
}
```

Q1

In C/C++ language the header file which is used to perform useful task and manipulation of character data is

- (d) cplex.h  
**ctype.h**  
(e) stdio.h  
(f) delay.h

Q2

The header file which is used for input and output is

- (e) maths.h  
(f) string.h

**iostream.h**

(g) ctype.h

Q3

Suppose `int multi[5][10]`; when we are using `**multi`, it means,

(e) Single dereferencing

(f) Single referencing

**Double referencing**

(g) Double dereferencing

Q4

To access the data members of structure \_\_\_\_ is used.

**dot operator (.)**

(e) \* operator

(f) € operator

(g) None of given.

Q5

The precedence of \* is higher than dot operator (.) operator

**True**

(c) False

Q6

Which of the following is the starting index of an array in C++?

**0**

(d) 1

(e) -1

(f) any number

Q7

When a call to function statement is encountered,

**The control transfers**

**to its Prototype**

(e) The control transfers to its definition

(f) Control returns to statement following function call

(g) The compiler stops execution of whole program

Q8

A function must always return value.

(c) True

**False**

Q9

If an Array has 100 elements, what is allowable range of subscripts? (e) 0

– 99

1 – 99

**0 – 100**

1 – 100

Q10

If a pointer appears on left hand side of an assignment operator then right side of that assignment operator must be,

(d) Variable name

### **Address of variable**

- (e) Variable value
- (f) Constan

t Q11

Editors are used to compile the code.

- (c) True

**False**

Q12

Which bitwise operator returns false if both bits are 1?

- (d) AND
- (e) XOR
- (f) NOT

**OR**

Q13

What does !(7) evaluate to in decimal where ! is a NOT operator?

**7**

- (e) 8
- (f) 9
- (g) 1

0 Q14

Structures cannot be passed as Function Parameters

- (c) True

**False**

Q15

When break statement is encountered in a loop body it,

**Transfers the control outside from current loop**

- (d) Transfers the control outside from current program
- (e) Enforces the next iteration of loop
- (f) Generates compile time

error. Q16

A union is a user-defined data type that contains only\_\_\_from its list of members at a time.

- (e) One object

**Two objects**

- (f) Three objects
- (g) None of the given options

Q17

Why programming is important? Describe in ONE line. (1)

Q18

Which bitwise operator returns true if both bits are different and returns false if both bits are same? (1)

Q19

Why we close a file after use? (2)

Q20

Can you use an assignment operator to assign the value of one C-string to another? (3)

Q21

The statement `int Name [2][2];` define a 2x2 array, Write the code which read data from keyboard for this array. (5)

Q22

Write a program which reads a text file “**PlayersInfo.txt**” residing in the current directory. Open the file **PlayersInfo.txt** in read mode and assign these values to the struct **Player**; assume order of the data in the file to be exactly the same as the order of struct attributes. The struct **Player** has following attributes

- i) Name
- ii) Height
- iii) Age
- iv) Score
- v) Game

After reading the file and assigning values to the struct, in the end close the file **PlayersInfo.txt**. (10)

Q23

Q24

Write a program to convert upper case letter to lower case letter. (5)

Write a program to convert lower case letter to upper case letter. ( 5)

Q2

Write a program to convert upper case letter to lower case letter.

(5) Ans:

```
#include <iostream>
```

```
using std::cout;
```

```
using std::cin;
```

```
using std::endl;
```

```

#include <stdlib.h>

int main()

{

 char inputString[100];
 char lowerCase[100];

 cout<<"Please enter a string (maximum 100 characters): ";
 gets(inputString);

 int i=0;

 for(i=0; i<strlen(inputString); i++)

 {

 lowerCase[i] = tolower(inputString[i]);

 }

 lowerCase[i]='\0';
 puts(lowerCase);
 system("PAUSE");
 return 0;

}

```

Write a program to convert lower case letter to upper case letter. ( 5)

Ans:

```
#include <iostream>

using
std::cout;

using
std::cin;

using
std::endl;

#include
<stdlib.h>

int main()

{

 char
 inputString[
 100]; char
 upperCase[1
 00];

 cout<<"Please enter a string (maximum 100
 characters): "; gets(inputString);

 int i=0;

 for(i=0; i<strlen(inputString); i++)

 {

 upperCase[i] = toupper(inputString[i]);

 }

 upperCase[i]
 ='\0';
```

```
puts(upperC
ase);
system("PA
USE");
return 0;
}
```

# **CS201**

## **GRAND QUIZ FILE (NEW)**

### **ALMOST 800 + MCQS**

### **BY NIMRA KHAN**

1 --- is used for input file or for reading from the file.

**Tellg()**

2. Following is the declaration of a ----- array. `int arr[2][2][2]`

**3-dimensional**

3. `!(x < 3)` means in C++ that

**X is greater than 3**

4. Which of the following options is correct to compare the declaration of the function?

----- (parameter1,parameter2)

{

// code to be executed

}

**Void functionNAME**

5. Ofstream is used for -----

**Output file stream**

6. By default, the starting index of an array in c++ is \_\_\_\_\_

**0**

7. How we declare an array of integer data whose size is 20 with

array name "count".

**Int count [20];**

8. Which of the following is a reserved word for data types in c++?

**Int**

9. C was developed with the help of ----- language

**BCPL and B**

10. An object of ----- class can be created for saving the

current position of any opened file before reading from the file.

**Streampos**

11. C++ is a ----- language.

**High level**

12. In while loop, the loop counter is initialized at -----

**Before entering the loop**

13. What is the output of the following code if the 3rd case is true

```
Switch (var)
{
 Case „a“;
 Cout<<”apple”<<endl;
}
```

**Mango**

14 ----- character is used to indicate the end of the string.

**Null**

15. Which of the following is used to perform bit-wise exclusive OR operation?

|

16. If  $x = 11$ , and  $x \% 3 = 2$ , then value of  $x$  will be-----

**2**

17. A variable of character data type occupies ----- bytes in memory.

**1**

18. Null character is represented by ----- in c++?

**\0**

19. In computer system there are mainly ----- type of softwares.

**2**

20. There is pointer variable named ptr of type float. Which type of variable address can be stored by ptr?

**Float**

21. A character is stored in the memory in -----

**Byte**

22. X is an integer variable; what does x++ mean,

**Add 1 in x value**

23. When we call a function, its control -----

**Transfers to its definition**

24. '\0' takes ----- space in the memory.

**1 byte**

25 -----will be used for enclosing function statements into a block?

**{ }**

**26. How can we declare an array of characters whose size is**

12 with array name "country"?

**Char country[12];**

**27. Following is the declaration of a 2d array, how many columns are declared for this array.**

Int arr[3][2];

**2**

**28. If we assign 2.06721 to an integer variable x, what will be the output if we print x using cout << statement?**

**2**

**29. We can access a global variable ----**

**From anywhere in the program**

**30. We use ----- to include all those functions which are located in different files.**

**Linker**

31. A character is stored in the memory in ----

**Byte**

32. Which is not true regarding structure?

**We can declare array of structure**

33. Mr. ahmad sleeps thirty hours every day. This sentence is \_\_\_\_

**Grammatically correct but illogical**

34 --will explain the function of a program.

**Comments**

35. Char \*\*argv can be read as ----

**Pointer to pointer to char**

36. What will be output of the following code? Stringname[4]=  
,"ali","bilal","omer",Ayesha"-; Cout<< name [0];

**Ali**

37. The data type before a function name represents its,

**Return type**

37. The expression  $((a+b) == 10) \parallel ((x+y) == 20)$  is true only if?

**Either  $((a+b) ==)$  OR  $((x + y) == 20)$  is true**

38. = operator is used for ----

**Assigning values to variables**

**39. What is the use of ios::trunc mode?**

**To truncate an existing file to zero**

**40. X is an integer variable; what does  $x++$  mean,**

**Add 1 in x value**

**41. Structures use ----- allocation.**

**Heap**

**42. What will be the size of the following character array?**

`Char name[] = "adeel";`

**6**

**43. A ----- is an array of characters that can store number of  
character specified.**

**String**

44. -- statement is used to terminate the processing of a  
particular case and exit from switch structure.

## Break

45. If  $y = 20$  and  $z = y ++$ , then the value of  $z$  will be-----

2146. If

Int a =

17; int

b = 3;

double c = a/b;

what will be value of variable c?

## 5

**47. What will be the output of the following code?**

```
String grade [3] = {"a", "b", "c"}; Grade[2]
```

```
= "F";
```

```
Cout<< grade[2];
```

## F

**48. Declaration of two-dimensional array of integers having 3 rows and 5 columns is following?**

**Int arr [3][5];**

**49.  $A = a + 1$ ; can be written as ----**

$A++ = 1$ ;

**50. A record is a group of related**

Fields

51. What will be the result of arithmetic expression  $6 + 48 / 4 * 3$ ?

**42**

52. ----- construct is used to execute a set of instructions

when a condition is true and other set when it is false?

**If else**

53. If an array has 50 elements, what is the allowable range of subscripts?

**0 – 49**

54. If `int a = 50;` then the value of `a /= 3;` will be?

**16**

55. Which of the following option is/are correct about array?

**Only one type of elements can store in an array**

56. Whenever some number is added in an array name, it will jump as many as the added number?

**Rows**

57 ----- return true if c is a digit or a letter and false otherwise.

Int isalnum(int c)

**58. Which of the following is the default mode of the opening using of ofstream class?**

**ios::trunc**

**59. In which language the compiler of C language is written?**

**C**

**60. While loop executes at least?**

**One time**

**61. String are accessed by variable of type----**

**Char**

**62. Which of the following is the correct operator to compare the values of two variables?**

**= =**

**63. Not operator is a -----**

**Single input operator**

**64. Pointer points to a particular-----**

**Memory address**

**65. A function is a block of statements that can be defined once and used ---in the program.**

**As many times as user wants**

66 ----- function is used to find the number of characters in a string?

**Strlen()**

**67. != operator is used to check whether the operand on the left-hand side is to the operand on the right-hand-side.**

**Not equal**

**68. A while loop may execute ----- or more times.**

**0**

**69. How we declare an array of integer data whose size is 20 with array name "count".**

**Int count[20];**

**70. A variable declared within a code block becomes ----- variable for that block.**

**Local**

**71. If a file is opened with ios::out mode, then -----**

**We can write in this file**

72. A function's prototype is written --- the function call.

**Before**

73. Loop is mostly used to perform repetition of tasks for a known number of times.

**For**

74. In shifting operations if zero is inserted at the least most bit, the negative number will become a ----- number.

**Positive**

75. It is the job of ----- to transfer the executable code from hard disk to  
main memory.

**Loader**

76. RAID stands for -----

**Redundant array of inexpensive devices**

77. Array indexes start from ----

**0**

78. In call by reference, ----- of a variable is passed to the called function.

**Address**

79. How many times the following do-while loop will execute?

```

Intk=
10;
Do
{
Cout << "statements" << endl; ?

```

**6**

**80. The name of a location in memory is called ----**

**Variable**

81-- function is used to find the number of character in a string.

**Strlen()**

82. Which statement is used to close a file?

**Myfile.close();**

83 ----- allow us to have a memory location and use it as int or char interchangeably.

**None of the given**

84. While handling files, one can have ----- options

**All of the given**

85 ----- function is used to determine the next position to write a character while writing into a file.

**Tellp()**

**86. Which is the correct syntax to define a structure named 'student'?**

**Struct student{**

**//variables**

**};**

**87. <and> both are ----- operators.**

**Logical**

Identify the logical error in the following if condition 88. If ((i<5) & & (i>15)) ?

**Condition is always false**

**89. Not operator is a ----**

**Single input operator**

**90. The use of \* sign in first and second statements is called --- and ----  
of pointer.**

Int\*ptr=

&x;

Cout<<\*ptr;

?

**Referencing, dereferencing**

**91. C is widely known as development language of ----- operating system.**

**Linux**

**92. If x is declared as an integer, what will be the result of the given expression?**

$X = 2 + 3 * 5 \% 2 + 3 / 2$  ??

**4**

**93. What will be the correct syntax of declaration of the following statements?**

“ptr is a pointer to const int” ??

**Const int \*ptr**

**94. After evaluating the expression**

**95.  $X = 10 + 3 * 4 / 2 - 1$ , the value of x will be?**

**15**

**96. We can read, write and manipulate the same file using----**

**Fstream.h**

**97. Return type of a function that does not return any value must be----**

**Void**

**98. Pointers store the----**

**Memory address**

99 -- operator uses the sign^.

**Exclusive OR**

100. To get the value stored at a memory address, we use the----

**Dereferencing operator**

101. Telling a computer what to do is called ----

**Programming**

102. Compiler translates high level language program into----- language code.

**Machine**

103. Each array declaration must provide the information about all these things except----

**The location of the first element to be stored**

104. EOF is used to check for the----- of a file when a file is being read.

**End**

105. The result of  $4\%3$  will be?

**1**

106. When the logical operator AND (&&) combines two expressions exp1 and exp2 then the result will be true only?

**When both exp1 and exp2 are true**

107. 'multi\*3+\*3+' element can also be accessed by -----

**\*(\*(multi + 3) + 3)**

108. What is output of following code?

```
Int data [10] = {10,20,30,40,50,60,70,80,90,100};
```

```
Int*xptr=&data[4]; Int
```

```
*xptr =&data [6];
```

```
Cout<<xptr+yptr++;??
```

**120**

109. Which of the following is not the subcategory of the system software?

**None of the above**

110. We get  $12|8=$ ----- where "|" is (bit-wise OR)?

**12**

111 --- is a substitute of multiple if statement.

**If..elseif statement**

112. Find out the logical error in following lines of code. If (x=10)

```
Cout << "x is 10"; ?
```

**Assignment operator should not be used for comparison**

113. Which is not true regarding structure?

### c) we can't declare pointers to structure

114-----are used to compile the code.

### Compiler

115. #include

<iostream> #include

<cstring> Using

namespace std; Int

main()

```
{
 Charstr1[10]="hello"; Char
 str2[10]="world";? 10
```

116. A----- structurespecifiesthatanactionistoberepeatedwhilesome  
condition remains true.

### Repetition

117. Inwhichlanguage the compiler of clanguage is written?

**C**

118. Characters string are terminated by -----

**\0**

119. A function is a block of statements that can be defined once and used-  
-- in the program.

### As many times as user wants

120 ----- is used for output during the process of writing to the file.

### Seekp()

In case of two-dimensional array, if you add some number in a two-dimensional name, it will jump

to the next----

## **Row**

121. In while loop the condition is checked ----- the execution of loop.

## **Before**

122. Array indexes start from ----

**0**

123. Pointers store the----

## **Memory address**

124. Which step is not performed during a proper program design?

## **Write details about the performance of program**

125. We can say that a pointer is a variable that holds the ----- address.

## **Memory**

126. When accessing a structure memory, the identifier to the left of the dot operator is the name of ----

## **Structure variable**

127. From the following; which one is the range of random number generator rand() ?

**0 – 32767**

128. To read command-line arguments, the main() function itself must be given arguments.

**2**

129. Which operator has the highest precedence? (\*, -, /, +)?

**\* and / both have the same highest precedence**

130. Suppose we have `int y[10];`

To access the 4<sup>th</sup> element of the array we write----

**Y[3];**

131. What will be the value of the variable `output` in the given piece of code? `Double output = 0;`

`Output = (2 + 2) * 4 + / (4 - 2);`

**17**

132. What is the output of the following statement? `int i =`

`2.5; do { cout << i * 2; } while (i > 3 && i < 10);` error

**Searching is easier when an array is already sorted  
True**

133. If an array has 100 elements, what is allowable range of subscripts?

**0 - 99**

134. While developing a program; should we think about the user interface?

//handouts main

**Yes**

135. A character is stored in the memory in \_\_\_\_

**String**

136. These functions `seekg()` and `seekp()` requires an argument of type \_\_\_\_\_ to let them how many bytes to move forward or backward.

**Int**

dereferencing operator is represented by \_\_\_\_

**\***

\_\_\_\_\_transfers the executable code from main memory to hard disk.

### **Loader**

When the logical operator OR (||) combine two expressions exp1 and exp2 then the result will be false only, //bitwise “or” applies here....

### **When both exp1 and exp2 are false**

suppose we have `int y[10]`; To access the 4th element of the array we write \_

### **none of given**

we have opened a file stream myfile for reading (getting), `myfile.tellg()` gives us the current get position of the file pointer. It returns a whole number of type \_\_\_\_\_

### **int**

`ofstream` is used for \_\_\_\_\_

### **Output file stream**

A structure is a \_\_\_\_\_ under a single name.  
collection of

### **Variables**

In C/C++ the string constant is enclosed ...corrected r red one

### **In curly braces**

The size of `int` data type is 4 bytes

In Flow chart process is represented by

Rectangle

If the break statement is missed in switch statement then, // a break

statement with no label always completes abruptly, the reason being a break with no label. If no switch, while, do, or for statement in the immediately enclosing method,

constructor or initializer encloses the break statement, a compile-time error occurs?

**This may cause a logical error**

When we are using const keyword with a variable x then initializing it at the time of declaration is, ?

**Optional**

Which of the following is the correct way to assign an integer value 5 to element of a matrix say 'm' at second row and third column?

**m[2][3] = 5 ;**

How many total elements must be in two-dimensional array of 3 rows and 2 columns?

**6**

Which of the following is the correct statement for the following declaration?

**\*ptr is a pointer to const int**

What will be the correct syntax to assign an array named arr of 5 elements to a pointer

ptr?

**\*ptr = arr[5] ;**

C is a/an language

**object oriented**

**The variables having a name, type and size are just like empty boxes.**

**True**

What's wrong with this for loop? for (int k = 2, k <=12, k++)

**the commas should be semicolons**

Most efficient method of dealing with structure variables is to define the structure globally

**True**

There are mainly      types of software.//

**Two**

When a pointer is incremented, it actually jumps the number of memory addresses ?

**According to data type**

setw is a parameterized manipulator.

**True**

if (a>b && a>c) then the condition will be true  
only if Both a>b and a>c are

**true**

In while loop the loop counter must be initialized,  
Before entering the

**loop**

\_\_\_\_\_ of a variable means the  
locations within a program from where it  
can be accessed.

**Reference**

Switch support data.

**Integer Character Float**

**Both integer and character**

A function is a block of statements that  
can be defined once and used\_\_in the  
program.

**As many times as user wants**

\_\_\_\_\_ statement is used to

**terminate the processing of a  
particular case and exit from**

switch structure.

**break**

The most suitable data type for number 325.25 is\_.

**float**

X is an integer variable; what does X++ means,

**Add 1 in X value**

The operators ++ and -- are used to  
increment or decrement the value of a  
variable by\_\_\_\_\_

**1**

Which of the following loops checks the test  
condition at the end of the loop?

**Do-While**

The operator to free the allocated memory  
using new operator is

\_\_\_\_\_. **delete**

suppose we have `int y[10]`; To access the 4th element of the array we write\_

---

**y[3];**

From the following; which one is the range of Random number generator function `rand()`?

**0 – 32767**

From following; which one is the correct syntax of an array initialize: Array size is 10 and it is of double data type to value 0?

**double arr[10] = {0.0};**

The string in the array is terminated by a\_\_\_\_\_

**null**

In C/C++; by default arguments are passed by\_\_\_\_\_to a function.

**Value**

char name [] = "Hello World"; In the above statement, a memory of characters will be allocated

**12**

\_Keyword is used to return some value from a function.

**return**

Which of the function call is call by value for the following function prototype? float add(int);

**add(x);**

The increment of a pointer depends on its \_.

**data type**

The ASCII code of null character is \_\_\_\_\_

**000**

Pointers are a special type of in which a memory address is

**store**

Transpose of a matrix means that when we interchange rows and columns \_\_\_\_\_

**the first row becomes the first column**

Pointers store the \_\_\_\_\_  
**value of a variable**

Which of the following function call is “call by reference” for the following function prototype?

**int add (int \*); add(&x);**

The name of the array is a constant pointer which contains the memory address of the of the array.

**first element**

array name always contains the memory address of the of the array

**first element**

At the , we try to break up the problem into functional units

**design phase**

Syntax of a union is identical to \_.

**structure**

A union is a user-defined data type that contains only \_--from its list

of members at a time.

**one object**

Which of the given file handling modes are similar in their functionality ?

**ios::trunc, ios::ate**

Which one of the given option is not a mode for reading/writing the data from a file?

**Trunk**

In C/C++, the header file used for string manipulation is \_\_.

**string.h**

A 2D array multi[5][10] can be accessed using the array name as \*\*multi, this technique is called \_\_.

**Double dereferencing**

C language was evolved from \_\_\_\_\_ language(s).

**B and BCPL**

The \_\_\_\_\_ statement forces the immediate next iteration of the loop.

**Continue**

\_\_\_\_\_ are very good tools for code reuse.

**Functions**

When an argument is passed by reference, \_\_\_\_\_

**the function accesses the argument's original value in the calling program.**

\_\_\_\_\_ executes all the lines before error and stops at the line which contains the error.

**Intrepreter**

Analysis is the \_\_\_\_\_ step in designing a program.

### **First**

Which of the following is the correct function call, having array named student of 10 elements as a parameter?

**addRecord(student);**

When an array element is passed to a function, it is passed by

\_\_\_\_\_.

### **Reference**

What is the correct syntax to declare an array of size 10 of int data type?

**int name[10] ;**

If we want to copy an array, both arrays need to be of same \_\_.

### **Data type and size**

Which of the following is the correct syntax to print multiple values or variables in a single command using cout?

**cout << "H" << x << "\n";**

Which one of the statements given below is equivalent to  $x \% = 5$ ?

**$x = x \% 5$**

The condition in loop should be a(n)\_\_\_\_\_.

### **Boolean Expression**

Which structure is represented by following flow chart ?

### **if structure**

Stream which sends data to the program is called\_\_\_\_\_.

## Input stream

Syntax of a union is identical to \_\_\_\_\_.

## Structure

When accessing a structure member, the identifier to the left of the dot operator is the name of \_\_\_\_\_.

## a structure variable

The parameter passed to isdigit() function is \_\_\_\_\_.

## a character variable

TWAIN stands for \_\_\_\_\_.

## Technology Without An Interesting Name

\_\_\_\_\_ translates high level language program into machine language code.

## Compiler

From the option given below, what will be the correct output of the code? for (int a = 1; a <= 3; a++)

```
cout << a*3-1 << " ";
```

## 2 5 8

If the return type of a function is void then it means that it will \_\_\_\_\_.

## return any type of data return no data

The functions that have already been defined as a part of the language are called \_\_\_\_\_.

### **built-in functions**

Given that `doubleArray` is an array of 7 integers. Which of the below is the correct statement to access the 5th element of the array?

### **`doubleArray[4]`**

The symbol which is used to represent a process in a flow chart is \_\_\_\_\_.

### **Rectangle**

When an array is passed to a function, by default it will be passed by \_\_\_\_.

### **Reference**

What is the range of Random number generator function `rand()`?

### **0 – 32767**

The first character of a variable name must be an alphabet or \_\_\_\_\_.

### **Underscore**

In \_\_\_\_, the first pointer stores the address of the some other pointer, which contains the address of any variable.

### **single dereference**

Initialization of variable at the time of definition is ----- practice.

## **Good Programming**

Which looping process checks the test condition at the end of the loop?

## **do-while**

You can use a \_\_\_\_\_ statement to transfer control elsewhere out of the nested structure.

## **Goto**

Which of the following array is the most suitable for storing a matrix structure?

## **Single-dimensional array**

The Microsoft Word document (.doc) is a kind of \_\_\_\_\_.

## **Random Access File**

A union is a user-defined data type that contains only \_\_\_\_\_ from its list of members at a time.

## **one object**

In C/C++, \_\_\_\_\_ are the doors through which data can enter and come out of a program.

## **Streams**

Which one of the given options is not a mode for reading/writing the data from a file?

## **Trunk**

C++ views each file as a sequential stream of \_\_\_\_\_.

## **Bytes**

In 2D array, the technique used to access the array elements is called \_\_\_\_\_

## **Double dereferencing**

From the options given below, identify the correct option which is used for calling the function float square (int &);

**square (num);**

Identify the correct option which is used for calling the function float area (int).

**area(num) ;**

Which kind of error occurs in the code below to access the variable num in the main function?

```
int num
=10; int
main(){
int x;
x = num ;
}
```

**No error**

A program statement that invokes a function is called \_\_\_\_\_.

**function call**

What will be the output of the following code segment? main(){

```
int a =
10 ; int
b =0; int
*x=&a;
b =
(*x)+3 ;
cout << a << " , " << b ;
}
```

**1**

**0**

**,**

**1**

**3**

When the logical operator AND (&&) combines two expressions exp1 and exp2 then the result will be true only

**When both exp1 and exp2 are true**

For which array, the size of the array should be one more than the number of elements in an array?

**char**

What is the range of Random number generator function rand()?

**0 – 32767**

To assign a value to a character type variable, \_\_\_\_\_ are used around the value.

**single quotes ' '**

In C/C++, the #include is called \_\_\_\_\_.

**Preprocessor Directive**

The number 544.53 must be stored in \_\_\_\_\_ data type.

**Float**

Use of the \_\_\_\_\_ statement violates the rules of structured programming.

**Goto**

In Flow Chart, flow of control is represented by \_\_\_\_\_.

**Arrow**

What will be the correct syntax for initialization of pointer ptr of type int with variable x?

**int \*ptr = &x ;**

Syntax of a union is identical to \_\_\_\_\_.

**structure**

A union is a user-defined data type that contains only \_\_\_\_\_ from its list of members at a time.

**one object**

Which of the given file handling modes are similar in their functionality ?

**ios::trunc, ios::ate**

Which one of the given options is not a mode for reading/writing the data from a file?

**Trunc**

Stream which sends data to the program is called \_\_\_\_\_.

**Input stream**

Syntax of a union is identical to \_\_\_\_\_.

**Structure**

When accessing a structure member, the identifier to the left of the dot operator is the name of \_\_\_\_.

**a structure variable**

The parameter passed to isdigit() function is \_\_\_\_\_.

**a character variable**

TWAIN stands for \_\_\_\_\_.

**Technology Without An Interesting Name**

\_\_\_\_\_ translates high-level language program into machine language code.

**Compiler**

From the option given below, what will be the correct output of the code? for (int a = 1; a <= 3; a++)

```
cout<<a*3-1<<" "; 2
```

**5 8**

If the return type of a function is void then it means that it will \_\_\_\_\_.

**return no data**

The functions that have already been defined as a part of the language are called

\_\_\_\_\_.

**built-in functions**

Given that doubleArray is an array of 7 integers. Which of the below is the correct statement to access the 5th element of the array?

**doubleArray[4]**

The symbol which is used to represent a process in a flow chart is

\_\_\_\_\_. **Rectangle**

When an array is passed to a function, by default it will be passed by \_\_\_\_.

**Reference**

The first character of a variable name must be an alphabet or

\_\_\_\_\_.

**Underscore**

In \_\_\_\_, the first pointer stores the address of the some other pointer, which contains the address of any variable.

**single dereference**

Initialization of variable at the time of definition is ----- practice.

**Good Programming**

Which looping process checks the test condition at the end of the loop?

**do-while**

You can use a \_\_\_\_\_ statement to transfer control elsewhere out of the nested structure.

**Goto**

An object of \_\_\_\_\_ class can be created for saving the current position of any opened file

**before reading from the file Stream**

The value returned by the eof() function is of \_\_\_\_\_ type.

**Character**

If a file is not properly closed in a program then what is the status of the program ?

**The program will terminate normally**

The parameter passed to isdigit() function is \_\_\_\_\_.

**a character variable**

In C/C++, the header file used for string manipulation is\_\_\_\_\_.

**string.h**

Arrays are best data structures\_\_\_\_\_.

**for relatively permanent collections of data**

Each array declaration must provide the information about all these things except

\_\_\_\_\_.

**Number of elements of the array**

The first step for developing a program is\_\_\_\_\_.

**Linking Analysis**

\_\_\_\_\_ transfers the executable code from hard disk to main memory.

**Loader**

\_\_\_\_\_ operator is used to pass the address of a variable to the called function.

**&**

In call by reference, \_\_\_\_\_ of a variable is passed to the called **function. Address**

\_\_\_\_\_ of a function is also known as signature of a function.

## **Declaration**

Preprocessor program performs its function before \_\_\_\_\_ phase takes place. **Compiling**

Which of the following is the correct function call, having array named student of 10 elements as a parameter?  
**addRecord(student[10]);**

The remainder (%) operator is a \_\_\_\_\_ operator.

## **Arithmetic**

Which of the following array is the most suitable for storing a matrix structure?

## **Two-dimensional array**

\*.doc is \_\_\_\_\_ by type.

## **Random Access File**

Which of the following is NOT a preprocessor directive?

## **#define**

The return type of operator function must always be void.

## **False**

Question No: 4 (Marks: 1) - Please choose one What does (\*this) represents?

## **A value of the data member**

The statement cin.get(); is used to,

## **Read a character from keyboard**

When we do dynamic memory allocation in the constructor of a class, then it is necessary to provide a destructor.

**True**

Overloaded new operator function takes parameter of type size\_t and returns

**void pointer**

The second parameter of operator functions for << and >> are objects of the class for which we are overloading these operators.

**True**

C++ is a case-sensitive language

**True**

To include code from the library in the program, such as iostream, a directive would be called up using this command.

► **#include <iostream.h>**

A template function must have only generic data types.

► **True**

Template class can not have static variables.

► **False**

What will be the correct syntax to assign an array named arr of 5 elements to a pointer ptr?

► **\*ptr = arr[5];**

What will be the correct syntax to access the value of fourth element of an array using pointer ptr?

► **Both 1 and 3**

If most significant bit of un-signed number is 1 then it represents a positive number.

► **True**

If there is a symbol (& sign) used with the variable name followed by data type then it refersto\_\_\_\_andif&isbeingusedwithvariablenamethenitrefersto\_\_\_\_.

► **Reference variable, address of variable**

We can also do conditional compilation with preprocessor directives.

► **True**

The default value of a parameter can be provided inside the \_\_\_\_\_

► **function definition**

Classesdefinedinsideotherclassesarecalled\_\_\_\_\_classe

► **nested**

What purpose do classes serve?

► **All of the given options**

Everyclasscontains\_\_\_\_\_.

► **Constructor**

newoperatorisusedtoallocatememoryfromthefreestoreduring

► **None of the given options**

When an object of a class is defined inside another class then,

▶ **Destructor of inner object will be called first**

It is possible to define a class within another class.

▶ **True**

New and Delete are also used with \_\_\_\_\_ and data types as well.

▶ **Class, Objects**

With New keyword, data types and class members are initialized with meaningful values instead of garbage.

▶ **True**

C language is developed by

▶ **Dennis Ritchie**

Which of the following choice is not an example of an int datatype?

▶ **-4.0**

In flow chart, the symbol used for decision making is,

▶ **Diamond**

Switch statement deals with,

▶ **Integer and character data**

Default case in switch statement is,

▶ **Optional**

\*.doc is \_\_\_\_\_ by type.

▶ **Sequential File**

Member function tellg() returns the current location of the \_\_\_\_\_

pointer.

▶ **tellptr()**

Dealing with structures and functions passing by reference is the most economical method

▶ **True**

In C/C++ all character strings are terminated with,

▶ **Null character**

Word processor is <http://vustudents.ning.com>

▶ **Application software**

Which of the following can not be a variable name?

▶ **10area**

Which looping process is best, when the number of iterations is known?

▶ **while**

By default an array of characters is passed by value to a function

▶ **False**

Which of the following operator is used to access the address of a variable?

▶ **& operator**

The name of an array represents address of first location of array element.

▶ **False**

Compiler is a

▶ **Application Software**

If Num is an integer variable then Num++ means,

► **Add 1 with Num**

For one byte there are \_\_\_\_\_

combinations of values that can be stored in computer.

►  **$2^8$**

In C/C++ language the header file which is used to perform useful task and manipulation of character data is

► **ctype.h**

Default case in switch statement is,

► **Optional**

When break statement is encountered in switch statement, it

► **Exits from switch statement**

What will be the result of arithmetic expression  $6 + 27 / 3 * 3$ ?

► **33**

What is the correct syntax to declare an array of size 10 of int data type?

► **int name[10] ;**

How many dimensions does n-dimensional array has?

► **n dimensions**

What will be the correct syntax to access the value of fourth element of an array using pointer ptr?

► **ptr[3]**

Which of the following values C++ uses to represent true and false?

► **1 and 0**

Declaring structures does not mean that memory is allocated.

▶ **True**

For which array, the size of the array should be one more than the number of elements in an array?

▶ **char**

If a variable is passed by value to a function and the function makes some changes to that variable then it

▶ **does not affect the original variable**

In C/C++ the #include is called,

▶ **Preprocessor Directive**

Loops are \_\_\_\_\_ Structure.

▶ **Repetition**

the sizeof operator is used to determine the size of \_\_\_\_\_.

▶ **data type**

Structures use \_\_\_\_\_ allocation.

▶ **Stack**

Which header file must be included to use the functions tolower() and toupper()?

▶ **ctype.h**

In Analysis, we try to have a \_\_\_\_\_

▶ **Precise problem statement**

Pointers store the \_\_\_\_\_

► **memory address**

\_\_\_\_\_ Return true if c is a digit and false otherwise.

**int isdigit( int c )**

eof is used to check for the \_\_\_\_\_ of file when a file is being read.

**End**

The statement `cout << yptr` will show the \_\_\_\_\_ the yptr points to.

**memory address**

Which of the following is true about streams?

**A & B**

While handling files, one can have \_\_\_\_\_ options

**All of the given**

If an array has 50 elements, what is the allowable range of subscripts?

**0 – 49**

A structure is a collection of \_\_\_\_\_ under a single name.

**variables**

If stream is used for \_\_\_\_\_

**Input file stream**

The ASCII code of null character is \_\_\_\_\_

**000**

dereferencing operator is represented by \_\_\_\_\_

**\***

Whenever some number is added in an array name, it will jump as many \_\_\_\_\_ as the added number.

**rows**

To access the data members of structure, \_\_\_\_\_ is used.

**Dot operator**

Which of the following is true about streams?

- A. It is a sequence of bytes
- B. It is an ordered sequence
- C. All bytes can go through the stream simultaneously
- D. Bytes that enters first into the stream will go out at last

**A and B**

We use \_\_\_\_\_ to open a file or stream for extraction

**in**

char\*\*argv can be read as \_\_\_\_\_.

**pointer to pointer to char**

From following; which one is the correct syntax of an array initialize: Array size is 10 and it is of double data type to value 0?

**double arr[10] = {0.0};**

Pointers work by pointing to a particular\_\_\_\_\_

**data type**

\_\_\_\_\_function give the position of the next character to be read from that file.

**tellg()**

aFile. seekg (-10L, ios:cur)

**backward direction**

An object of \_\_\_\_\_class can be created for saving the current position of any opened file before reading from the file.

**Stream**

Structures are syntactically defined with the word\_\_\_\_\_.

**struct**

\_\_\_\_\_allow us to have a memory location and use it as int or char interchangeably

**union**

Question # 9 of 10 ( Start time: 12:25:41 PM ) Total Marks: 1

A hierarchy of classes which are used to deal with console and disk files are called \_\_\_\_.

**IO classes**

\_\_\_\_\_function is used to determine the next position to write a character while writing into a file.

**tellp()**

For binary member operators, operands on the \_\_\_\_\_ drives (calls) the operation.

**Left**

We cannot increment \_\_\_\_\_.

**references**

We can \_\_\_\_\_ pointer.

**all of the given**

We can \_\_\_\_\_ references.

**None of the given**

What will be the correct syntax for the following function call?

**add(x);**

An instance of a class is called \_\_\_\_\_.

**object**

The \_\_\_\_\_ is called automatically when an object destroys

**destructor**

The destructor is used to \_\_\_\_\_.

**deallocate memory**

\_\_\_\_\_ data is not accessible by non-member functions or outside classes.

**private**

Member functions of the class \_\_\_\_\_ main program.

**are accessible from**

Overloading means :

**Using the same name to perform multiple tasks or different tasks depending on the situation.**

The main advantage of function overloading is \_\_\_\_\_.

**The program becomes more readable**

You cannot overload the \_\_\_\_\_ operator.

**? :**

In C++, a variable can be declared anywhere in the program this will increase \_\_\_\_.

**efficiency**

Memory allocated from heap or free store \_\_\_\_\_.

**cannot be returned back unless freed explicitly using free and delete operators**

We cannot use \_\_\_\_\_ pointer for storing and reading data from it.

**'NULL**

The dynamic memory allocation uses \_\_\_\_\_ whereas static memory allocation uses \_\_\_\_\_.

**heap , stack**

What will be the output of the given code? #include#defineMAX(A,B)((A)>(B)?(A):(B)) void main() {  
 inti,  
 x,y;  
 x =  
 23;  
 y =45;  
 i = MAX( x++, y++ );  
 // Side-effect: // larger value incremented twice cout << "x = " << x << " y  
 = " << y << '\n';  
 }

**x=24 y=47**

NULLhasbeendefinedin\_\_\_\_\_header file.

**Stdlib.h**

Symbolic constant PI can be defined as:

**#define PI 3.14**

Thefriendfunctionofaclasscanhaveaccess\_\_\_\_\_.

**to the private data members**

C++wasdevelopedby\_\_\_\_\_.

**Bejarne Stroustrup**

Oncethe\_\_\_\_\_arecreated,theyexistforthelifetimeoftheprogram.

**static variables**

Encapsulation means \_\_\_\_\_.

**that the data of a class can be accessed from outside**

An address is a \_\_\_\_\_, while a pointer is a \_\_\_\_\_.

**constant, variable**

The syntax of declaration of a function that returns the reference to an integer is \_\_\_\_\_.

**int & myfunc();**

Which one of the following is mandatory preprocessor directive for c++?

**#include**

The members of a class declared with the keyword `struct` are \_\_\_\_\_ by default.

**Public**

`getche()` is a \_\_\_\_\_ function and defined in \_\_\_\_\_ header file.

**built-in function , conio.h**

\_\_\_\_\_ operators are the ones that require two operands on both sides of the operator.

**Binary**

\_\_\_\_\_ will return the number of bytes reserved for a variable or data type.

**sizeof operator**

\_\_\_\_\_are not available in C language.

## **Library functions**

The members of a class declared without any keyword are \_\_\_\_\_ by default.

## **private**

For console input and output we use \_\_\_\_\_.

## **conio.h header file**

The name of the destructor is the same as that of a class proceeding with a \_\_\_\_.

## **~ sign**

A reference cannot be NULL it has to point a data type.

## **True**

A pointer is \_\_\_\_\_.

## **a variable for storing address**

Constructor is a special function, called whenever we \_\_\_\_\_.

## **instantiate an object of a class**

Symbolic constant PI can be defined as:

## **#define PI 3.14**

Object code is machine code but it is not \_\_\_\_\_ and \_\_\_\_\_.

## **relocatable, executable**

The default visibility for the data members of the class is private

## **accessible outside the class**

The \_\_\_\_\_ is called automatically when an object destroys.

## **destructor**

Constructor is special type of function : which has no

## **return type**

\_\_\_\_\_ variables are those that are defined outside of main.

## **Global**

Within the statement `obj1=obj2`; `obj1` will call the assignment operator function and `obj2` will be passed as an argument to function.

## **True**

When the compiler overload the assignment (`=`) operator by default then **Compiler does member wise assignment.**

It is possible to return an object from function using this pointer.

**True**

Overloaded assignment operator must be Member function of class

Let suppose

```
int a, b, c, d, e;
```

```
a = b = c = d = e = 42;
```

This can be interpreted by the compiler as `a = (b = (c = (d = (e = 42))))`;

In statement `a+b+c`, at first `b+c` is

executed first

Suppose `int i = 10`; then what is the output of `cout << oct << i`;

**12**

`ostream` is a \_\_\_\_\_ operator.

**Standalone**

\_\_\_\_\_ must be included to use stream manipulation in your code.

**Io manip**

\_\_\_\_\_ operators are the ones that require only one operator to work.

**Unary**

`Theend` and `flush` are \_\_\_\_\_.

**Manipulators**

When operator function is implemented as member function then return type of function is \_\_\_\_\_.

**Can be any data type**

When a variable is defined as static in a class then \_\_\_\_\_.

**Only one copy is created for all objects of this class**

Automatic variables are created on \_\_\_\_\_.

**Stack**

```
cout << i << " ";
```

```
cout << d << " ";
```

```
cout << f;
```

Above statements can be written within statement of one line as:

**cout << i << " " << d << " " << f;**

dec, hex, oct are all \_\_\_\_\_.

**Non-parameterized manipulators**

What will be the output of following statement?

```
cout << setfill('0') << setw(7) << 128;
```

**0000128**

Which of the following syntax is best used to delete an array of 5 objects named 'string' allocated using new operator.

**delete []string;**

If we have a program that writes the output data (numbers) to the disc, and if we collect the output data and write it on the disc in one write operation instead of writing the numbers one by one.

**In the above situation the area where we will gather the number is called Buffer**

The first parameter of operator function for << operator \_\_\_\_\_.

**Must be passed by reference**

The second parameter of operator function for >> operator must always be passed

**By reference**

The only operator that the compiler overloads for user define data type by default is-----

**Assignment (=) operator**

Consider the following code, the printed value will be converted into: int n=10;

```
cout << oct << n;
```

**Base 8**

\_\_\_\_\_ variables are defined in the main.

**Local**

ostream class is \_\_\_\_\_ and not under our control.

**built-in**

The memory allocation in C++ is carried out with the help of \_\_\_\_\_.

**new operator**

If B is designated as friend of A, B can access A's non-public members.

**A can access non-public members of B**

If the request of new operator is not fulfilled due to insufficient memory in the heap \_.

**the operator returns 0**

We should not use such variable names that are starting with \_\_\_\_

because in C++, there are lots of internal constants and symbolic names that start with it.

**double underscore**

The friend keyword provides access \_\_\_\_\_.

**in one direction only**

The malloc function takes \_\_\_\_\_ argument(s).

**one**

The constructor contains \_\_\_\_\_.

**no return type**

What will be the output of the following c++ code?

```
#include<iostream.h>
#define max 100
main()
{
#ifdef max
 Cout<<"Hellow;
}
```

**Error**

Once we have defined a symbolic constant value using #define, that value

\_\_\_\_\_

**during program execution cannot be changed**

The memory allocation functions return a chunk of memory with a pointer of type \_.

**ptr**

A class can be declared as a \_\_\_\_\_ of other class.

**member**

To avoid dangling reference, don't return\_\_\_\_\_.

**the reference of a local variable from the function**

Constructor is itself a\_\_\_\_\_ of C++ and\_\_\_\_\_.

**function, can be overloaded**

The parameter passed to isdigit() function is\_\_\_\_\_ variable.

**Integer**

char\*\*argv can be read as\_\_\_\_\_.

**pointer to pointer to char**

To read command-line arguments, the main() function itself must be given  
\_\_\_\_\_ arguments.

**2**

How many bytes an integer type pointer intPtr will jump in memory if the statement below is executed?

intPtr += 2;

**8**

The increment of a pointer depends on its\_\_\_\_\_.

**data type**

The statement cout<<yptr will show the\_\_\_\_\_ the yptr points to.

**memory address**

\_\_\_\_\_ is used as a dereferencing operator.

**\***

Transpose of a matrix means that when we interchange rows and columns \_.

**the first row becomes the first column**

Individual characters in a string stored in an array can be accessed directly using array \_.

**subscript**

We can define a matrix as \_\_\_\_\_ array.

**Multi dimensional**

A \_\_\_\_\_ is an array of characters that can store a number of characters

**specified. String**

Given a two-dimensional array of integers, what would be the correct way of assigning the value 6 to the element at third row and fourth column?

**array[2][3] = 6;**

\_\_\_\_\_ of a variable means the locations within a program from where it can be accessed.

**Visibility**

Which of the following function call is "call by reference" for the following function prototype?

**int add (int \*); add(&x);**

Which of the following function call is "call by reference" for the following function prototype?

**float add (float \*);**

**add(&x);**

Which of the function call is call by value for the following function prototype?

**float add(float);**

**add(x);**

Which of the function call is "call by value" for the following function prototype?

**float add(int);**

**add(x);**

Return type of a function that does not return any value must be \_\_\_\_\_.

**void**

\_\_\_\_\_ will be used for enclosing function statements into a block.

**{ }**

What is the output of the following code if the 2nd case is true switch (var){

case 'a': cout<<"apple"<<endl; case

'b':cout<<"banana"<<endl; case

'm':cout<<"mango"<<endl; default: cout<<"any

fruit"<<endl;

}

**Banana mango any fruit**

When the break statement is encountered in a loop's body, it transfers the control from the current loop.

**Outside**

What is the output of the following code if the 3rd case is true

switch (var) {

case 'a':cout<<"apple"<<endl; case

'b':cout<<"banana"<<endl; case

'm':cout<<"mango"<<endl; default: cout<<"any

fruit"<<endl;

}

**Mango any fruit**

What is the output of the following code, if the first case is true switch (var){  
case 'a':cout<<"apple"<<endl; case  
'b':cout<<"banana"<<endl; case  
'm':cout<<"mango"<<endl; default: cout<<"any  
fruit"<<endl;  
}

**Apple banana mango any fruit**

What will be the output of following code segment? for(int i=2;i<10;  
i++){  
if(i==5)continue;  
cout<<i<<" ";  
}

**2,3,4,6,7,8,9**

\_\_\_\_\_ statement is used to terminate the processing of a particular case and exit from switch structure.  
**break**

What will be the result of the expression j= i++; if initially j= 0 and i= 5?

**5**

What will be the result of the expression k = ++m; if initially k = 0 and m = 5?

**6**

How many times the following do-while loop will execute?

```
int k=10; do{cout<<"Statements"<<endl;k-=2;}while(k>0);
```

**6**

Which of the following loops checks the test condition at the end of the loop?

## Do-While

The operators ++ and -- are used to increment or decrement the value of a variable by \_.

1

How many times the following loop will execute?

```
int j=3; while(j>0){cout<<"Statements"<<endl; j-=2;}
```

2

A \_\_\_\_ structure specifies that an action is to be repeated while some condition remains true.

## Control

!(x>3) means in C++ that x is less

**than or equal to 3**

When the logical operator && combines two expressions then the result will be true only when the both expressions are \_\_\_\_\_.

**true**

<and> both are \_\_\_\_\_ operators.

## Relational

What will be the value of variable "input" if the initial value of input is 67?

```
if(input >= 50)
 input = input + 1;
if(input <= 75)
 input = input + 2;
```

else

input = input - 1;

**70**

!( x < 3) means in C++ that

**x is greater than or equal to 3**

!=operator is used to check whether the operand on the left-hand-side is \_\_\_\_\_ to the operand on the right-hand-side.

**Not equal**

When the if statement consists more than one statement then enclosing these statement in curly braces is,

**Must**

The most suitable data type for number 325.25 is \_\_\_\_\_.

**Float**

What will be the result of arithmetic expression  $6 + 48 / 4 * 3$ ?

**42**

Which of the following will be the most appropriate data type to store the value 63.547?

**Float**

In the given expression which operator will be evaluated first?  $10 + (6 / 2) - 2 * 3$ ?

**/**

What will be the value of the variable output in the given piece of code? `double output = 0;`

`output = (2 + 2) * 4 + 2 / (4 - 2);`

**17**

It is the job of \_\_\_\_\_ to transfer the executable code from hard disk to

## main memory. Loader

In computers system there are mainly \_\_\_\_\_ type of softwares.

**2**

\_\_\_\_\_ will explain the function of a program.

## Comments

if(a>b&& a>c) then the condition will be true only if Both a>b and a>c are

**true**

A variable of character data type occupies \_\_\_\_\_ byte(s) in memory.

**1**

We must include the header file \_\_\_\_\_ to convert the value of one type into another type using built-in functions

**. string.h**

A function is a block of statements that can be defined once and used \_\_\_\_\_ in the program.

**As many times as user wants**

Select the correct way to assign the address of first element of array to pointer?

**int \*ptr = &data;**

Consider the following code segment. What will be the output of following code?

```
int addValue (int *a){ int b
= (*a) + 2; return b;
}
main()
{ int x
```

```
= 6;
cout<<addValue(&x)<<" "; cout<<x;
}
```

**6,8,6**

Herethecodeisgivenbelow.Youhavetoidentifytheprobleminthecode. while(i < 10) && (i > 24))

**the test condition is always false**

Thecorrectsyntaxofdo-whileloopis\_\_\_\_\_.

**do {statements;} while (condition);**

Matrixisdefinedas\_\_\_\_\_.

**Multi-dimensional array**

In programming, comments are used to explain the functioning of the \_\_\_\_\_.

**Program**

OperatingSystemisatypeofa/an\_\_\_\_\_.

**system software**

Fromtheoptionsgiven,youneedtochoosetheoptionwhichistruefor the given code.

```
for (int i = 1; i>0; i++) {
/*loop code*/
}
```

**the test condition is always true**

Which of the following values are used in C/C++ to represent true and false?

**1 and 0**

'While' loop may execute \_\_\_\_\_ or more times.

**zero**

Body of any function is enclosed within \_\_\_\_\_.

**{ }**

What will be the correct syntax for initialization of a pointer ptr with string "programming"?

**char \*ptr = "programming";**

Which one of the given options is not a mode for reading/writing the data from a file?

**Get**

Which of the following operators is used to access the value of variable pointed by a pointer?

**\* operator**

In case of single referencing, the value of the \_\_\_\_\_ is the address of the \_\_\_\_\_.

**pointer, variable**

The remainder (%) operator is a \_\_\_\_\_ operator.

**Arithmetic**

What will be the output of following code? int x=10;

cout<<"x="<<x;

**x=10**

The purpose of using cout<<isto \_\_\_\_\_.

**Display information on the screen**

Which of the following data types will be assumed if no data type is specified with constant?

## **short**

When an array element is passed to a function, it is passed by \_\_\_\_\_.

## **reference**

While programming, it is good to provide an easy to understand and easy to use interface; this programming skill is called \_\_\_\_\_.

## **usability**

\_\_\_\_\_ executes all the lines before error and stops at the line which contains the error.

## **Interpreter**

Which of the following is the correct syntax to access the value of first element of an array using pointer ptr?

**\*ptr[0]**

C is a/an \_\_\_\_\_ language.

## **function oriented**

\_\_\_\_\_ of a function is also known as signature of a function.

## **Declaration**

\_\_\_\_\_ are very good tools for code reuse.

## **functions**

If any break statement is missed in switch statement then \_\_\_\_\_.

**this may cause a logical error**

A 2D array multi[5][10] can be accessed using the array name as \*\*multi, this technique is called \_\_\_\_\_.

## **Double dereferencing**

In C/C++, the default command line arguments passed to the main function are \_\_\_\_.

**int argc, char \*\*argv**

A record is a group of related \_\_\_\_.

**Fields**

The Microsoft Word document (.doc) is a kind of \_\_\_\_.

**Random Access File**

NULL character is used to indicate the \_\_\_\_ of string.

**End**

How many dimensions does n-dimensional array has?

**n dimensions**

Which of the following function call is "call by reference" for the following function prototype?

**func(&num);**

The loop which is most suitable to be used when the number of iterations is known is called \_\_\_\_.

**for**

In C/C++, the string constant is enclosed in \_\_\_\_.

**double quotes " "**

In order to get the right most digit of a number, we divide this number by 10 and take \_\_.

**Its remainder**

What is the correct syntax to declare an array of size 10 of int data type?

**int name[10];**

How many bytes of memory are occupied by array 'str'? char str[] =  
"programming";

**11**

Suppose that an integer type pointer contains a memory address 0x22f230. What will be the new memory address if we increment this pointer by one?

**0x22f234**

Which of the following if missing would result in infinite recursion in case of recursive function?

**Recursive call**

Whenever we use a library function or a predefined object or macro, we need to use a \_\_.

**header file**

Switch statement deals with \_\_\_\_\_ type of data.

**Both Integer and Character**

Both compiler and \_\_\_\_\_ are used to translate program into machine language code.

**Interpreter**

TWAIN stands for \_\_\_\_\_.

**Technology Without An Interesting Name**

The parameter passed to isdigit() function is \_\_\_\_\_.

**a character variable**

C++ view each file as a sequential stream of \_\_\_\_\_.

**Bytes**

Structure is a collection of \_\_\_\_\_ under a single name.

**only variables**

The default mode for writing into a file using ofstream object is \_\_\_\_\_.

**out**

The memory address of the first element of an array is called \_\_\_\_\_.

**base address**

We want to access array in random order which of the following approach is better?

**Pointer**

The \_\_\_\_\_ structure is a multiple-selection construct which makes the code more efficient and easy to read and understand.

**switch**

Which of the following is not a reserved word in C/C++? **Sum**

To access rand(), which library is required to be included in program? **stdlib.h**

What is the highest legal index for the following array?

**int arr[4]**

**3**

Wordprocessor is a type of a/an\_\_\_\_\_.

**application software**

Identify the correct option which is used for calling the function float area (int).

**area(int num);**

The\_\_\_\_\_statement allows us to select from multiple choices based on a set of fixed values for a given expression. **switch**

C is widely known as a development language of\_\_\_\_\_operating system.

**Unix**

To convert the value of one type into another type using built-in functions, we include\_\_\_\_\_header file.

**stdlib.h**

The keyword\_\_\_\_\_is used to get some value back from a function.

**return**

The function seekg() takes\_\_\_\_\_parameter(s).

**2**

The function write() takes\_\_\_\_\_as parameter(s).

**String and no. of bytes to be written**

When the logical operator AND (&&) combines two expressions exp1 and exp2 then the result will be true only\_\_\_\_\_.

**When both exp1 and exp2 are true**

Syntax of a union is identical to \_\_\_\_\_.

**structure**

In Flow Chart, flow of control is represented by \_\_\_\_\_.

**Arrow**

There can be \_\_\_\_\_ 'default' statement(s) in any switch structure.

**2**

The condition in loop should be a(n) \_\_\_\_\_.

**Arithmetic Expression**

How many nested loops would be required to manipulate n-dimensional array?

**2n**

Which of the following is not an example of int data type?

**-4**

We should use \_\_\_\_\_ for clarity and to force the order of evaluation in an expression.

**parenthesis ()**

Which of the following is the starting index of an array in C++?

**0**

The statement `x+=y` can be interpreted as \_\_\_\_\_.

**Adding the value of the x to the value of the y and storing the result in x**

Given a 2D array of integers, what would be the correct way of assigning the value 5 to the element at second row and third column?

**m[1][2] = 5;**

Array is a data structure that stores \_\_\_\_\_.

**Variables**

A program statement that invokes a function is called \_\_\_\_\_.

**function call**

The compiler of C language is written in \_\_\_\_\_ language.

**C**

Which function is used to locate the first occurrence of a character in any string?

**strlen()**

To access the data members of structure, \_\_\_\_\_ is used.

**Dot operator**

In the following nested For Loop, which loop will run most number of times?

```
for(int i = 0; i < 5; i++)
{
 for(int k = 0; k < 5; k++)
 {

 }
}
```

**Depends upon the statements in the inner loop's body**

Structure uses \_\_\_\_\_ allocation.

**Heap**

\_\_\_\_\_function gives the position of the next character to be read from that file.

**tellp()**

What will be the size of the following character array? `char name[] = "Adeel";`

**5**

Function prototype is written,

**Before call of that function**

Which one of the following languages has been used to write the compiler of "C" language?

**C**

A hierarchy of classes which are used to deal with console and disk files are called \_\_\_\_.

**Stream classes**

\_\_\_\_\_stop execution at the line that contains error(s) in the code.

**Interpreter**

C++ is a \_\_\_\_\_ language.

**High level**

How many elements are stored in the following? `int matrix [4][5];`

**20**

\_\_\_\_\_is a substitute of multiple if statement.

**if. . .else if statement**

if

int sum = 54;

Then the value of the following statement is sum = sum -3;

**51**

What will be the correct syntax for declaration of the following statement? "ptr is a constant pointer to an integer"

**const int \*myptr;**

\_\_\_\_\_ operator is used to pass the address of a variable in call by reference method.

**&**

\_\_\_\_\_ data type can operate on modulus operator.

**int**

Whenever some number is added in an array name, it will jump as many \_\_\_\_\_ as the added number.

**rows**

Suppose that an integer type pointer contains a memory address 0x22f220. What will be the new memory address if we increment this pointer by one?

**0x22f224**

\_\_\_\_\_ is the pointer which determines the position in a file from where the next read operation occurs.

**put**

what is the correct context to declare an array of 15 elements of type float?

**float array [14]**

suppose we have a file with name "mylife.txt". what will be the correct  
Syntax of file opening for input?

**my file (—myfile.txtl. ios::in)**

which of the following will be the most appropriate data type to store the value 63.5 47?

**Float**

----- will explain the function of a program

**Comments**

these functions Seekg() and seekp() requires an argument of type -----

**-- to let them how many bites to move forward or backward Int**

all elements of an array must be of ----- data type

**Same**

what will be the result of arithmetic expression  $6 + 48 / 4 * 3$ ?

**42**

----- allow us to have a memory location and use it as int or char interchangeably

**none of the given**

accessing a structure member the identifier to the left of the dot operator is the name of -----

**structural variable**

<and> both are ----- operators

## Arithmetic

what will be the output of the following code segment? For(int=2; i<10; i++){

If ( i == 5 )

Continue;

cout<<i<<“.”;

}

**2,3,4,5,6,7,8,9**

----- returns true if it is a digit or a letter and false otherwise **Int isalnum (int c)**

if a ---- find an error, it stops translating without generating any executable code

## Compiler

the ASCII code of null character is-----

**000**

if X=9; then the expression x%=2; will calculate the value of x as-----

**- 1**

the original value means unchanged in -----

## Call by Value

which of the following is used to create an account stream?

**Ofstream**

how many times the following loop will execute? Int j = 3;

While (j > 0)

{

Cout << "statements" << endl;

J - = 2;

}

**2**

Use of word processor for writing the code of a program is not good because -----

**It has less feature for writing code of a program**

how many bytes and integer type pointer intptr will jump in memory if the statement below is executed?

Intptr

+= **2**;

**8**

avoid loop may execute ----- or more Times

**0**

raid stands for -----

**redundant array of inexpensive dices**

in a file is opened with iOS:: out mode, then -----

**a new file is created**

first step in designing a program is **Analysis**

what will be the result of the expression  $J = i++$

If initially  $j=0$  and  $i=5$ ?

**suppose we have `int y [10];`**

to access the 4th element of the array we write-----

**`Y [3];`**

A ----- structure specifies that an action is to be repeated while

**some condition remains true Control**

the most suitable data type for number 325.25 is-----

**float**

Which one of the following should be placed immediately either before or after the noun?

**Modifiers of Nouns**

Titles typically include both.....

**a number and a description**

For what purposes are colons used?

**All of the above**

The name of the array is a constant pointer which contains the memory address of the array.

**first element**

What will be the correct syntax for the following function call?

**float add (int &); add(x);**

In C/C++; by default arguments are passed by \_\_\_\_\_ to a function.

**Value**

<, <=, >, >= are called \_\_\_\_\_ operators.

**Relational**

char name [ ] = "Hello World" ; In the above statement, a memory of \_\_\_\_\_ characters will be allocated

**12**

While developing a program; should we think about the reusability?

**Yes**

C is widely known as development language of \_\_\_\_\_ operating system.

**Unix**

Which of the following function call is "call by reference" for the following function prototype?

**int add (int \*); add(&x);**

If a function has not been declared before its definition,

**It is a None of these**

While loop executes at least----

## Zero time

A \_\_\_\_\_ is a precise sequence of steps to solve a particular  
**problem. Program**

In while loop the loop counter must be initialized,  
**Before entering the loop**

```
int
cou
nter
= 0 ;
whil
e(co
unter
<
15)
cout
<<
count
er<<
“”,
cou
nter
++;
```

**Infi  
nite  
loo  
p**

Why we use "cin"?

**To read data from keyboard**

Loops are \_\_\_\_\_ Structure

## **Repetition**

**Topic 13 to 18 here**

1. In an n-dimensional array , we use \_\_\_\_\_ for loop(s) to populate the  
array n
2. In 3 dimensional array, we use \_\_\_\_\_ for loop(s) to populate the array  
3
3. In 2 dimensional array, we use \_\_\_\_\_ for loop(s) to populate the array  
2
4. In 1 dimensional array, we use \_\_\_\_\_ for loop(s) to populate the array  
1
5. A group is a record of related \_\_\_\_\_  
Fields
6. \_\_\_\_\_ contains functions for manipulations of  
character data ctype.h
7. ifstream inFile refers \_\_\_\_\_  
Object for reading from a  
file
8. ofstream outFile refers \_\_\_\_\_  
object for writing to a file
9. A character is stored in the memory in \_\_\_\_\_  
Byte
10. A string which is assigned to a string variable is terminated by

Null character

11. When an identifier is declared with keyword const

then, Its value cannot be changed

12. If stream is used for \_\_\_\_\_

Input file stream

13. Which of the following option is correct about array

Only one type of elements can be stored in an array

14. Suppose that an integer type pointer contains a memory address 0x221220. What will be the new memory address if we increment this pointer by one

0x221224

15. In the case of pointer to pointer or \_\_\_\_\_ the first pointer contains the address of the second pointer, which contains the address of the variable, which contains the desired value

Double dereference

16. The data in the memory is \_\_\_\_\_

Volatile

17. Suppose we have a file with name myfile.text What will be the correct syntax of file opening for input

myfile,open("myfile.text", ios::in);

18. In analysis, we try to have a \_\_\_\_\_

Precise problem statement

19. \_\_\_\_\_ Returns true if c is a digit or a letter and false

otherwise int isalnum ( int c)

20. \_\_\_\_\_ Returns true if c is a digit and false

otherwise int isdigit(int c)

21. \_\_\_\_\_ Returns true if c is a letter and false otherwise `Int isalpha(int c)`
22. \_\_\_\_\_ Returns true if c is a hexadecimal digit character and false otherwise  
`Int isxdigit(int c)`
23. \_\_\_\_\_ Returns true if c is a lowercase letter and false otherwise `Int islower(int c)`
24. \_\_\_\_\_ Returns true if c is an uppercase letter and false otherwise  
`Int isupper(int c)`
25. \_\_\_\_\_ if c is an uppercase letter, tolower returns c as a lowercase letter. Otherwise, tolower returns the argument unchanged  
`Int tolower (int c)`
26. \_\_\_\_\_ if c is an lowercase letter, toupper returns c as an uppercase letter. Otherwise, toupper returns the argument unchanged  
`Int toupper(int c)`
27. Returns true if c is a white-space character newline ('\n'), space (' '), from feed ('\f'), carriage return ('\r'), horizontal tab ('\t'), or vertical tab ('\v') and false otherwise  
`Int isspace(int c)`
28. \_\_\_\_\_ returns true if c is a control character and false otherwise `Int isctrl(int c)`
29. \_\_\_\_\_ Returns true if c is a printing character other than a space, a digit, or a letter and false otherwise  
`Int ispunct (int c)`
30. Return true value if c is a printing character including space (' ') and false otherwise  
`Int isprint (int c)`

31. Returns true if c is a printing character other than space(' ') and false

otherwise `int isgraph(int c)`

32. The function `tolower` and `toupper` are \_\_\_\_\_

**Conversion function**

33. Which header file must be included to use the function `tolower()` and `toupper()`

`Ctype.h`

34. Which mode of file is used to open a file or stream for

**insertion Out**

35. Which mode of file is used to open a file or stream for extraction

**In**

36. Append rather than truncate an existing file. Each insertion will be written to the end of the file

**App**

37. Discards the file's contents if it

**exists. Trunk**

38. Opens the file without truncating, but allows data to be written anywhere

**in the file Ate**

39. Treat the file as binary rather than text. A binary file has data stored in internal formats, rather than readable text format

**Binary**

40. Following is the declaration of a \_\_\_\_ array. `int`

**`arr[2][2]` 2 dimensional**

41. \_\_\_\_\_ are conventional names of the command line parameters of the `'main()'` function.

**`'argc'` and `'argv'`**

42. What will be the correct syntax for declaration of the following statement 'ptr is a constant pointer to an integer'

`Int*const ptr;`

43. What will be the correct syntax for declaration of the following statement 'ptr is a pointer to const in'

`const int *ptr.`

44. Transpose of a matrix means that when we interchange rows and columns\_\_\_\_\_

The first row becomes the first columns

45. Char dest[50] = using strncat function”;

Char src[50] = this part is added and this is ignored;  
Strncat(dest, src, 19);  
Cout <<dest;

Using strncat function , this part is added

46. Base address is the memory address of \_\_\_element of an

array. 1<sup>st</sup>

47. While handling files, one can have\_\_\_\_\_options

Read the file, write in some file, read and write in the some file...all

48. String are accessed by variable of

type Char

49. For the following statement, how much memory will be allocated to my array[0] Char\*myarray[] = {“arif”,”sadia”}

5 bytes

50. To get the value stored of a memory address we use the \_\_\_\_\_

**Deferencing operator**

51. Data means \_\_\_\_\_

**The value of whatever the data points to**

52. If a file is opened ios::out, then \_\_\_\_\_

**A new file is created**

53. If we want to append into the file, the mode will be \_\_\_\_\_

**ios::app**

54. If we want to write anywhere in the file, the mode is \_\_\_\_\_

**ios::app**

55. Following is the declaration of a 2D array, how many row are declared for this array `int arr[3][2]`

**3**

56. Following is the declaration of a 2D array, how many columns are declared for this array `int arr[3][2]`

**2**

57. In case of two dimensional array, if you add some number in a two-dimensional array name, it will jump to the next\_\_.

**Rows**

58. Which of the following is true about streams? It is a sequence of bytes

It is an ordered sequence

All bytes can go through the stream simultaneously Bytes

that enters first into the stream will go out at last

**A and B**

59. Which of the following is a correct way to assign the address of the first element of array is pointer? `Int data[10]`

**`Int*ptr=data[0];`**

60. A \_\_\_\_\_ is an array of characters that can store number of character  
**specified String**

61. A string which is assigned to a string variable is terminated by \_\_\_\_.  
**Null character**

62. How many bytes an integer type pointer int ptr will jump in  
memory if the statement below is executed? Int ptr +=2

**8**

63. Give a two dimensional array of integers, what would be the  
correct way of assigning the value 6 to the element at third row and  
fourth columns

**Array[2][3]=6;**

64. Following is an array of \_\_\_\_\_ rows and \_\_\_\_\_, int

**num[3][4]; 3 and 4**

65. What will be the output of the following code

**segment? Char myName[]="student ID";**

**Int n=sizeof(myName);**

**Cout<<n;**

**11**

66. If we have a pointer on the left hand side then the right hand side  
should have a/an\_

**Address**

67. The parameter passed to isdigit() function is\_

\_\_\_\_\_ **variabl**

**e Character or integer**

68. Ofstream is used for \_\_\_\_\_

**Output file stream**

69. A hierarchy of classes which are used to deal with console and  
disk files are called \_\_\_\_\_

Stream classes

70. '\0' takes \_ space in the memory.

1 byte... not sure

71. Which statement is used to close a file

Myfile.close()

72. Which statement is used to open

a file My file.open()

73. We must include the header file\_\_to convert the value of one type into another type using built-in-function.

String.h

74. Pointer points to a particular\_\_\_\_\_.

Memory address

75. Ifstream is used for\_\_\_\_\_

Input file stream

76. Ofstream is used for

Output file stream

77. To get the value stored at a memory address, we use the\_\_.

Deferencing operator

78. To read command-line arguments, the main() function itself must be given \_ arguments.

2

79. When an identifier is declared with keyword const

then, Its value cannot be changed

80. Constructor is itself\_\_\_\_\_of C++ and \_\_\_\_\_

Function, can be overloaded

81. In analysis, we try to have a \_\_\_\_\_

Precise problem statement

82. In c++ a variable can be declare anywhere in the program this will

increase Readability

83. The name of the array is a constant pointer which contains the memory address of the \_\_\_\_\_ of the array”

First element

84. What is output of following code?

```
Int data[10]=[10,20,30,40,50,60,70,80,90,100];
Int xptr=&data[4];
Int*yptr=&data[6];
Cout<<xptr+xptr++
```

120

85. C++ offers \_\_\_\_\_ levels of data access control inside a class

Three

86. The function calloc takes two arguments, first argument is the \_ and the second argument is the \_\_\_\_

Required space in terms of numbers, size of space

87. The statement cout<<yptr will show \_\_\_\_\_ the yptr points

to Memory address

88. The use of \*sign in first and second statements is called \_\_\_\_\_ and \_ of pointer. Int \* ptr = &x

```
Cout<<*ptr
```

Referencing dereferencing

89. What will be the output of following program? Int

```
data[10]=[10,20,30,40,50,60,70,80,90,100];
Int*yptr=&data[5];
Cout<<++yptr
```

61

90. Character string are terminated by \_\_\_\_\_

`'\0'`

91. There is a pointer variable named ptr of type float. Which type of variable address can be stored by ptr.

`int`

92. `char **argv` can be read as

array of pointers to char

93. `eof` is used to check for the `_of` file when a file is being

read end

94. suppose we have `int y[10]`; To access the 4th element of the array we write\_

`y[4];`  
`y[3];`  
`y[2];`  
none of given

95. From the following; which one is the range of Random number generator function `rand()`?

0 – 32768  
1 – 32768  
0 – 32767  
1 – 32767

96. From following; which one is the correct syntax of an array initialize:  
Array size is 10 and it is of double data type to value 0?

`arr[10] = {0.0};`  
`double arr[10]= 0.0;`

```
double arr[10] = {0.0};
double arr[] = 0.0;
```

97. The string in the array is terminated by a\_\_

zero  
nil  
**null**  
one

98. In C/C++; by default arguments are passed by\_ to a function.

Reference  
**Value**  
Type Data

99. `char name [] = "Hello World"` ; In the above statement, a memory of\_ characters will be allocated

12

100. \_\_\_\_\_ Keyword is used to return some value from a function.

break  
**return**  
continue  
goto

101. Which of the function call is call by value for the following function prototype? `float add(int);`

`add(&x);`  
**`add(x);`**  
`add(int x);`  
`add(*x);`

102. The increment of a pointer depends on its\_\_\_\_\_.

variable  
value  
**data type**  
None of the given

103. The ASCII code of null character is\_\_\_\_\_

000

104. Pointers are a special type of \_\_\_\_\_in which a memory address is stored

variables

Location

Characters

None of the given

105. Transpose of a matrix means that when we interchange rows and columns\_\_\_\_

the first row becomes the Last column

the first row becomes the first column

the Last row becomes the first column

the first column becomes the first row

63. Pointers store the \_\_\_\_\_

value of a variable

memory address

characters

None of the given

106. Which of the following function call is “call by reference” for the following function prototype? int add (int \*);

add(&x);

add(int x);

add(x);

add(\*x);

107. The name of the array is a constant pointer which contains the memory address of the\_\_\_of the array.

first element Last

element second

element None of

the given

108.        array name always contains the memory address of the \_\_\_\_\_ of the array

entire elements last  
element first element  
None of the given

109.        At the \_\_\_\_\_, we try to break up the problem into functional

units analysis phase  
design phase  
Implementation phase None  
of the given

# **CS201 For final term**

## **UPDATED QUIZ 2 2021**

### **PROVIDE BY ORANGE MONKEY**

Declaring structures does not mean that memory is allocated.

Example:

- ▶ True
- ▶ False

structures do not occupy any memory until it is associated with the structure variable

Identifier is a name that can be given to variables, labels and functions.

- ▶ True
- ▶ False

An 'Identifier' means any name that the user creates in his/her program. These names can be of variables, functions and labels

If a class A declares itself a friend of class B and a class B declares itself a friend of class C then

- ▶ Class A is also a friend of class C.
- ▶ Class B is also a friend of class A.
- ▶ Class A is also a friend of class C if A declares C as its friend.
- ▶ Class A is also a friend of class C if C declares A as its friend.

If we want a two-way relationship, OtherClass will have to declare ClassOne as a friend class, resulting in a complete two-way relationship

Which of the following statement is best regarding declaration of friend function?

- ▶ Friend function must be declared after public keyword.
- ▶ Friend function must be declared after private keyword.
- ▶ Friend function must be declared at the top within class definition.
- ▶ It can be declared anywhere in class as these are not affected by the public and private keywords.

*Friend is a very strong statement. It is too strong to be affected by public or private we can put it anywhere in the class*

A pointer is a special type of variable that contain \_\_\_\_\_

- ▶ Memory Address
- ▶ Data values
- ▶ Both Values and Memory
- ▶ None of given of options

*Pointer is a special type of variable that contains a memory address.*

When memory for a program is allocated at run time then it is called \_\_\_\_\_

- ▶ static memory allocation
- ▶ dynamic memory allocation
- ▶ stack memory allocation
- ▶ virtual memory allocation

When we create an object of the class at run time, it will allocate memory according to our requirement. So there is no waste of memory and the situations in which we want to store large data in small memory or vice versa are prevented. So we do dynamic memory allocation inside these classes.

What purpose do classes serve?

- ▶ Data encapsulation
- ▶ Providing a convenient way of modeling real-world objects
- ▶ Simplifying code reuse
- ▶ All of the given options

Which of the following function cannot be overloaded?

- ▶ Member functions
- ▶ Utility functions
- ▶ Constructor

► **Destructor**

*The destructors can be summarized as The destructors cannot be overloaded. The destructors take no arguments. The destructors don't return a value*

The following prototype of unary operator function indicates that it is\_\_\_\_\_.

Date operator++(int )

► **Member functions of post increment operator**

- Member functions of pre increment operator
- Non-member functions of post increment operator
- Non-member functions of pre increment operator

Overloading Unary Operators

//Preincrement operator overloaded as a member function. Date

Date::operator++()

```
{
 helpIncrement();
 return *this; // value return; not a reference return
}
```

// Postincrement operator overloaded as a member function.

// Note that the dummy integer parameter does not have a  
// parameter name.

Date Date::operator++(int)

```
{
 Date temp = *this; helpIncrement();

 //return non-incremented, saved, temporary object
 return temp; // value return; not a reference return
} // This paper was solved by vuzs Team and meant for hosting
at vuzs otherwise its stolen contents
```

Static variable which is defined in a function is initialized\_\_\_\_\_.

► **Only once during its life time**

- Every time the function call
- Compile time of the program
- None of the above

*Once the static variables are created, they exist for the life of the program. They do not die. In the*

member initialize list, the data members are initialized,

► **From left to right**

- From right to left
- In the order in which they are defined within class
- None of the given options

If we do not indent the code properly it will \_\_\_\_\_

- ▶ Be a syntax error
- ▶ Be a logical error
- ▶ **Not be an error at all**
- ▶ None of the given options

we Indent the code for better readability and understanding

Truth tables are used for analyzing \_\_\_\_\_.

- ▶ **logical expressions**
- ▶ arithmetic expressions
- ▶ both logical and arithmetic expressions
- ▶ none of the given options.

The truth tables are very important. These are still a tool available for analyzing logical expressions.

Static memory allocation is also known as \_\_\_\_\_

- ▶ Dynamic allocation
- ▶ **Compile time allocation**
- ▶ Run time allocation
- ▶ None of the given options

This type of memory static allocation. It is also known as compile time allocation.

If an array of objects is allocated, *operator new[]* is called to allocate memory for the whole array, and then the constructor is called for each element of the array.

- When a single object is deleted, the destructor for the object is called first, and then *operator delete* is called to free the memory occupied by the object.
- When an array of objects is deleted, the destructor for each element of the array object is called first, and then *operator delete[]* is called to free the memory occupied by the array.

**Question No: 4 ( Marks: 1 ) - Please choose one**

We can delete an array of objects without specifying [] brackets if a class is not doing dynamic memory allocation internally

- **True**
- False

Although, this is good to deallocate an array of objects without specifying array operator ([]) as there is no dynamic memory allocation occurring from inside the Date class. But this is a bad practice.

**Question No: 5 ( Marks: 1 ) - Please choose one**

The declarator of Plus (+) member operator function is

· **Class-Name operator + (Class-Name rhs)**

- Operator Class-Name + ( )
- Operator Class-Name + ( rhs)
- Class-Name operator + (

) Page 371,373 example

are here

Complex operator +

(Complex & ); Complex  
operator + (parameter-list);  
The syntax of the prototype of the overloaded  
operator function is: return-type operator operator-  
symbol (parameter-list);  
operator is the keyword here. An example of this will  
be as follows: Complex operator + (Complex & );

**Question No: 6 ( Marks: 1 ) - Please choose one**

The second parameter of operator functions for << and >> are  
objects of the class for which we are overloading these operators

- True (not sure)
- False

**Question No: 7 ( Marks: 1 ) - Please choose one**

Which of the following is correct way to initialize a variable x of int type with value 10?

- int x ; x = 10 ;
- int x = 10 ;
- int x, x = 10;
- x = 10 ;

variable already created in question only it is asking for initialization.

**Question No: 8 ( Marks: 1 ) - Please choose one**

Default mechanism of function calling in case of array is \_\_\_\_\_ and in case of variable is \_\_\_\_\_

- Call by value, call by reference
- Call by reference, call by reference
- Call by reference, call by value
- Call by value, call by value

**Question No: 9 ( Marks: 1 ) - Please choose one**

What does STL stand for?

- Source template library
- Standard template library
- Stream template library
- Standard temporary library

STL stands for Standard Template Library

**Question No: 10 ( Marks: 1 ) - Please choose one**

Skill(s) that is/are needed by programmers \_\_\_\_\_

- Paying attention to detail
- Think about the reusability
- Think about user interface

**· All of the given options**

Programming is an important activity as people life and living depends on the programs one  
make. Hence while programming one should

- o Paying attention to detail
- o Think about the reusability.
- o Think about user interface
- o Understand the fact the computers are stupid
- o Comment the code

liberally

**Question No: 11 ( Marks: 1 ) - Please choose one**

For which array, the size of the array should be one more than the number of elements in an  
array?

- int
- double
- float
- char

**Question No: 12 ( Marks: 1 ) - Please choose one**

new and delete are \_\_\_\_\_ whereas malloc and free are \_\_\_\_\_

- Functions, operators
- Classes, operators

## · Operators, functions

· Operators, classes

new and delete are operators in c++

C functions like **malloc()** and **free()** functions can also be used from within C++ code

### Question No: 13 ( Marks: 1 ) - Please choose one

The prototype of friend functions must be written \_\_\_\_\_ the class and its definition must be written \_\_\_\_\_

· inside. inside the class

## · inside, outside the class

· outside, inside the class

· outside, outside the class

### Question No: 14 ( Marks: 1 ) - Please choose one

Friend function of a class are \_\_\_\_\_ of a class.

## · Non-member functions not sure

· Friend functions

· Any function outside class

· None of the given

options

### Question No: 15 ( Marks: 1 ) - Please choose one

If overloaded plus operator is implemented as non-member function then which of the following statement will be true for the statement given below?

obi3 = obi1 + obi2 :

## · obj2 will be passed as an argument to + operator whereas obj1 will drive the + operator

· obj1 will drive the + operator whereas obj2 will be passed as an argument to + operator

· Both objects (obj1, obj2) will be passed as arguments to the + operator

· Any of the objects (obj1, obj2) can drive the + operator

c3 = c1 + c2 ; In the above statement ( c3 = c1 + c2; ), c1 is the object that is calling or driving the + operator. c2 object is being passed as an argument to the + operator. So c1 and c2 objects are added by the + operator and resultant

### Question No: 16 ( Marks: 1 ) - Please choose one

Which one of the following is the declaration of overloaded pre-increment operator implemented as member function?

· Class-name operator +() ;

· Class-name operator +(int) ;

## · Class-name operator ++() ;

· Class-name operator

++(int) ; **Overloading**

**Unary Operators**

```
// Preincrement operator overloaded as a
member function. Date Date::operator++()
{
 helpIncrement();
 return *this; // value return; not a reference return
}
```

```
// Postincrement operator overloaded as a member function.
// Note that the dummy integer parameter does not have a
// parameter name.
Date Date::operator++(int)
{
 Date temp =
 *this;
 helpIncrement
 ();
}
```

```
// return non-incremented, saved,
temporary object return temp; // value
return; not a reference return
}
```

**Question No: 17 ( Marks: 1 ) - Please choose one**

For cin, the source is normally a \_\_\_\_\_ and destination can be \_\_\_\_\_

- File, native data type
- Disk, user-defined type
- **Keyboard, variable**
- File, user-defined type

For cin, the source is normally keyboard and the destination can be an ordinary variable i.e. native-data type variable. It could be some area of memory or our own data type, i.e. object for which we h

**Question No: 18 ( Marks: 1 ) - Please choose one**

We can do condition compilation with pre processor directives.

- **True**
- False

All the preprocessor directives start with the sharp sign (#). We can also do conditional compilation with it.

**Question No: 19 ( Marks: 1 ) - Please choose one**

The programs, in which we allocate static memory, run essentially on \_\_\_\_\_

- Heap
- System Cache
- None of the given options
- **Stack**

The programs, in which we allocate static memory, run essentially on stack

**Question No: 20 ( Marks: 1 ) - Please choose one**

A template function must have at least or more arguments

- Zero
- **One**
- Two
- Three

The function arguments must contain at least one generic data type. Normal function declaration is: return\_type function\_name(argument\_list)

**Question No: 21 ( Marks: 1 ) - Please choose one**

The default value of a parameter can be provided inside the \_\_\_\_\_

- function prototype
- function definition
- **both function prototype or function definition**
- none of the given options

The default value of a parameter is provided inside the function prototype or function definition.

**Question No: 22 ( Marks: 1 ) - Please choose one**

While calling function, the arguments are assigned to the parameters from \_

- **left to right**
- right to left

- no specific order is followed
- none of the given options

While calling function, the arguments are assigned to the parameters from left to right.

**Question No: 23 ( Marks: 1 ) - Please choose one**

When an operator function is defined as member function for a binary Plus (+) operator then the number of argument it take is/are

- Zero
- **One**

- Two
- N arguments

### Operators as member functions

Aside from the operators which must be members, operators may be overloaded as member or non-member functions. The choice of whether or not to overload as a member is up to the programmer. Operators are generally overloaded as members when they:  
change the left-hand operand, or

1. require direct access to the non-public parts of an object.

When an operator is defined as a member, the number of explicit parameters is reduced by one, as the calling object is implicitly supplied as an operand. Thus, binary operators take one explicit parameter and unary operators none. In the case of binary operators, the left hand operand is the calling object, and no type coercion will be done upon it.

#### Question No: 24 ( Marks: 1 ) - Please choose one

new operator allocates memory from free store and return \_\_\_\_\_

#### · A pointer

- A reference
- An integer
- A float

#### Question No: 25 ( Marks: 1 ) - Please choose one

With user-defined data type variables (Objects), assignment can produce \_\_\_\_\_ self

#### · Syntax error not sure

- Logical error
- Link error
- Non of the given options

#### Question No: 26 ( Marks: 1 ) – Write Simple Program

Assignment operator is used to initialize a newly declared object from existing object

#### · True

- False

#### Question No: 27 ( Marks: 1 ) – Briefly define/Justify

When an object of a class is defined inside an other class then,

- Constructor of enclosing class will be called first

#### · Constructor of inner object will be called first

- Constructor and Destructor will be called simultaneously
- None of the given options

• **A class can contain instances of other classes as its data members.** • It is a way of reusing the code when we contain objects of our already written classes into a new class.

- The inner data members of the object are constructed and then the object itself.

The order of destruction of an object is reverse to this construction order, where the outer object is destroyed first before the inner data members.

- Initializer list is used to initialize the inner objects at the construction time.

• In C++, we can have structures or classes defined inside classes. Classes defined within other classes are called nested classes.

#### Question No: 28 ( Marks: 1 ) – Brief answer required

In the member initializer list, the data members are initialized,

#### · From left to right

- From right to left
- In the order in which they are defined within class
- None of the given options

**Question No: 29 ( Marks: 1) - Brief answer required**

"new" and "delete" keywords are \_\_\_\_\_ in C++ language

- Built-in- Function
- **Operators**
- Memory Allocation Function
- None of the given options

In C/C++ if we define an array of size eight (8) i.e. int Arr [8]; then the last element of this array will be stored at,

► Arr[0]

- Arr[8]
- **Arr[7]**
- Arr[-1]

Question No: 2( M - 1 ) .

When an array is passed to a function then default way of passing this array is,

- By data
- **By reference**
- By value
- By data type

Question No: 3( M - 1 ) .

Array is a data structure which store

- Memory addresses
- Variables
- Data Type
- **Data**

Question No: 4( M - 1 ) .

We can also create an array of user define data type.

- **True**
- False

Question No: 5( M - 1 ) .

When we define an array of objects then,

- Destructor will call once for whole array
- **Destructor will call for each object of the array**
- Destructor will never call
- Depends on the size of array

Question No: 6( M - 1 ) .

What is the sequence of event(s) when allocating memory using new operator?

- ▶ Only block of memory is allocated for objects
- ▶ Only constructor is called for objects
- ▶ **Memory is allocated first before calling constructor**
- ▶ Constructor is called first before allocating memory

Question No: 7( M - 1 ) .

We can delete an array of objects without specifying [] brackets if a class is not doing dynamic memory allocation internally.

- ▶ **True**
- ▶ False

Question No: 8( M - 1 ) .

The second parameter of operator functions for are objects of the class for which we are overloading these operators.

- ▶ **True**
- ▶ False

Question No: 9( M - 1 ) .

Which of the following is correct way to initialize a variable x of int type with value 10?

- ▶ int x ; x = 10 ;
- ▶ int x = 10 ;
- ▶ int x, x = 10;
- ▶ **x = 10 ;**

Question No: 10 ( M - 1 ) .

Default mechanism of function calling in case of array is\_\_\_\_\_and in case of variable is\_

\_\_\_\_\_.

- ▶ Call by value, call by reference
- ▶ Call by referene, call by reference
- ▶ **Call by reference, call by value**
- ▶ Call by value, call by value

Question No: 11 ( M - 1 ) . What does STL stand for?

- ▶ Source template library
- ▶ **Standard template library**
- ▶ Stream template library
- ▶ Standard temporary library

Question No: 12 ( M - 1 ) . Skill(s) that is/are needed by programmers\_\_\_\_\_.

- ▶ Paying attention to detail
- ▶ Think about the reusability
- ▶ Think about user interface
- ▶ **All of the given options**

Question No: 13 ( M - 1 ) . For which array, the size of the array should be one more than the number of elements in an array?

- ▶ int
- ▶ double
- ▶ float
- ▶ **char**

Question No: 14 ( M - 1 ) . new and delete are\_\_whereas malloc and free are\_\_\_\_\_.

- ▶ Functions, operators
- ▶ Classes, operators
- ▶ **Operators, functions**
- ▶ Operators, classes

Question No: 15 ( M - 1 ) . Friend functions are\_\_of a class.

- ▶ Member functions
- ▶ Public member functions
- ▶ Private member functions
- ▶ **Non-member functions**

Question No: 16 ( M - 1 ) . The prototype of friend functions must be written\_\_\_\_\_the class and its definition must be written \_\_\_\_\_

- ▶ inside, inside the class
- ▶ **inside, outside the class**
- ▶ outside, inside the class
- ▶ outside, outside the class

Question No: 17 ( M - 1 ) . If overloaded plus operator is implemented as non-member function then which of the following statement will be true for the statement given below?

- obj3 = obj1 + obj2 ;
- ▶ obj2 will be passed as an argument to + operator whereas obj2 will drive the + operator
  - ▶ **obj1 will drive the + operator whereas obj2 will be passed as an argument to + operator**
  - ▶ Both objects (obj1, obj2) will be passed as arguments to the + operator
  - ▶ Any of the objects (obj1, obj2) can drive the + operator

Question No: 18 ( M - 1 ) . Which one of the following is the declaration of overloaded pre-increment operator implemented as member function?

- ▶ Class-name operator +() ;
- ▶ Class-name operator +(int) ;

- ▶ **Class-name operator ++() ;**
- ▶ Class-name operator ++(int) ;

Question No: 19 ( M - 1 ) .

For cin, the source is normally a\_\_and destination can be\_\_.

- ▶ File, native data type
- ▶ Disk, user-define type
- ▶ **Keyboard, variable**
- ▶ File, user-define type

Question No: 20 ( M - 1 ) .

We can also do conditional compilation with preprocessor directives.

- ▶ **True**
- ▶ False

Question No: 21 ( M - 1 ) .

The programs, in which we allocate static memory, run essentially on \_\_\_\_\_

- ▶ Heap
- ▶ System Cache
- ▶ None of the given options
- ▶ **Stack**

Question No: 22 ( M - 1 ) .

The default value of a parameter can be provided inside the \_\_\_\_\_

- ▶ function prototype
- ▶ function definition
- ▶ **both function prototype or function definition**
- ▶ none of the given options.

Question No: 23 ( M - 1 ) .

While calling function, the arguments are assigned to the parameters from\_\_\_\_\_.

- ▶ **left to right.**
- ▶ right to left
- ▶ no specific order is followed
- ▶ none of the given options.

Question No: 24 ( M - 1 ) .

When an operator function is defined as member function for a binary Plus (+) operator then the number of argument it take is/are.

- ▶ Zero
- ▶ **One**
- ▶ Two
- ▶ N arguments

Question No: 25 ( M - 1 ) .

With user-defined data type variables (Objects), self assignment can produce\_\_\_\_\_.

- ▶ Syntax error
- ▶ **Logical error**
- ▶ Link error
- ▶ Non of the given options

Question No: 26 ( M - 1 ) .

Assignment operator is used to initialize a newly declared object from existing object.

- ▶ **True**
- ▶ False

Question No: 27 ( M - 1 ) .

When an object of a class is defined inside an other class then,

- ▶ Constructor of enclosing class will be called first
- ▶ **Constructor of inner object will be called first**
- ▶ Constructor and Destructor will be called simultaneously
- ▶ None of the given options

Question No: 28 ( M - 1 ) .

In the member initializer list, the data members are initialized,

- ▶ From left to right
- ▶ From right to left
- ▶ **In the order in which they are defined within class**

► None of the given options

**Question No: 4** ( M - 1 ) .

Which of the following is NOT a preprocessor directive?

- ▶ #define
- ▶ #line
- ▶ #undef

#### list of preprocessors

• #include • #include "filename" • #define • #undef • #ifdef • #ifndef • #if • #else • #elif • #endif • #error • #line • #pragma • #assert

**Question No: 5** ( M - 1 ) .

The stream objects **cin** and **cout** are included in which header file?

- ▶ iostream.h
- ▶ fstream.h
- ▶ istream.h
- ▶ ostream.h

<http://www.vuzs.info/> **Question No:**

**6** ( M - 1 ) .

Overloaded delete operator function takes the same parameter as an argument returned by new operator function.

- ▶ True
- ▶ False

The same pointer that is returned by the new operator, is passed as an argument to the delete operator. These rules apply to both, if operators (new and delete) are overloaded as member or non-member operators (as global operators).

**Question No: 7** ( M - 1 ) .

When an array of object is created dynamically then there is no way to provide parameterized constructors for array of objects.

- ▶ True
- ▶ False

if we are allocating an array of objects, there is no way to pass arguments to objects' constructors. Therefore it is required that the objects that are stored in such an array have a no-argument constructor.

**Question No: 8** ( M - 1 ) .

C is widely known as development language of \_\_\_\_\_ operating system.

- ▶ Linux
- ▶ Windows
- ▶ Unix
- ▶ Mac OS

In the start C became widely known as the development language of the UNIX operating system, and the UNIX operating system was written by using this C language. The C language is so powerful that the compiler of C and other various operating systems are written in C.

<http://vuzs.net>

**Question No: 9** ( M - 1 ) .

Computer can understand only machine language code.

- ▶ True
- ▶ False

**Question No: 10** ( M - 1 ) .

We can not define a function as a friend of a Template class.

- ▶ True
- ▶ False

Class templates can have friends. A class or class template, function, or function template can be a friend to a template class. Friends can also be specializations of a class template or function template, but not partial specializations.

**Question No: 11** ( M - 1 ).

What will be the value of 'a' and 'b' after executing the following statements?

a = 3;

b = a++;

- ▶ 3, 4
- ▶ 4, 4
- ▶ 3, 3
- ▶ **4, 3**

**Question No: 12** ( M - 1 ).

Consider the following code segment. What will be the output of following code? int addValue (int

\*a){

int b = (\*a) + 2;

return b ;

}

main () { int x

=6 ;

cout << x << “,” ;

cout << addValue(&x) << “,” ;

cout << x ;

}

▶ **6,8,6**

- ▶ 6,6,8
- ▶ 6,8,8
- ▶ 6,6,6

**Question No: 13** ( M - 1 ).

\_\_\_\_\_ is used to trace the logic of the program and correct the logical errors.

- ▶ Compiler
- ▶ Editor
- ▶ Linker
- ▶ **Debugger**

**Question No: 14** ( M - 1 ).

new and delete are \_\_\_\_\_ whereas malloc and free are \_\_\_\_\_.

- ▶ Functions, operators
- ▶ Classes, operators
- ▶ **Operators, functions**
- ▶ Operators, classes

Hence, we can call new and delete operators, P# 342

we have allocated a memory space for our use by malloc function. P# 285

**Question No: 15** ( M - 1 ).

Like member functions, \_\_\_\_\_ can also access the private data members of a class.

- ▶ Non-member functions
- ▶ **Friend functions**
- ▶ Any function outside class
- ▶ None of the given options

**Question No: 16** ( M - 1 ).

Which situation would require the use of a non-member overloaded operator?

- ▶ The overloaded operator is an Assignment operator.

- ▶ The left most operand is an object of a class.
- ▶ **The left operand is built-in data type.**
- ▶ The operator returns a reference.

(100% confirmed by Rainbowbright)

Question No: 17 ( M - 1 ).

The stream insertion and stream extraction operators are already overloaded for\_.

- ▶ User-defined data types
- ▶ **Built-in data types**
- ▶ User-defined and built-in data types
- ▶ None of the given options

Question No: 18 ( M - 1 ).

If we define an identifier with the statement **#define PI 3.1415926** then during the execution of the program the value of PI\_

- 
- ▶ can not be replaced
  - ▶ None of the given options
  - ▶ **Remain constant.**
  - ▶ can be changed by some operation

Question No: 19 ( M - 1 ) . vuzs

Assignment operator is----- associative.

- ▶ **right**
- ▶ left
- ▶ binary
- ▶ unary

You can assign values to several variables in a single statement. For example, the following code sets the contents of apples and oranges to the same value:

apples = oranges = 10;

The assignment operator is right associative, so this statement executes by first storing the value 10 in oranges and then storing the value in oranges in apples, so it is effectively

apples = (oranges = 10);

<http://www.vuzs.info/> Question No: 20

( M - 1 ).

When ever dynamic memory allocation is made in C/C++, it is freed\_\_\_\_\_.

- ▶ **Explicitly**
- ▶ Implicitly
- ▶ Both explicitly and implicitly
- ▶ None of the given options

(Rainbowbright, vuzs. jul2011)

Question No: 21 ( M - 1 ).

The appropriate data type to store the number of rows and columns of the matrix is\_\_\_\_\_.

- ▶ float
- ▶ **int**
- ▶ char
- ▶ none of the given options.

Question No: 22 ( M - 1 ).

Which of the following function do NOT initialize the chunk of memory to all zero?

- ▶ calloc() function
- ▶ Both malloc() and calloc()
- ▶ None of the above
- ▶ **malloc() function**

The **malloc** function differs from **calloc** in the way that the space allocated by **malloc** is not initialized and contains any values initially.

**Question No: 23** ( M - 1 ) .

The function free() returns back the allocated memory got thorough calloc and malloc to \_\_\_\_\_.

- ▶ stack
- ▶ **heap**
- ▶ stack and heap
- ▶ None of the given options

**Question No: 24** ( M - 1 ) .

width() is member function of \_\_\_\_\_

- ▶ cin object
- ▶ cout object
- ▶ **Both cin and cout object**
- ▶ None of the given option

**Question No: 25** ( M - 1 ) .

Templates are not type safe.

- ▶ True
- ▶ **false**

Templates are type-safe. This is because the types that templates act upon are known at compile time, so the compiler can perform type checking before errors occur.

**Question No: 26** ( M - 1 ) .

A Matrix can be composed of ints, floats or doubles as their elements. Best way is to handle this ,

- ▶ Write a separate class to handle each
- ▶ **Use templates**
- ▶ Use strings to store all types
- ▶ None of the given options

**1. For binary member operators, operands on the \_\_\_\_\_ drives (calls) the operation.**

**Left**

**2. We cannot increment \_\_\_\_\_.**

**references**

**3. We can \_\_\_\_\_ pointer.**

**all of the given**

**4. We can \_\_\_\_\_ references.**

**None of the given**

**5. What will be the correct syntax for the following function call?**

float add (int &);

add(x);

**6. An instance of a class is called\_\_\_\_\_.**

object

**7. The\_\_\_\_\_is called automatically when an object destroys**

destructor

**8. The destructor is used to\_\_\_\_\_.**

1. deallocate memory

**9. \_\_\_\_\_data isn't accessible by non-member functions or outside classes.**

private

**10. Member functions of the class\_\_\_\_\_main program.**

1. are accessible from

**11. Overloading means :**

1. Using the same name to perform multiple tasks or different tasks depending on the situation.

**12. The main advantage of function overloading is \_\_\_\_\_.**

1. The program becomes more readable

**13. You cannot overload the\_\_\_\_\_operator.**

1. ? :

**14.** In C++, a variable can be declared anywhere in the program this will increase\_\_\_\_\_.

1. efficiency

**15.** Memory allocated from heap or free store\_\_\_\_\_.

1. cannot be returned back unless freed explicitly using free and delete operators

**16.** We cannot use\_\_\_\_\_pointer for storing and reading data from it.

1. 'NULL

**17.** The dynamic memory allocation uses\_\_\_\_\_whereas static memory allocation uses\_\_\_\_\_.

1. heap , stack

**18.** What will be the output of the given code?

```
#include <iostream>
#define MAX(A, B) ((A) > (B) ? (A) : (B))
void main() {
 int i, x, y; x = 23;
 y = 45;
 i = MAX(x++, y++);
 // Side-effect: // larger value incremented twice
 cout << "x = " << x << " y = " << y << "\n";
}
```

1. x=24 y=47

**19.** NULL has been defined in\_\_\_\_\_header file.

1. Stdlib.h

**20.** Symbolic constant PI can be defined as:

1. #define PI 3.14

**21.** The friend function of a class can have access\_\_\_\_\_.

1. to the private data members
22. C++ was developed by\_\_\_\_\_.
1. BejarneStroustrup
23. Once the\_\_\_\_\_are created, they exist for the life time of the program.
1. static variables
24. Encapsulation means\_\_\_\_\_.
1. that the data of a class can be accessed from outside
25. An address is a\_\_\_\_\_, while a pointer is a\_\_\_\_\_.
1. variable, constant
26. The syntax of declaration of a function that returns the reference to an integer is\_\_\_\_\_.
1. int &myfunc();
2. int myfunc();
3. int myfunc() &;
4. integer &myfunc();
27. Which one of the following is mandatory preprocessor directive for c++?
1. #undef
2. #include
3. #undef
4. All of the given
28. The members of a class declared with the keyword struct are\_\_\_\_\_by default.
1. static
2. Private
3. protected
4. public

**29.** `getche()` is a \_\_\_\_\_ function and defined in \_\_\_\_\_ header file.

1. user-define function ,conio.h
2. built-in function ,conio.h
3. built-in function, stlib.h
4. built -in function, iostream.h

**30.** \_\_\_\_\_ operators are the ones that require two operands on both sides of the operator.

1. Double
2. Tow sided
3. Binary
4. None of the given

**31.** \_\_\_\_\_ will return the number of bytes reserved for a variable or data type.

1. sizeof operator
2. free operator
3. void pointer
4. new operator

**32.** \_\_\_\_\_ are not available in C language.

1. User defned functions
2. Built in functions
3. Library functions
4. Inline functions

**33.** The members of a class declared without any keyword are \_\_\_\_\_ by default.

1. protected
2. private
3. public
4. constant

**34.** For console input and output we use \_\_\_\_\_.

1. `conio.h` header file
2. `stdlib.h` header file
3. `process.h` header file
4. `getch.h` header file

**35.** The name of the destructor is the same as that of a class proceeding with a\_\_\_\_\_.

1. &sign
2. # sign
3. @ sign
4. ~ sign

**36.** A reference cannot be NULL it has to point a data type.

1. True
2. False

**37.** A pointer is\_\_\_\_\_.

1. the address of a variable
2. an indication of the variable to be accessed next
3. a variable for storing address
4. the data type of an address variable

**38.** Constructor is a special function, called whenever we\_\_\_\_\_.

1. create a function
2. instantiate an object of a class
3. destroy an object
4. create a class

**39.** Symbolic constant PI can be defined as:

1. `#define PI 3.14;`
2. `#define PI 3.14`
3. `#define PI=3.14`
4. `# include pi=3.14`

**40.** Object code is machine code but it is not\_\_\_\_\_and\_\_\_\_\_.

1. relocatable, executable
2. faster, efficient
3. compiled, debugged
4. tested, compiled

**41. The default visibility for the data members of the class is**

1. private
2. protected
3. public
4. accessible outside the class

**42. The \_\_\_\_\_ is called automatically when an object destroys.**

1. destructor
2. constructor
3. main program
4. default constructor

**43. Constructor is special type of function :**

1. which has no return type
2. which returns NULL pointer
3. which returns zero
4. which returns integer type data

**44. \_\_\_\_\_ variables are those that are defined outside of main.**

1. Local
2. Dynamic
3. Global
4. Static

**45. Within the statement obj1=obj2; obj1 will call the assignment operator function and obj2 will be passed as an argument to function.**

1. True

2. False

**46. When the compiler overload the assignment (=) operator by default then**

1. Class members are not assigned properly
2. Compiler does not allow default assignment operator
3. Compiler does member wise assignment.
4. None of the given

**47. It is possible to return an object from function using this pointer.**

1. True
2. False

**48. Overloaded assignment operator must be**

1. Member function of class
2. Non-member function of class
3. Friend function of class
4. Global function

**49. Let suppose**

**int a, b, c, d, e;**

**a = b = c = d = e = 42;**

This can be interpreted by the compiler as

1. (a = b = (c = (d = (e = 42))));
2. a = (b = (c = (d = (e = 42))));
3. a = b = (c = (d = (e = 42)));
4. (a = b) = (c = d) = (e = 42);

**50. In statement a+b+c, at first**

1. a+b is executed first
2. b+c is executed first
3. All executed at the same time
4. None of the given

**51. Suppose int i = 10; then what is the output of  
cout<<oct<<i;**

1. 10
2. 11
3. 12
4. 13

**52. ostream is a \_\_\_\_\_ operator.**

1. dependent
2. member
3. standalone
4. None of the given

**53. \_\_\_\_\_ must be included to use stream manipulation in your code.**

1. conio.h
2. iostream
3. stdlib.h
4. iomanip

**54. \_\_\_\_\_ operators are the ones that require only one operator to work.**

1. Unit
2. Unary
3. Single
4. None of the given

**55. The endl and flush are \_\_\_\_\_.**

1. Functions
2. Operators
3. Manipulators
4. Objects

**56. When operator function is implemented as member function then return type of function \_\_\_\_\_.**

1. Must be an object of same class
2. Must be user-defined data type
3. Must be built-in data type

4. Can be any data type

**57.** When a variable is defined as static in a class then \_\_\_\_\_.

1. Separate copy of this variable is created for each object
2. Only one copy is created for all objects of this class
3. A copy of this variable is created for only static objects
4. None of the given

**58.** Automatic variables are created on \_\_\_\_\_.

1. Heap
2. Free store
3. Static storage
4. stack

59. `cout<<i<< " "; cout<< d  
<<" "; cout<< f;`

Above statements can be written within statement of one line as:

1. `cout<<i<<" "<< d " "<< f << ;`
2. `cout<<i<<<< d <<<< f <<;` 3. `cout<<i<<" "<< d <<" "<< f;`
4. `cout<<i<< " "<< d <<" " f<<;`

**60.** dec, hex, oct are all \_\_\_\_\_.

1. Member functions
2. Objects of input/output streams
3. Parameterized manipulators
4. Non-parameterized manipulators

**61.** What will be the output of following statement?

`cout<<setfill('0')<<setw(7)<< 128;`

1. 0128128
2. 0000128

- 3. 1280000
- 4. 0012800

**62.** Which of the following syntax is best used to delete an array of 5 objects named 'string' allocated using new operator.

- 1. delete string;
- 2. delete []string;
- 3. delete string[];
- 4. delete string[5];

**63.** If we have a program that writes the output data(numbers) to the disc, and if we collect the output data and write it on the disc in one write operation instead of writing the numbers one by one.

In the above situation the area where we will gather the number is called

- 1. Heap
- 2. Stack
- 3. Buffer
- 4. Cache

**64.** The first parameter of operator function for << operator

\_\_\_\_\_.

- 1. Must be passed by value
- 2. Must be passed by reference
- 3. Can be passed by value or reference
- 4. Must be object of class

**65.** The second parameter of operator function for >> operator must always be passed

- 1. By reference
- 2. Function takes no argument
- 3. By value
- 4. None of the given

**66. The only operator that the compiler overloads for user define data type by default is**

1. Plus (+) operator
2. Minus (-) operator
3. Assignment (=) operator
4. Equal (==) operator

**67. Consider the following code, the printed value will be converted into:**

```
int n=10; cout<<oct<<n;
```

1. Base 8
2. Base 2
3. Base 10
4. Decimal number system

**68. \_\_\_\_\_variables are defined in the main.**

1. Global
2. Dynamic
3. Local
4. All

**69. ostream class is \_\_\_\_\_and not under our control.**

1. user-defined
2. built-in
3. both user-defined and built-in
4. None of the given

**70. The memory allocation in C++ is carried out with the help of\_**  
**\_\_\_\_\_.**

1. NULL pointer
2. new operator
3. dot operator
4. + operator

**71. If B is designated as friend of A, B can access A's non- public members.**

1. B cannot access private member of A
2. B cannot access protected member of A
3. A can access non-public members of B
4. A cannot access B

**72. If the request of new operator is not fulfilled due to insufficient memory in the heap\_\_\_\_\_.**

1. the new operator returns 2
2. the new operator returns 1
3. the operator returns 0
4. free operator returns nothing

**73. We should not use such variable names that are starting with\_\_\_\_because in C++, there are lots of internal constants and symbolic names that start with it.**

1. upper case alphabets
2. lower case alphabets
3. double underscore
4. None of the given

**74. The friend keyword provides access\_\_\_\_\_.**

1. in one direction only
2. in two directions
3. to all classes
4. to the data members of the friend class only

**75. The malloc function takes\_\_\_\_\_argument(s).**

1. two
2. three
3. four
4. one

**76. The constructor contains\_\_\_\_\_.**

1. return type
2. no return type
3. objects
4. classes

**77.** What will be the output of the following c++ code?

```
#include<iostream.h>
#define max 100
main()
{
 #ifdef max
 Cout<<"Hellow;
}
```

1. Hello
2. "Hellow"
3. Max is 100
4. Error

**78.** Once we have defined a symbolic constant value using #define, that value\_\_\_\_\_during program execution

1. can be changed
2. cannot be changed
3. varies
4. becomes zero

**79.** The memory allocation functions return a chunk of memory with a pointer of type\_\_\_\_\_.

1. integer
2. float
3. ptr
4. void

**80.** A class can be declared as a\_\_\_\_\_of other class.

1. member
2. member function
3. friend

4. part

**81.** To avoid dangling reference, don't return\_\_\_\_\_.

1. the reference of a local variable from the function
2. the reference of a global variable from the function
3. the reference of a static variable from the function
4. the reference of a private data member from the function

**82.** Constructor is itself a\_\_\_\_\_of C++ and\_\_\_\_\_.

1. class, can be overloaded
2. function, cannot be overloaded
3. function, can be overloaded
4. object, can not be initialized

**83.** The parameter passed to isdigit() function is \_\_\_\_\_ variable.

1. Character
2. Boolean
3. Integer
4. Float

**84.** char \*\*argv can be read as\_\_\_\_\_.

1. pointer to pointer
2. pointer to char
3. pointer to pointer to char
4. None of the given

**85.** To read command-line arguments, the main() function itself must be given\_\_\_\_\_arguments.

1. 1
2. 2
3. 3
4. 4

**86.** How many bytes an integer type pointer intPtr will jump in memory if the statement below is executed?

intPtr += 2 ;

- 1. 2
- 2. 4
- 3. 8
- 4. 12

**87.** The increment of a pointer depends on its\_\_\_\_\_.

- 1. variable
- 2. value
- 3. data type
- 4. None of the given

**88.** The statement `cout<<ypr` will show the \_\_\_\_\_  
theypr points to.

- 1. Value
- 2. memory address
- 3. variable
- 4. None of the given

**89.** \_\_\_\_\_is used as a dereferencing operator.

- 1. \*
- 2. +
- 3. -
- 4. None of the above

**90.** Transpose of a matrix means that when we interchange rows and columns\_\_\_\_\_.

- 1. the first row becomes the Last column
- 2. the first row becomes the first column
- 3. the Last row becomes the first column
- 4. the first column becomes the first row

**91.** Individual characters in a string stored in an array can be accessed directly using array\_\_\_\_\_.

- 1. superscript
- 2. script
- 3. subscript

4. value

**92. We can define a matrix as \_\_\_\_\_ array.**

1. Sorted
2. Unsorted
3. Single dimensional
4. Multi dimensional

**93. A \_\_\_\_\_ is an array of characters that can store number of character specified.**

1. Char
2. String
3. Multidimensional array
4. Data type

**94. Given a two dimensional array of integers, what would be the correct way of assigning the value 6 to the element at third row and fourth column?**

1. `array[3][4] = 6 ;`
2. `array[2][4] = 6 ;`
3. `array[4][3] = 6 ;`
4. `array[2][3] = 6 ;`

**95. \_\_\_\_\_ of a variable means the locations within a program from where it can be accessed.**

1. Data type
2. Visibility
3. Value
4. Reference

**96. Which of the following function call is "call by reference" for the following function prototype? `int add (int *)`;**

1. `add(&x);`
2. `add(int x);`
3. `add(x);`

4. `add(*x);`

**97.** Which of the following function call is "call by reference" for the following function prototype? `float add(float *);`

1. `add(&x);`

2. `add(float x);`

3. `add(x);`

4. `add(*x);`

**98.** Which of the function call is call by value for the following function prototype?

`float add(float);`

1. `add(&x);`

2. `add(x);`

3. `add(float x);`

4. `add(*x);`

**99.** Which of the function call is "call by value" for the following function prototype?

`float add(int);`

1. `add(&x);`

2. `add(x);`

3. `add(int x);`

4. `add(*x);`

**100.** Return type of a function that does not return any value must be \_\_\_\_\_.

1. `char`

2. `int`

3. `void`

4. `double`

**101.** \_\_\_\_\_ will be used for enclosing function statements into a block.

1. `" "`

2. `()`

3. `[]`

4. `{ }`

**102. What is the output of the following code if the 2nd case is true**

```
switch (var) {
case 'a': cout<<"apple"<<endl; case
'b':cout<<"banana"<<endl; case
'm':cout<<"mango"<<endl; default: cout<<"any
fruit"<<endl;
}
```

1. banana

2. banana

any fruit

3. banana

mango

any fruit

4. None of the given

**103. When the break statement is encountered in a loop's body, it transfers the control \_\_\_\_\_ from the current loop.**

1. Inside

2. Outside

3. To break statement

4. To continue statement

**104. What is the output of the following code if the 3rd case is true**

```
switch (var) {
case 'a':cout<<"apple"<<endl; case
'b':cout<<"banana"<<endl; case
'm':cout<<"mango"<<endl; default: cout<<"any
fruit"<<endl;
}
```

1. mango
2. mango  
any fruit
3. apple
4. None of the given

**105. What is the output of the following code, if the first case is true**

```
switch (var) {
case 'a':cout<<"apple"<<endl; case
'b':cout<<"banana"<<endl; case
'm':cout<<"mango"<<endl; default: cout<<"any
fruit"<<endl;
}
```

1. apple
2. apple  
any fruit
3. apple  
banana  
mango  
any fruit
4. none of above

**106. What will be the output of following code segment?**

```
for (int i = 2; i<10; i++) { if (i == 5)
continue; cout<<i<< "," ;
}
```

1. 2,3,7,8,9
2. 2,3,4,6,7,8,9
3. 2,3,4
4. 4,6,7,8,9

**107. \_\_\_\_\_statement is used to terminate the processing of a particular case and exit from switch structure.**

1. if
2. goto
3. break
4. continue

**108.** What will be the result of the expression  $j = i++$ ; if initially  $j = 0$  and  $i = 5$ ?

1. 0
2. 5
3. 6
4. 4

**109.** What will be the result of the expression  $k = ++m$ ; if initially  $k = 0$  and  $m = 4$ ?

1. 0
2. 5
3. 6
4. 4

**110.** What will be the result of the expression  $k = ++m$ ; if initially  $k = 0$  and  $m = 5$ ?

1. 0
2. 5
3. 6
4. 4

**111.** How many times the following do-while loop will execute?

```
int k = 10; do { cout<< "Statements" <<endl; k -= 2; } while(k>0);
```

1. 4
2. 5
3. 6
4. 7

**112.** Which of the following loops checks the test condition at the end of the loop?

1. While
2. Do-While
3. For
4. Nested Loop

**113.** The operators ++ and -- are used to increment or decrement the value of a variable by\_\_\_\_\_.

1. 3
2. 2
3. 1
4. 4

**114.** How many times the following loop will execute?

```
int j = 3; while(j > 0) { cout<< "Statements" <<endl; j -= 2; }
```

1. 0
2. 1
3. 2
4. 3

**115.** A\_\_\_\_\_structure specifies that an action is to be repeated while some condition remains true.

1. Control
2. Logical
3. Repetition
4. Relational

**116.** !( x> 3) means in C++ that

1. x is greater than 3
2. x is less than or equal to 3
3. x is less than 3
4. x is equal to 3

**117.** When the logical operator && combines two expressions then the result will be true only when the both expressions are\_.

1. Logical
2. Arithmetic
3. true
4. false

**118.** < and > both are\_\_\_\_\_operators.

1. Arithmetic
2. Relational
3. Logical
4. Mathematical

**119.** What will be the value of variable “input” if the initial value of input is 67?

```
if(input >= 50) input = input
+ 1; if(input <= 75) input =
input + 2; else
input = input - 1;
```

1. 68
2. 69
3. 70
4. 66

**120.**  $!(x < 3)$  means in C++ that

1. x is less than 3
2. x is greater than or equal to 3
3. x is greater than 3
4. x is equal to 3

**121.**  $!=$  operator is used to check whether the operand on the left-hand-side is \_\_\_\_\_ to the operand on the right-hand-side.

1. Less than or equal
2. Greater than or equal
3. Not equal

4. Approximately equal to

**122.** When the if statement consists more than one statement then enclosing these statement in curly braces is,

1. Not required
2. Good programming
3. Relevant
4. Must

**123.** The most suitable data type for number 325.25 is

\_\_\_\_\_.

1. char
2. int
3. short
4. float

**124.** What will be the result of arithmetic expression  $6+48/4*3$ ?

1. 10
2. 40.5
3. 42
4. 41

**125.** Which of the following will be the most appropriate data type to store the value 63.547?

1. Integer
2. Character
3. Short
4. Float

**126.** In the given expression which operator will be evaluated first?  $10 + (6 / 2) - 2 * 3$ ?

1. +
2. -
3. /

4. \*

**127.** What will be the value of the variable output in the given piece of code?

```
double output = 0;
output = (2 + 2) * 4 + 2 / (4 - 2);
```

1. 15
2. 17
3. 12
4. 11

**128.** It is the job of \_\_\_\_\_ to transfer the executable code from hard disk to main memory.

1. interpreter
2. Debugger
3. Linker
4. Loader

**129.** In computer systems there are mainly \_\_\_\_\_ type of softwares.

1. 1
2. 2
3. 3
4. 4

**130.** \_\_\_\_\_ will explain the function of a program.

1. Comments
2. Debugger
3. Compiler
4. Linker

**131.** if (a>b && a>c) then the condition will be true only if

1. Both a>b and a>c are true
2. a>b is false and a>c is true
3. a>b is true and a>c is false

4. Both  $a > b$  and  $a > c$  are false

**132.** A variable of character data type occupies \_\_\_\_\_ byte(s) in memory.

1. 1

2. 2

3. 4

4. 8

**133.** We must include the header file \_\_\_\_\_ to convert the value of one type into another type using built-in functions.

1. conio.h

2. stdlib.h

3. string.h

4. iostream.h

**134.** A function is a block of statements that can be defined once and used \_\_\_\_\_ in the program.

1. One time

2. Two times

3. Three times

4. As many times as user wants

**135.** Select the correct way to assign the address of first element of array to pointer?

1. `int *ptr = &data[1];` 2.

`int *ptr = &data;`

3. `int *ptr = data;`

4. `int *ptr = data[0];`

**136.** Consider the following code segment. What will be the output of following code?

```
int addValue (int *a){ int b = (*a)
+ 2; return b;
}
```

```
main() { int x
= 6;
cout<<addValue(&x)<<" "; cout<<x;
}
```

1. 6,8,6
2. 6,6,8
3. 6,8,8
4. 6,6,6

**137.** Here the code is given below. You have to identify the problem in the code.

```
while(i < 10) && (i > 24))
```

1. the logical operator && cannot be used in test condition
2. the while loop is an exit-condition loop
3. the test condition is always true
4. the test condition is always false

**138.** The correct syntax of do-while loop is \_\_\_\_\_.

1. (condition) while; do {statements};
2. {statements;} do-while();
3. while(condition); do {statements};
4. do {statements;} while (condition);

**139.** Matrix is defined as \_\_\_\_\_.

1. Single dimensional array
2. Multi-dimensional array
3. Vector product
4. Scalar product

**140.** In programming, comments are used to explain the functioning of the \_\_\_\_\_.

1. Debugger
2. Editor
3. Program

4. Linker

**141. Operating System is a type of a/an\_\_\_\_\_.**

1. application software
2. system software
3. computer language
4. interpreter

**142. From the options given, you need to choose the option which is true for the given code.**

```
for (int i = 1; i>0; i++) {
/*loop code*/
}
```

1. the logical operator && cannot be used in a test condition
2. the while loop is an exit-condition loop
3. the test condition is always false
4. the test condition is always true

**143. Which of the following values are used in C/C++ to represent true and false?**

1. 1 and 0
2. 1 and -1
3. 11 and 00
4. any numerical value

**144. 'While' loop may execute\_\_\_\_\_or more times.**

1. three
2. zero
3. two
4. one

**145. Body of any function is enclosed within\_\_\_\_\_.**

1. { }
2. ( )
3. [ ]

4. " "

**146.** What will be the correct syntax for initialization of a pointer ptr with string "programming"?

1. char ptr = 'programming';
2. char \*ptr = "programming";
3. char \*ptr = 'programming';
4. \*ptr = "programming";

**147.** Which one of the given option is not a mode for reading/writing the data from a file?

1. in
2. out
3. trunc
4. get

**148.** Which of the following operators is used to access the value of variable pointed by a pointer?

1. \* operator
2. -> operator
3. && operator
4. &operator

**149.** In case of single dereferencing, the value of the \_\_\_\_\_ is the address of the \_\_\_\_\_.

1. pointer, variable
2. pointer, constant
3. variable, pointer
4. constant, pointer

**150.** The remainder (%) operator is a \_\_\_\_\_ operator.

1. Logical
2. Arithmetic
3. Relational
4. Conditional

**151. What will be the output of following code? int x = 10;**  
`cout<<"x="<<x;`

1. 10
2. "x=10"
3. x=10
4. 10=x

**152. The purpose of using cout<< is to\_\_\_\_\_.**

1. Display information on the screen
2. Read the data from keyboard
3. Read the data from a file
4. Write into a file

**153. Which of the following data types will be assumed if no data type is specified with constant?**

1. short
2. float
3. int
4. double

**154. When an array element is passed to a function, it is passed by\_.**

1. reference
2. data type
3. value
4. data

**155. While programming, it is good to provide an easy to understand and easy to use interface; this programming skill is called\_\_\_\_\_.**

1. scalability
2. usability
3. reliability
4. sustainability

**156.** \_\_\_\_\_executes all the lines before error and stops at the line which contains the error.

1. **Intrepreter**
2. Compiler
3. Linker
4. Debugger

**157.** Which of the following is the correct syntax to access the value of first element of an array using pointer ptr?

1. ptr[0]
2. \*(ptr+1)
3. ptr[1] 4. **\*ptr[0]**

**158.** C is a/an\_\_\_\_\_language.

1. low level
2. object based
3. object oriented
4. **function oriented**

**159.** \_\_\_\_\_of a function is also known as signature of a function.

1. Definition
2. **Declaration**
3. Calling
4. Invoking

**160.** \_\_\_\_\_are very good tools for code reuse.

1. operators
2. loops
3. **functions**
4. variables

**161.** If any break statement is missed in switch statement then\_\_\_\_\_.

1. compiler will give error
2. this may cause a logical error
3. no effect on program
4. program stops its execution

**162.** A 2D array `multi[5][10]` can be accessed using the array name as `**multi`, this technique is called \_\_\_\_\_.

1. Single referencing
2. Single dereferencing
3. Double referencing
4. Double dereferencing

**163.** In C/C++, the default command line arguments passed to the `main` function are \_\_\_\_\_.

1. `float argc, char **argv`
2. `int argc, char **argv`
3. `int *argc, char *argv`
4. `int argc, float **argv`

**164.** A record is a group of related \_\_\_\_\_.

1. Data
2. Fields
3. Bytes
4. Files

**165.** The microsoft word document (.doc) is a kind of \_\_\_\_\_.

1. Sequential File
2. Random AccessFile
3. Binary Access File
4. Executable File

**166.** NULL character is used to indicate the \_\_\_\_\_ of string.

1. Start
2. End

3. Begin
4. Middle

**167. How many dimensions does n-dimensional array has?**

1. n dimensions
2. 2n dimensions
3. (n+1) dimensions
4. (n-1) dimensions

**168. Which of the following function call is "call by reference" for the following function prototype?**

1. func(int &num);
2. func(&num);
3. func(\*num);
4. func(num);

**169. The loop which is most suitable to be used when the number of iterations is known is called\_\_\_\_\_.**

1. for
2. while
3. do-while
4. all looping processes require that the iterations be known.

**170. In C/C++, the string constant is enclosed in\_\_\_\_\_.**

1. curly braces { }
2. parentheses ( )
3. single quotes ' '
4. double quotes " "

**171. In order to get the right most digit of a number, we divide this number by 10 and take\_\_\_\_\_.**

1. Its remainder
2. Its quotient
3. Its divisor
4. The number

**172.** What is the correct syntax to declare an array of size 10 of int data type?

1. int [10] name;
2. name[10] int;
3. int name[10];
4. int name[];

**173.** How many bytes of memory are occupied by array 'str'? char str[]  
= "programming";

1. 10
2. 11
3. 12
4. 13

**174.** Suppose that an integer type pointer contains a memory address 0x22f230. What will be the new memory address if we increment this pointer by one?

1. 0x22f231
2. 0x22f234
3. 0x22f226
4. 0x22f238

**175.** Which of the following if missing would result in infinite recursion in case of recursive function?

1. Recursive call
2. Base case
3. Function parameters
4. Local variables

**176.** Whenever we use a library function or a predefined object or macro, we need to use a\_\_\_\_\_.

1. source file
2. object file
3. header file
4. exe file

**177.** Switch statement deals with\_\_\_\_\_type of data.

1. Integer
2. Float
3. Character
4. Both Integer and Character

**178.** Both compiler and\_\_\_\_\_are used to translate program into machine language code.

1. debugger
2. linker
3. loader
4. interpreter

**179.** TWAIN stands for\_\_\_\_\_.

1. Technology With An Interesting Name
2. Technology Without An Informative Name 3.
- Technology Without An Interesting Name
4. Technology With An Informative Name

**180.** The parameter passed to isdigit() function is\_\_\_\_\_.

1. a character variable
2. a boolean variable
3. an integer variable
4. a character string

**181.** C++ views each file as a sequential stream of \_\_\_\_\_.

1. Bits 2.
- Bytes
3. Numbers
4. Words

**182.** Structure is a collection of\_\_\_\_\_under a single name.

1. only functions

2. only variables
3. both functions and variables
4. only data types

**183.** The default mode for writing into a file using ofstream object is \_\_\_\_.

1. out
2. bin
3. app
4. ate

**184.** The memory address of the first element of an array is called\_\_.

1. floor address
2. foundation address
3. first address
4. base address

**185.** We want to access array in random order which of the following approach is better?

1. Pointer
2. Array index
3. Both pointers and array index are better
4. Matrix

**186.** The\_\_\_\_\_structure is a multiple-selection construct which makes the code more efficient and easy to read and understand.

1. multiple-if
2. switch
3. if-else
4. else-if

**187.** Which of the following is not a reserved word in C/C++?

1. int
2. float

- 3. double
- 4. sum

**188.** To access rand(), which library is required to be included in program?

- 1. conio.h
- 2. stdio.h
- 3. stdlib.h
- 4. iostream.h

**189.** What is the highest legal index for the following array? int arr[4]

- 1. 4
- 2. 3
- 3. 2
- 4. 1

**190.** Word processor is a type of a/an\_\_\_\_\_.

- 1. operating system
- 2. application software
- 3. device driver
- 4. utility software

**191.** Identify the correct option which is used for calling the function float area (int).

- 1. area(&num);
- 2. area(num);
- 3. area(int num);
- 4. area(\*num);

**192.** The\_\_\_\_\_statement allows us to select from multiple choices based on a set of fixed values for a given expression.

- 1. switch
- 2. break
- 3. continue

4. goto

**193. C is widely known as development language of \_\_\_\_\_operating system.**

1. Windows

2. Unix

3. Mac OS

4. Linux

**194. To convert the value of one type into another type using built-in functions, we include\_\_\_\_\_header file.**

1. conio.h

2. stdlib.h

3. iostream.h

4. string.h

**195. The keyword\_\_\_\_\_is used to get some value back from a function.**

1. return

2. break

3. continue

4. goto

**196. The function seekg() takes\_\_\_\_\_parameter(s).**

1. 0

2. 1

3. 2

4. 3

**197. The function write() takes\_\_\_\_\_as parameter(s).**

1. String of pointer type

2. String of variable lengths, no. of bytes to be read and flags

3. Pointer array of characters and a delimiter 4.

String and no. of bytes to be written

**198.** When the logical operator AND (&&) combines two expressions exp1 and exp2 then the result will be true only \_\_\_\_\_.

1. When both exp1 and exp2 are true
2. When both exp1 and exp2 are false
3. When exp1 is true and exp2 is false
4. When exp1 is false and exp2 is true

**199.** Syntax of a union is identical to \_\_\_\_\_.

1. structure
2. class
3. function
4. loop

**200.** Which one of the symbol is used to represent a decision in a flow chart?

1.

2. Rectangle correct

3.

4.

**201.** In Flow Chart, flow of control is represented by \_\_\_\_\_.

1. Rectangle
2. Circle
3. Diamond
4. Arrow

**202.** There can be \_\_\_\_\_ 'default' statement(s) in any switch structure.

- 1. 1
- 2. 2
- 3. 3
- 4. n

**203.** The condition in loop should be a(n)\_\_\_\_\_.

- 1. Constant Expression
- 2. Boolean Expression
- 3. Primary Expression
- 4. Arithmetic Expression

**204.** How many nested loops would be required to manipulate n-dimensional array?

- 1. n
- 2.  $n + 1$
- 3.  $n - 1$
- 4.  $2n$

**205.** Which of the following is not an example of int data type?

- 1. 0
- 2. -32
- 3. 65531
- 4. -4

**206.** We should use\_\_\_\_\_for clarity and to force the order of evaluation in an expression.

- 1. brackets []
- 2. parenthesis ()
- 3. curly braces {}
- 4. quotation marks " "

**207.** Which of the following is the starting index of an array in C++?

- 1. 0
- 2. 1

- 3. -1
- 4. 2

**208.** The statement `x += y` can be interpreted as\_\_\_\_\_.

- 1. Adding the value of the x to the value of the y and storing the result in x
- 2. Adding the value of the y to the value of x, store the result in y
- 3. Adding the value of the x to the value of x, store the result in x
- 4. Adding the value of the y to the value of y, store the result in x

**209.** Given a 2D array of integers, what would be the correct way of assigning the value 5 to the element at second row and third column?

- 1. `m[2][3] = 5;`
- 2. `m[3][2] = 5;`
- 3. `m[1][2] = 5;`
- 4. `m[2][3] = '5';`

**210.** Array is a data structure that stores\_\_\_\_\_.

- 1. Memory addresses
- 2. Variables
- 3. Data type
- 4. Data

**211.** A program statement that invokes a function is called\_\_\_\_\_.

- 1. function declaration
- 2. function call
- 3. function definition
- 4. function prototype

**212.** If a function has been declared but not defined before its function call then it is termed as\_\_\_\_\_.

1. logical error
2. syntax error
3. run time error
4. program time error

**213. The compiler of C language is written in \_\_\_\_\_ language.**

1. JAVA
2. BASIC
3. FORTRAN
4. C

**214. Which one of the below functions is not included in ctype.h header file?**

1. isdigit(int c)
2. isxdigit(int c)
3. tolower(int c)
4. getdigit(int c)

**215. Which function is used to locate the first occurrence of a character in any string?**

1. strchr()
2. strstr()
3. strtok()
4. strlen()

**216. To access the data members of structure, \_\_\_\_\_ is used.**

1. Logical operator
2. Dereference operator
3. Dot operator
4. Address operator

**217. In the following nested For Loop, which loop will run most number of times?**

```
for(inti = 0; i < 5; i++)
{
```

```
for(int k = 0; k < 5; k++)
{

}
}
```

1. Outer loop
2. Inner loop
3. Both loops run equal number of times
4. Depends upon the statements in the inner loop's body

**218.** Structure use \_\_\_\_\_ allocation.

1. Queue
2. Heap
3. Cache
4. Stack

**219.** \_\_\_\_\_ function give the position of the next character to be read from that file.

1. tellp()
2. tellg()
3. seekg()
4. seekp()

**220.** What will be the size of the following character array? char name[] = "Adeel";

1. 5
2. 6
3. 4
4. 7

**221.** Function prototype is written,

1. Within main function
2. After the return statement in main
3. Before the return statement in main
4. Before call of that function

**222. Which one of the following languages has been used to write the compiler of "C" language?**

1. Java
2. Fortran
3. Basic
4. C

**223. A hierarchy of classes which are used to deal with console and disk files are called\_\_\_\_\_.**

1. Stream classes
2. Simple classes
3. Binary classes
4. IO classes

**224. \_\_\_\_\_stops execution at the line that contains error(s) in the code.**

1. Compiler
2. Debugger
3. Interpreter
4. Linker

**225. C++ is a \_\_\_\_\_language.**

1. High level
2. Low level
3. Machine
4. Assembly language

**226. How many elements are stored in the following? int matrix [4][5];**

1. 9
2. 20
3. 25
4. 10

**227. \_\_\_\_\_is a substitute of multiple if statement.**

1. if. .elseif statement
2. Continue statement
3. Break statement
4. Default statement

**228. if**

int sum = 54;

Then the value of the following statement is sum = sum - 3;

1. 52
2. 50
3. 51
4. 57

**229. What will be the correct syntax for declaration of the following statement?**

"ptr is a constant pointer to an integer"

1. const \* int myptr;
2. const int \*myptr;
3. int const \*ptr;
4. int \*const ptr;

**230. \_\_\_\_\_operator is used to pass the address of a variable in call by reference method.**

1. %
2. +
3. @
4. &

**231. \_\_\_\_\_data type can operate on modulus operator.**

1. int
2. float
3. char
4. double

**232. Whenever some number is added in an array name, it will jump as many\_\_\_\_\_as the added number.**

1. rows
2. value
3. column
4. bytes

**233.** Suppose that an integer type pointer contains a memory address 0x22f220. What will be the new memory address if we increment this pointer by one?

1. 0x22f221
2. 0x22f222
3. 0x22f223
4. 0x22f224

**234.** \_\_\_\_\_ is the pointer which determines the position in a file from where the next read operation occurs.

1. put
2. seek
3. get
4. tell

To access the data members of structure\_\_\_\_\_is used.

► dot operator (.)

- \* operator
- à operator
- None of given.

► One

- 
- Two
  - N arguments

The declarator of Plus (+) member operator function is

- ▶ **Class-Name operator + (Class-Name rhs)**
- ▶ operator Class-Name + ( )
- ▶ operator Class-Name + ( rhs)
- ▶ Class-Name operator + ( )

**Question No: 7 ( M - 1 ) .**

Friend function of a class is\_\_\_\_\_.

- ▶ Member function
- ▶ **Non-member function**
- ▶ Private function
- ▶ Public function

**Question No: 8 ( M - 1 ) .**

We can also create an array of user define data type.

- ▶ **True**
- ▶ False

**Question No: 9 ( M - 1 ) .**

What is the sequence of event(s) when deallocating memory using delete operator?

- ▶ Only block of memory is deallocated for objects
- ▶ Only destructor is called for objects
- ▶ **Memory is deallocated first before calling destructor**
- ▶ Destructor is called first before deallocating memory

**Question No: 10 ( M - 1 ) .**

Deleting an array of objects without specifying [] brackets may lead to memory leak

- ▶ **True**
- ▶ False

**Question No: 11 ( M - 1 ) .**

Which of the following data type(s) can operate on modulus operator '%'?

- ▶ float, int
- ▶ float, double
- ▶ **int**
- ▶ char

**Question No: 12 ( M - 1 ) .**

Array is passed by value to a function by default.

► True

► False

**Question No: 13 ( M - 1 ) .**

With template function, the compiler automatically detects the passed data and generates a new copy of function using passed data.

► True

► False

**Question No: 14 ( M - 1 ) .**

What will be the correct syntax to initialize all elements of two-dimensional array to value 0?

► `int arr[2][3] = {0,0} ;`

► `int arr[2][3] = {{0},{0}} ;`

► `int arr[2][3] = {0},{0} ;`

► `int arr[2][3] = {0} ;`

**Question No: 15 ( M - 1 ) .**

When an operator function is define as member function then operand on the left side of operator must be an object.

► True

► False

**Question No: 16 ( M - 1 ) .**

break statement can be used outside a loop or switch statement.

► True

► False

**Question No: 17 ( M - 1 ) .**

The keyword\_\_\_\_\_is used to return some value from a function.

► return

► break

► continue

► goto

**Question No: 18 ( M - 1 ) .**

Every data member or function inside the structure is\_\_\_\_\_by default whereas everything declared inside a class is\_\_\_\_\_by default.

- ▶ private, public
- ▶ **public, private**
- ▶ private, protected
- ▶ public, protected

**Question No: 19 ( M - 1 ) .**

Which of the following is true for the C++ statement given below?

**int &ref = val ;**

- ▶ It creates a synonym for variable 'val'
- ▶ It creates an alias for variable 'val'
- ▶ It's a new name for variable 'val'

▶ **All of the given options**

**Question No: 20 ( M - 1 ) .**

If overloaded plus operator is implemented as non-member function then which of the following statement will be true for the statement given below?

**obj3 = obj1 + obj2 ;**

- ▶ obj2 will be passed as an argument to+ operator whereas obj2 will drive the + operator
- ▶ obj1 will drive the + operator whereas obj2 will be passed as an argument to + operator
- ▶ **Both objects (obj1, obj2) will be passed as arguments to the + operator**
- ▶ Any of the objects (obj1, obj2) can drive the + operator

**Question No: 21 ( M - 1 ) .**

Unary operator implemented as member function takes\_\_\_\_\_arguments whereas non-member function takes\_\_\_\_\_arguments.

- ▶ One, zero
- ▶ **Zero, one**
- ▶ One, two
- ▶ Two, one

**Question No: 22 ( M - 1 ) .**

The input/output streams cin and cout are\_\_\_\_\_therefore have\_\_\_\_\_.

- ▶ Structures, function
- ▶ Objects, member functions
- ▶ Functions, objects

▶ None of the given options

**Question No: 23 ( M - 1 ) .**

If a symbolic constant has been defined, it will be an error to define it again.

▶ True

▶ False

**Question No: 24 ( M - 1 ) .**

Every class contains\_\_\_\_\_.

▶ Constructor

- ▶ Destructor
- ▶ Both a constructor and a destructor
- ▶ None of the given options

**Question No: 25 ( M - 1 ) .**

new and delete keywords are\_\_\_\_\_in C++ language.

- ▶ Built-in- Function
- ▶ Operators

▶ Memory Allocation Function

▶ None of the given options

**Question No: 26 ( M - 1 ) .**

Consider the following code segment.

```
class M {
public:
 M &operator+(const M &);
...
};
```

`p + q` **▶** `//code of line implies that p.operator+(q)`

...

Let assume if p and q are class objects then function is implemented as \_\_\_\_\_

**▶ Member function**

- ▶ Non-member function
- ▶ Friend function
- ▶ None of the given options

**Question No: 27 ( M - 1 ) .**

Assignment operator is----- associative.

**▶ right**

- ▶ left
- ▶ binary
- ▶ unary

**Question No: 28 ( M - 1 ) .**

Static variable which is defined in a function is initialized\_\_\_\_\_.

**▶ Only once during its life time**

- ▶ Every time the function call
- ▶ Compile time of the program
- ▶ None of the above

**Question No: 29 ( M - 1 ) .**

We can not define a variable of user-defined data type in the class.

▶ True

**▶ False**

**Question No: 30 ( M - 1 ) .**

A constructor that will create a new object with a full copy of the other object, is copy is known as \_\_\_\_\_

▶ deep copy

▶ shallow copy

▶ constructor copy

▶ none of the options

**CS201**

Mid-Term

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